MATHEMATICS

QUADRATIC EQUATION

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QUADRATIC EQUATION

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BASED ON ROOTS & FORMATION OF QUADRATIC EQUATION

1. If α, β are roots of the equation $2x^2 - 35x + 2 = 0$, then the value of $(2\alpha - 35)^3 . (2\beta - 35)^3$ is equal to :

(A) 1 (B) 8 (C) 64 (D) None of these

यदि α, β समीकरण $2x^2 - 35x + 2 = 0$ के मूल है, तो $(2\alpha - 35)^3 . (2\beta - 35)^3$ का मान बराबर है :

(A) 1 (B) 8 (C) 64 (D) इनमें से कोई नहीं

Ans. C

Sol. $2x^2 - 35x + 2 = 0$

α, β are the roots

$\alpha\beta = 1$

$\Rightarrow 2\alpha^2 - 35\alpha + 2 = 0$

$\alpha(2\beta - 35) = -2$

$$2\alpha - 35 = \frac{-2}{\alpha}$$

$2\beta^2 - 35\beta + 2 = 0$

$\beta(2\beta - 35) = -2$

$$2\beta - 35 = \frac{-2}{\beta}$$

Now, $(2\alpha - 35)^3 (2\alpha - 35)^3$

$$= \left(\frac{-2}{\alpha}\right)^3 \cdot \left(\frac{-2}{\beta}\right)^3 = \frac{-8}{\alpha^3}\frac{-8}{\beta^3} = \frac{64}{(\alpha\beta)^3} = \frac{64}{1^3} = 64$$

2. If the roots of the equation $ax^2 + x + b = 0$ be real and distinct, then the roots of the equation $x^2 - 4\sqrt{ab}\,x + 1 = 0$ will be :

(A) Rational (B) Irrational (C) Real (D) Imaginary

यदि समीकरण $ax^2 + x + b = 0$ के मूल वास्तविक तथा भिन्न हों, तो समीकरण $x^2 - 4\sqrt{ab}\,x + 1 = 0$ के मूल होंगे :

(A) परिमेय (B) अपरिमेय (C) वास्तविक (D) काल्पनिक

Ans. D

Sol. $ax^2 + x + b = 0$ for $x^2 - 4\sqrt{ab}x + 1 = 0$

$\because D > 0$ $D = 16ab - 4$

$1 - 4ab > 0$ $= 4(4ab - 1)$

$4ab - 1 < 0$ $= 4 \times (< 0)$

$D < 0$

$\therefore$ roots will be imaginary

3. If α and β are the root of $ax^2 + bx + c = 0$, then the value of $\left\{\frac{1}{a\alpha+b}+\frac{1}{a\beta+b}\right\}$ is :

(A) $\frac{a}{bc}$ (B) $\frac{b}{ca}$ (C) $\frac{c}{ab}$ (D) None of these

यदि α तथा β समीकरण $ax^2 + bx + c = 0$ के मूल हों, तो $\left\{\frac{1}{a\alpha+b}+\frac{1}{a\beta+b}\right\}$ का मान होगा :

(A) $\frac{a}{bc}$ (B) $\frac{b}{ca}$ (C) $\frac{c}{ab}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $ax^2 + bx + c = 0$
α, β are the roots.

$$\alpha+\beta=\frac{-b}{a}, \alpha\beta=\frac{c}{a}$$

$$\Rightarrow a\alpha^2+b\alpha+c=0$$

$$\alpha(a\alpha+b)+c=0$$

$$a\alpha+b=\frac{-c}{\alpha}$$

$$a\beta^2+b\beta+c=0$$

$$\beta(a\beta+b)=-c$$

$$a\beta+b=\frac{-c}{\beta}$$

Now, $\frac{1}{(a\alpha+b)}+\frac{1}{(a\beta+b)}=\frac{\alpha}{-c}+\frac{\beta}{-c}=\frac{(\alpha+\beta)}{-c}=\frac{+b}{a\cdot(+c)}=\frac{b}{ac}$

4. Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10}-2a_8}{3a_9}$ is :

[JEE(Main)Feb.2021]

माना $x^2 - 6x - 2 = 0$ के मूल α तथा β हैं। यदि $n \geq 1$ के लिए $a_n = \alpha^n - \beta^n$ है, तो $\frac{a_{10}-2a_8}{3a_9}$ का माना है :

[JEE(Main)Feb.2021]

(A) 1 (B) 3 (C) 4 (D) 2

Ans. D

Sol. $$\frac{\alpha^{10}-\beta^{10}-2\left(\alpha^8-\beta^8\right)}{3\left(\alpha^9-\beta^9\right)}=\frac{\alpha^8\left(\alpha^2-2\right)-\beta^8\left(\beta^2-2\right)}{3\left(\alpha^9-\beta^9\right)}$$

$$=\frac{\alpha^8(6\alpha)-\beta^8(6\beta)}{3\left(\alpha^9-\beta^9\right)}=\frac{6\left(\alpha^9-\beta^9\right)}{3\left(\alpha^9-\beta^9\right)}=\frac{6}{3}=2$$

$\because\ \alpha^2=6\alpha-2=0$

$\alpha^2-2=6\alpha$

$\beta^2-2=6\beta$

5. If one root of the equation $x^2+px+12=0$ is 4, while the equation $x^2+px+q=0$ has equal roots, then the value of 'q' is :

यदि समीकरण $x^2+px+12=0$ का एक मूल 4 हो, जबकि समीकरण $x^2+px+q=0$ के मूल समान हों, तो 'q' का मान होगा :

(A) 49/4 (B) 12 (C) 3 (D) 4

Ans. A

Sol. $x^2+px+12=0$
4, β are the roots.
$\therefore\ 16+4p+12=0$
$p=-7$
$\because$ Equation $x^2+px+q=0$ has equal roots.
α, α are the roots.
$\therefore\ x^2-7x+q=0$
α, α are ithe roots
$2\alpha=-7$

$$\alpha=\frac{-7}{2}$$

& $\alpha\cdot\alpha=q$
$q=\alpha^2$

$$q=\frac{49}{4}$$

6. The value of a for which the sum of the squares of the roots of the equation $x^2-(a-2)x-a-1=0$ assume the least value is :

समीकरण $x^2-(a-2)x-a-1=0$ के मूलों के वर्गों के योग को न्यूनतम होने के लिए a का मान है :

(A) 1 (B) 0 (C) 3 (D) 2

Ans. A

Sol. $x^2-(a-2)x-a-1=0$
$x^2-(a-2)-(a+1)=0$

α, β are the roots

$\alpha + \beta = (a - 2)$

$\alpha\beta = -(a + 1)$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (a - 2)^2 + 2(a + 1)$

$= a^2 + 4 - 4a + 2a + 2$

$= a^2 - 2a + 6$

for minimum of a, $\frac{-b}{2a} = \frac{2}{2} = 1$

7. If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals :

यदि समीकरण $x^2 - bx + c = 0$ के मूल दो क्रमागत पूर्णांक हैं, तो $b^2 - 4c$ का मान है :

(A) –2 (B) 3 (C) 2 (D) 1

Ans. D

Sol. $x^2 - bx + c = 0$

$\alpha + \alpha + 1 = b$

$2\alpha + 1 = b$

$\alpha = \frac{(b-1)}{2}$

& $\alpha \cdot (\alpha + 1) = c$

$\frac{(b-1)}{2}\left(\frac{b-1}{2} + 1\right) = c$

$\frac{(b-1)(b+1)}{4} = c$

$b^2 - 1 = 4c$

$\Rightarrow b^2 - 4c = 1$

8. Let a,b,c be real numbers with $a \neq 0$ and α, β be the roots of the equation $ax^2 + bx + c = 0$. Express the roots of $a^3x^2 + abcx + c^3 = 0$ in terms of α, β : **[JEE-2001]**

(A) $\alpha\beta^2$ (B) $\alpha^2\beta$ (C) both (A) and (B) (D) None of these

माना a,b,c वास्तविक संख्याऐं है एवं $a \neq 0$ तथा α, β समीकरण $ax^2 + bx + c = 0$ के मूल हैं, तो समीकरण $a^3x^2 + abcx + c^3 = 0$ के मूलों को α, β के पदों में व्यक्त होंगे : **[JEE-2001]**

(A) $\alpha\beta^2$ (B) $\alpha^2\beta$ (C) (A) तथा (B) दोनों (D) इनमें से कोई नहीं

Ans. C

Sol. $\because \quad ax^2 + bx + c = 0 \begin{matrix}\nearrow \alpha \\ \searrow \beta\end{matrix} \quad \Rightarrow \quad \alpha + \beta = -\frac{b}{a} \quad \Rightarrow \quad \alpha\beta = \frac{c}{a}$

$\because$ Let $a^3x^2 + (a\ b\ c)\ x + c^3 = 0 \begin{matrix}\nearrow \gamma \\ \searrow \delta\end{matrix}$

$\therefore \quad \gamma + \delta = -\frac{abc}{a^3} = \left(-\frac{b}{a}\right)\left(\frac{c}{a}\right) = (\alpha\ \beta)\ (\alpha + \beta) = \alpha^2\beta + \alpha\beta^2$ ....(i)

$\because \quad \gamma\ \delta = \left(\frac{c}{a}\right)^3 = (\alpha\beta)^3 = (\alpha^2\beta)\ (\alpha\beta^2)$ ....(ii)

From (i) and (ii) we can say that $\gamma = \alpha^2\beta$ and $\delta = \alpha\beta^2$ and $\gamma = \alpha\beta^2$ and $\delta = \alpha^2\beta$

9. If one root of the equation $x^2 + px + q = 0$ is the square of the other, then : **[JEE-2004]**

यदि समीकरण $x^2 + px + q = 0$ का एक मूल दूसरे मूल का वर्ग है तो : **[JEE-2004]**

(A) $p^3 + q^2 - q(3p + 1) = 0$ (B) $p^3 + q^2 + q(1 + 3p) = 0$

(C) $p^3 + q^2 + q(3p - 1) = 0$ (D) $p^3 + q^2 + q(1 - 3p) = 0$

Ans. D

Sol. $x^2 + px + q = 0$

$\alpha + \alpha^2 = -p$ ...(i)

$\alpha^3 = q$

$\alpha = q^{\frac{1}{3}}$

$\therefore q^{\frac{1}{3}} + q^{\frac{2}{3}} = -p$

$q^{\frac{2}{3}} + q^{\frac{1}{3}} + p = 0$

$q^{\frac{1}{3}} + p = -q^{\frac{2}{3}}$

cubing both sides

$\left(q^{\frac{1}{3}} + p\right)^3 = -q^2$

$q + p^3 + 3pq^{\frac{1}{3}}\left(q^{\frac{1}{3}} + p\right) = -q^2$

$q + p^3 + 3pq^{\frac{1}{3}}\left(-q^{\frac{2}{3}}\right) = -q^2$

$q + p^3 - 3pq + q^2 = 0$

$p^3 + q^2 + q\ (1 - 3p) = 0$

10. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is :

[JEE-2010]

माना p तथा q वास्तविक संख्याए इस प्रकार है कि $p \neq 0$, $p^3 \neq q$ तथा $p^3 \neq -q$. यदि α व β दो अशून्य समिश्र संख्याए $\alpha + \beta = -p$ एवं $\alpha^3 + \beta^3 = q$ को संतुष्ट करती है तो द्विघात समीकरण जिसके मूल $\frac{\alpha}{\beta}$ तथा $\frac{\beta}{\alpha}$ है, होगी : **[JEE-2010]**

(A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$

(C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

Ans. B

Sol. Product = 1

$$\text{Sum} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta}$$

Since $\alpha^3 + \beta^3 = q \quad \Rightarrow \quad -p(\alpha^2 + \beta^2 - \alpha\beta) = q$

$$((\alpha + \beta)^2 - 3\alpha\beta) = -\frac{q}{p} \quad \Rightarrow \quad p^2 + \frac{q}{p} = 3\alpha\beta$$

$$\text{Hence sum} = \frac{\left\{p^2 - \frac{2}{3}\left(\frac{p^3+q}{p}\right)\right\}3p}{(p^3+q)} = \frac{p^3-2q}{p^3+q}$$

so the equation is $x^2 - \left(\frac{p^3-2q}{p^3+q}\right)x + 1 = 0$

$\Rightarrow \quad (p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$

11. The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root in common. The other roots of the first and second equations are integers in the ratio 4 : 3. Then the common root is :

द्विघातीय समीकरणों $x^2 - 6x + a = 0$ तथा $x^2 - cx + 6 = 0$ का एक मूल उभयनिष्ठ है। प्रथम तथा द्वितीय समीकरण के दूसरे मूल पूर्णांक हैं तथा 4 : 3 के अनुपात में है। तो उभयनिष्ठ मूल है :

(A) 4 (B) 3 (C) 2 (D) 1

Ans. C

Sol. $x^2 - 6x + a = 0$ $\qquad$ $x^2 - cx + 6 = 0$

$4\alpha\beta = a$ $\qquad$ $3\alpha\beta = 6$

$\alpha\beta = \frac{a}{4}$ $\qquad$ $\alpha\beta = 2$

$\Rightarrow \frac{a}{4} = 2 \Rightarrow a = 8$

$\Rightarrow$ Equation $x^2 - 6x + 8 = 0$; $x = 4, 2$

Equation (i) – (ii)

$(c - 6)x + (a - 6) = 0$

$$x = \frac{6 - a}{c - 6}$$

$$x = \frac{-2}{c - 6}$$

Now, $\frac{4}{(c-6)^2} + \frac{12}{(c-6)} + 8 = 0$

$4 + 12 (c - 6) + 8 (c - 6)^2 = 0$

$1 + 3 (c - 6) + 2 (c^2 + 36 - 12c) = 0$

$1 + 3c - 18 + 2c^2 + 72 - 24c = 0$

$2c^2 - 21c + 55 = 0$

$2c^2 - 10c - 11c + 55 = 0$

$2c (c - 5) - 11 (c - 5) = 0$

$$c = 5 \text{ or } \frac{11}{2}$$

$\therefore$ Equation $x^2 - 5x + 6 =$; $x = 2$, 3 (rejected)

and $x^2 - \frac{11}{2}x + 6 = 0$

$2x^2 - 11x + 12 = 0$

$2x^2 - 8x - 3x + 12 = 0$

$2x (x - 4) - 3(x - 4) = 0$

$(x - 4) (2x - 3) = 0$

$$x = 4, \frac{3}{2}$$

$\therefore$ Common root = 4

12. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\alpha/2$, 2β be the roots of the equation $x^2 - qx + r = 0$. Then the value of 'r' is : **[JEE-2007]**

यदि α, β समीकरण $x^2 - px + r = 0$ के मूल है तथा $\alpha/2$, 2β समीकरण $x^2 - qx + r = 0$ के मूल है तो 'r' का मान है : **[JEE-2007]**

(A) $\frac{2}{9}(p-q)(2q - p)$ (B) $\frac{2}{9}(q - p)(2p - q)$ (C) $\frac{2}{9}(q - 2p)(2q - p)$ (D) $\frac{2}{9}(2p-q)(2q - p)$

Ans. D

Sol. $x^2 - px + r = 0$

$\alpha + \beta = p$ ...(i)

$\alpha\beta = r$

and $x^2 - qx + r = 0$

$\frac{\alpha}{2} + 2\beta = q$

$\alpha + 4\beta = 2q$ ...(ii)

$\frac{\alpha}{2} \cdot 2\beta = r \Rightarrow \alpha\beta = r$

Now, (i) – (ii)

$-3\beta = p - 2q$

$\beta = \frac{(2a - p)}{3}$

$\Rightarrow \alpha = p - \beta$

$\alpha = p - \frac{(2q - p)}{3}$

$\alpha = \frac{3p - 2q + p}{3}$

$\alpha = \frac{(4p - 2q)}{3}$

$\because \alpha\beta = r$

$\frac{(4p - 2q)}{3} \cdot \frac{(2q - p)}{3} = r$

$r = \frac{2}{9}(2p - q)(2q - p)$

13. If α and β be the roots of the equation $(x - a)(x - b) = c$ and $c \neq 0$, then roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are :

(A) a and c (B) b and c (C) a and b (D) a+ b and b + c

यदि α तथा β समीकरण $(x - a)(x - b) = c$ एवं $c \neq 0$ के मूल हों, तो समीकरण $(x - \alpha)(x - \beta) + c = 0$ के मूल होंगे :

(A) a तथा c (B) b तथा c (C) a तथा b (D) a+ b तथा b + c

Ans. C

Sol. $(x - a)(x - b) = c$

$x^2 - (a + b)x + ab - c = 0$

$\alpha + \beta = (a + b)$

$\alpha\beta = (ab - c)$

Now, $(x - \alpha)(x - \beta) + c = 0$

$x^2 - (\alpha + \beta)x + \alpha\beta + c = 0$

$x^2 - (a + b)x + ab - c + c = 0$

$x^2 - ax - bx + ab = 0$

$(x - a)(x - b) = 0$

$x = a, b$

14. If $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$ then the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is :

(A) 19/3 (B) 25/3 (C) – 19/3 (D) None of these

यदि $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$ हो, तो $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ का मान है :

(A) 19/3 (B) 25/3 (C) – 19/3 (D) इनमें से कोई नहीं

Ans. A

Sol. If $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$

$\alpha^2 - 5\alpha + 3 = 0$ $\beta^2 - 5\beta + 3 = 0$

$\Rightarrow x^2 - 5x + 3 = 0$

$\alpha + \beta = 5$

$\alpha\beta = 3$

Now $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{25-6}{3} = \frac{19}{3}$

IDENTITY

15. The number of values of the pair (a, b) for which $a(x + a)^2 + b(x^2 - 3x - 2) + x + 1 = 0$ is an identity in x is :

(A) 0 (B) 1 (C) 2 (D) infinite

युग्म (a, b) के मानों की संख्या जिनके लिए समीकरण $a(x + a)^2 + b(x^2 - 3x - 2) + x + 1 = 0$, x में एक सर्वसमिका है :

(A) 0 (B) 1 (C) 2 (D) अनन्त

Ans. A

Sol. $a(x + a)^2 + b(x^2 - 3x - 2) + x + 1 = 0$

$a(x^2 + a^2 + 2ax) + (bx^2 - 3bx - 2b) + x + 1 = 0$

$(a + b)x^2 + x(2a^2 - 3b + 1) + (a^3 - 2b + 1) = 0$

$(a + b)x^2 + (2a^2 - 3b + 1)x + (a^3 - 2b + 1) = 0$

$\because$ Equation is an identity

$\therefore a + b = 0$

$a = -b$

$2a^2 - 3b + 1 = 0$

$2b^2 - 3b + 1 = 0$

$2b^2 - 2b - b + 1 = 0$

$2b(b - 1) - 1(b - 1) = 0$

$(b - 1)(2b - 1) = 0$

$b = +1, +\frac{1}{2}$

$a^3 - 2b + 1 = 0$

$-b^3 - 2b + 1 = 0$

$b^3 + 2b - 1 = 0$ No real roots

No common value of the pair (a, b).

16. If $(a^3 - 3a - 2)x^2 + (a^2 - a - 2)x = 2a^4 - 10a + k$, is true for all real values of x then the value of k is :

यदि $(a^3 - 3a - 2)x^2 + (a^2 - a - 2)x = 2a^4 - 10a + k$, x के सभी वास्तविक मानों के लिये सत्य (सर्वसमिका) हो, तो k का मान होगा :

(A) 6 (B) –12 (C) –6 (D) 12

Ans. B

Sol. Equation : $(a^3 - 3a - 2)x^2 + (a^2 - a - 2)x = 2a^4 - 10a + k$ is true $\forall x \in R$

$\therefore$ This is an identity

$\Rightarrow a^3 - 3a - 2 = 0$,

$a^3 + a^2 - a^2 - a - 2a - 2 = 0$

$(a + 1)(a^2 - a - 2) = 0$

$a = -1, -1, 2$

$a^2 - a - 2 = 0$,

$a^2 - 2a + a - 2 = 0$

$a(a - 2) + 1(a - 2) = 0$

$(a + 1)(a - 2) = 0$

$a = -1, 2$

$2a^4 - 10a + k = 0$

for $a = -1$

$2 + 10 + k = 0$

$k = -12$

for $a = 2$

$32 - 20 + k = 0$

$k = -12$

17. If a, b, c are three distinct real numbers then the equation

$$\frac{(x-b)(x-c)}{(a-b)(a-c)}a^2 + \frac{(x-c)(x-a)}{(b-c)(b-a)}b^2 + \frac{(x-a)(x-b)}{(c-a)(c-b)}c^2 - x^2 = 0 \text{ has :}$$

(A) exactly one root (B) exactly two roots (C) no root (D) None of these

यदि a, b, c तीन भिन्न वास्तविक संख्याऐं है तो $\frac{(x-b)(x-c)}{(a-b)(a-c)}a^2 + \frac{(x-c)(x-a)}{(b-c)(b-a)}b^2 + \frac{(x-a)(x-b)}{(c-a)(c-b)}c^2 - x^2 = 0$ के :

(A) केवल एक मूल होगा (B) केवल दो मूल होंगे (C) कोई मूल नहीं (D) इनमें से कोई नहीं

Ans. D

Sol. $\because a \neq b \neq c, a, b, c \in R$

$$\frac{(x-b)(x-c)}{(a-b)(a-c)}a^2 + \frac{(x-c)(x-a)}{(b-c)(b-a)}b^2 + \frac{(x-a)(x-b)}{(c-a)(c-b)}c^2 - x^2 = 0$$

for $x = a$

$$\frac{(a-b)(a-c)}{(a-b)(a-c)}a^2 + 0 + 0 - a^2 = 0$$

$a^2 - a^2 = 0$

for $x = b$

$$0 + \frac{(b-c)(b-a)}{(b-c)(b-a)}b^2 + 0 - b^2 = 0$$

$b^2 - b^2 = 0$

$0 = 0$

for x = c

$$0+0+\frac{(c-a)(c-b)}{(c-a)(c-b)}c^2-c^2=0$$

$c^2 - c^2 = 0$

$0 = 0$

$\because$ Quadratic Equation having 3 roots. Therefore, this is an identity.

CONDITION OF COMMON ROOTS

18. If one root of the equations $x^2 + 2x + 3k = 0$ and $2x^2 + 3x + 5k = 0$ is common then the value of k is :

(A) 1, 2 (B) 0, –1 (C) 1, 3 (D) None of these

यदि समीकरणों $x^2 + 2x + 3k = 0$ व $2x^2 + 3x + 5k = 0$ का एक ही उभयनिष्ठ मूल हो, तो k का मान होगा :

(A) 1, 2 (B) 0, –1 (C) 1, 3 (D) इनमें से कोई नहीं

Ans. B

Sol. $x^2 + 2x + 3k = 0$

& $2x^2 + 3x + 5k = 0$

$\because$ One root of the Equation (i) is common.

$$\therefore \alpha^2+2\alpha+3k=0$$

$$2\alpha^2+3\alpha+5k=0$$

$$\frac{\alpha^2}{10k-9k}=\frac{\alpha}{6k-5k}=\frac{1}{3-4}$$

$$\frac{\alpha^2}{k}=\frac{\alpha}{k}=\frac{1}{-1}$$

$\alpha = -k$

Now, $k^2 - 2k + 3k = 0$

$k^2 + k = 0$

$k = 0, -1$

19. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in R$, have a common root, then a : b : c is :

यदि समीकरणों $x^2 + 2x + 3 = 0$ तथा $ax^2 + bx + c = 0$, $a, b, c \in R$ का एक मूल उभयनिष्ठ है, तो a : b : c है :

(A) 1 : 2 : 3 (B) 3 : 2 : 1 (C) 1 : 3 : 2 (D) 3 : 1 : 2

Ans. A

Sol. $x^2 + 2x + 3 = 0$

$D = 4 - 12 = -8$

$\because$ $D < 0$

$\therefore$ root are imaginary and imaginary roots occur in conjugate pair.

But equations $x^2 + 2x + 3 = 0$, $ax^2 + bx + c = 0$ having one root in common.

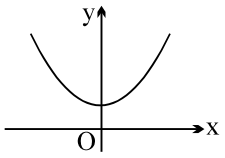
Now the equations have both roots in common.

$\therefore \frac{a}{1} = \frac{b}{2} = \frac{c}{3}$

a : b : c = 1 : 2 : 3

QUADRATIC POLYNOMIALAND GRAPH

20. The graph of a quadratic polynomial $y = ax^2 + bx + c$ (a, b, c $\in$ R) with vertex on y-axis is as shown in the figure. Then which one of the following statement is INCORRECT?



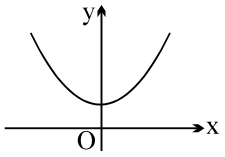
(A) Product of the roots of the corresponding quadratic equation is positive

(B) Discriminant of the quadratic equation is negative

(C) Nothing definite can be said about the sum of the roots, whether positive, negative or zero

(D) Both roots of the quadratic equation are purely imaginary

एक द्विघात बहुपद $y = ax^2 + bx + c$ (a, b, c $\in$ R) का आरेख जिसका शीर्ष y-अक्ष पर स्थित है चित्र में दर्शाया गया है। तो निम्नलिखित में से कौनसा कथन सत्य नही है?



(A) अनुकूल द्विघात समीकरण के मूलो का गुणन धनात्मक है

(B) द्विघात समीकरण का विविक्तकर ऋणात्मक है

(C) मूलो के योग के बारे में पूर्णतया कुछ भी नहीं कहा जा सकता कि वह धनात्मक, ऋणात्मक अथवा शून्य होगा

(D) द्विघात समीकरण के दोनों मूल काल्पनिक है

Ans. C

Sol. Roots are purely imaginary

i.e. $i\beta$ and $-i\beta$

$\therefore$ sum of roots = 0

incorrect (C)

product of roots $= -i^2\beta^2 = \beta^2 \Rightarrow$ product > 0 ; $\frac{c}{a} > 0 \Rightarrow$ c = + ve

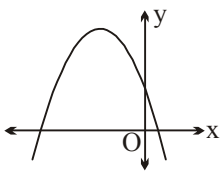
note that $-\frac{b}{2a} = 0 \Rightarrow b = 0$

hence $y = ax^2 + c$ when x > 0, y = c > 0

$\Rightarrow$ $y = ax^2 + c$ when c > 0

21. The graph of a quadratic polynomial $y = ax^2 + bx + c$ is as shown in the adjacent figure. Which of the following quantities is(are) positive?

संलग्न चित्र में एक द्विघात बहुपद $y = ax^2 + bx + c$ का आरेख दर्शाया गया है, तो निम्न में से कौनसी राशि/राशियाँ धनात्मक है?



(A) $b - c$ (B) bc (C) $c - a$ (D) ab^2

Ans. C

Sol. $y = ax^2 + bx + c$

$a < 0, D > 0, c > 0$

$$\frac{-b}{2a} < 0 \Rightarrow \frac{b}{2a} > 0$$

$\because\ a < 0 \Rightarrow b < 0$

(A) $b - c = (-ve) - (+ve) = -ve$ (wrong)

(B) $bc = (-)(+) = -ve$ (wrong)

(C) $c - a = (+) - (-) = +ve$ (right)

(D) $ab^2 = (-)\cdot(-)^2 = (-)(+) = -ve$ (wrong)

22. The distance between the highest point on the graph of $y = -(x-1)(x-3)$ and the lowest point on the graph of $y = x(x-4)$ is :

किसी आरेख $y = -(x-1)(x-3)$ के उच्चतम बिन्दु तथा ग्राफ $y = x(x-4)$ के न्यूनतम बिन्दु के बीच की दूरी होगी :

(A) 0 (B) 3 (C) 4 (D) 5

Ans. D

Sol. $y = -(x^2 - 4x + 3)$

$= -x^2 + 4x - 3$

$$\frac{-b}{2a} = \frac{-4}{2\times(-1)} = 2$$

$$\frac{-D}{4a} = \frac{4ac - b^2}{4a} = c - \frac{b^2}{4a} = -3 - \frac{16}{4\times(-1)} = -3 + 4 = 1$$

(2, 1) Heighest point

$y = x(x-4)$

$y = x^2 - 4x$

$$\frac{-b}{2a} = \frac{4}{2} = 2$$

$$\frac{-D}{4a} = c - \frac{b^2}{4a} = 0 - \frac{16}{4} = -4$$

Lowest point = (2, –4)

distance = $\sqrt{(2-2)^2+(-1-4)^2} = \sqrt{0+25} = \sqrt{25} = 5$

RANGE OF RATIONAL FUNCTIONS

23. The range of the values of $\frac{x}{x^2+4}$ for all real value of x is :

(A) $\frac{-1}{4} \le y \le \frac{1}{4}$ (B) $\frac{-1}{2} \le y \le \frac{1}{2}$ (C) $\frac{-1}{6} \le y \le \frac{1}{6}$ (D) None of these

x के सभी वास्तविक मानों के लिये $\frac{x}{x^2+4}$ के मानों का परिसर होगा :

(A) $\frac{-1}{4} \le y \le \frac{1}{4}$ (B) $\frac{-1}{2} \le y \le \frac{1}{2}$ (C) $\frac{-1}{6} \le y \le \frac{1}{6}$ (D) इनमें से कोई नहीं

Ans. A

Sol. Let $y = \frac{x}{x^2+4}$

$yx^2 + 4y - x = 0$

$yx^2 - x + 4y = 0$

$D \ge 0$

$1 - 16y^2 \ge 0$

$(1 + 4y)(1 - 4y) \ge 0$

$(4y + 1)(4y - 1) \le 0$

$-\frac{1}{4} \le y \le \frac{1}{4}$

24. If x is real, then maximum value of $\frac{3x^2+9x+17}{3x^2+9x+7}$ is :

यदि x एक वास्तविक संख्या है, तो $\frac{3x^2+9x+17}{3x^2+9x+7}$ का अधिकतम मान है :

(A) 41 (B) 1 (C) $\frac{17}{7}$ (D) $\frac{1}{4}$

Ans. A

Sol. Let $y = \frac{3x^2+9x+17}{3x^2+9x+7}$

$3yx^2 + 9xy + 7y - 3x^2 - 9x - 17 = 0$

$(3y-3)x^2 + (9y-9)x + (7y-17) = 0$
$D \geq 0,\ 3(y-1) \neq 0,\ y \neq 1$
$(9(y-1))^2 - 4 \times 3(y-1)(7y-17) \geq 0$
$81(y-1)^2 - 12(y-1)(7y-17) \geq 0$
$3(y-1)[27(y-1) - 4(7y-17)] \geq 0$
$(y-1)[27y - 27 - 28y + 68] \geq 0$
$(y-1)[-y + 41] \geq 0$
$(y-1)(y-41) \leq 0$
$1 \leq y \leq 41$
Maximum value = 41

ALWAYS POSITIVE & ALWAYS NEGATIVE BASED QUESTIONS

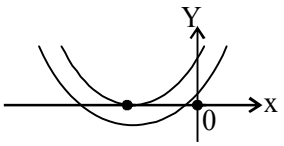
25. The set of values of 'p' for which the expression $x^2 - 2px + 3p + 4$ is negative for at least one real x is :

'p' के मानों का वह समुच्चय जिसके लिये व्यंजक $x^2 - 2px + 3p + 4$ कम से कम एक वास्तविक x के लिये ऋणात्मक हो, होगा :

(A) ϕ (B) $(-1, 4)$ (C) $(-\infty, -1) \cup (4, \infty)$ (D) $\{-1, 4\}$

Ans. C

Sol. $x^2 - 2px + 3p + 4$ is negative for at least one real x.
$D > 0$
$b^2 - 4ac > 0$
$4p^2 - 4 \cdot (3p + 4) > 0$
$p^2 - 3p - 4 > 0$
$p^2 - 4p + p - 4 > 0$
$p(p-4) + 1(p-4) > 0$
$(p+1)(p-4) > 0$
$p \in (-\infty, -1) \cup (4, \infty)$



26. If $x^2 + 2ax + 10 - 3a > 0$ for all $x \in R$, then :

यदि $x^2 + 2ax + 10 - 3a > 0$ सभी $x \in R$ के लिये हो तो :

(A) $-5 < a < 2$ (B) $a < -5$ (C) $a > 5$ (D) $2 < a < 5$

Ans. A

Sol. If $x^2 + 2ax + 10 - 3a > 0\ \forall x \in R$
$D < 0$
$4a^2 - 4(10 - 3a) < 0$
$a^2 - 10 + 3a < 0$
$a^2 + 3a - 10 < 0$
$a^2 + 5a - 2a - 10 < 0$
$a(a+5) - 2(a+5) < 0$
$(a+5)(a-2) < 0$
$-5 < a < 2$

GENERAL EQUATION OF SECOND DEGREE IN TWO VARIABLES

27. If $2x^2 + mxy + 3y^2 - 5y - 2$ have two rational factors then m is equal to :

(A) ± 7 (B) ± 6 (C) ± 5 (D) None of these

यदि $2x^2 + mxy + 3y^2 - 5y - 2$ के दो परिमेय गुणनखण्ड हो, तो m का मान होगा :

(A) ± 7 (B) ± 6 (C) ± 5 (D) इनमें से कोई नहीं

Ans. A

Sol. If $2x^2 + 3xy + 3y^2 - 5y - 2$ have two rational factors.

$a = 2, b = 3, h = \frac{m}{2}, g = 0, f = \frac{-5}{2}, c = -2$

$\because \Delta = 0$

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$$2\times3\times(-2) + 2\times0\times\frac{m}{2} - 2\times\left(-\frac{5}{2}\right)^2 - 3\times(0)^2 - (-2)\left(\frac{m}{2}\right)^2 = 0$$

$$-12 - 2\times\frac{25}{4} + 2\times\frac{m^2}{4} = 0$$

$$\frac{-24-25}{2} + \frac{m^2}{2} = 0$$

$m^2 = 49$

$m = \pm 7$

THEORY OF EQUATION INVOLVING HIGHER DEGREE

28. If $\alpha, \beta, \gamma, \delta$ are roots of $x^4 - 100x^3 + 2x^2 + 4x + 10 = 0$, then $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}$ is equal to :

यदि $\alpha, \beta, \gamma, \delta$ समीकरण $x^4 - 100x^3 + 2x^2 + 4x + 10 = 0$ के मूल है, तो $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}$ बराबर है :

(A) $\frac{2}{5}$ (B) $\frac{1}{10}$ (C) 4 (D) $-\frac{2}{5}$

Ans. D

Sol. $x^4 - 100x^3 + 2x^2 + 4x + 10$

$\alpha, \beta, \gamma, \delta$ are the roots.

$\alpha\beta\gamma + \alpha\beta\delta + \beta\gamma\delta + \alpha\gamma\delta = -4$

$\alpha\beta\gamma\delta = 10$

$$\Rightarrow \frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta} = \frac{\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{-4}{10} = \frac{-2}{5}$$

29. Let r_1, r_2 and r_3 be the solutions of the equation $x^3 - 2x^2 + 4x + 5074 = 0$ then the value of $(r_1 + 2)(r_2 + 2)(r_3 + 2)$ is :

माना समीकरण $x^3 - 2x^2 + 4x + 5074 = 0$ के मूल r_1, r_2 तथा r_3 है तो $(r_1 + 2)(r_2 + 2)(r_3 + 2)$ का मान है :

(A) 5050 (B) 5066 (C) – 5050 (D) – 5066

Ans. C

Sol. $x^3 - 2x^2 + 4x + 5074 = (x - r_1)(x - r_2)(x - r_3)$

put $x = -2$

$-8 - 8 - 8 + 5074 = -(2 + r_1)(2 + r_2)(2 + r_3)$

$\therefore \quad 5050 = -(2 + r_1)(2 + r_2)(2 + r_3)$

$(2 + r_1)(2 + r_2)(2 + r_3) = -5050$ Ans.

LOCATION OF ROOTS

30. The number of integral values of m, for which the roots of $x^2 - 2mx + m^2 - 1 = 0$ will lie between –2 and 4 is :

m के पूर्णांक मानों की संख्या जिसके लिये समीकरण $x^2 - 2mx + m^2 - 1 = 0$ के मूल –2 तथा 4 के मध्य स्थित है, होगी :

(A) 2 (B) 0 (C) 3 (D) 1

Ans. C

Sol. $x^2 - 2mx + m^2 - 1 = 0$

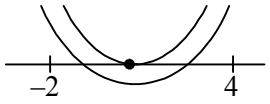
$D \geq 0$

$4m^2 - 4(m^2 - 1) \geq 0$

$m \in R$

$$-2 < \frac{-b}{2a} < 4$$

$$-2 < \frac{2m}{2} < 4$$



$-2 < m < 4$

$m \in (-2, 4) = -1, 0, 1, 2, 3$

$f(-2) > 0$

$4 + 4m + m^2 - 1 > 0$

$m^2 + 4m + 3 > 0$

$(m + 1)(m + 3) > 0$

$m \in (-\infty, -3) \cup (-1, \infty)$

$m = \ldots\ldots -5, -4, 0, 1, 2, 3, 4, \ldots\ldots$

$f(4) > 0$

$16 - 8m + m^2 - 1 > 0$

$m^2 - 8m + 15 > 0$

$m^2 - 5m - 3m + 15 > 0$

$m(m - 5) - 3(m - 5) > 0$

$(m - 3)(m - 5) > 0$

$m \in (-\infty, 3) \cup (5, \infty)$
$m = ..., -2, -1, 0, 1, 2, 6, 7, ...$
final answer
$m = 0, 1, 2$
Integral values of $m = 3$

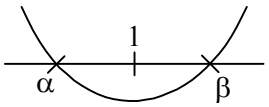
31. If α, β are the roots of the quadratic equation $(p^2 + p + 1)x^2 + (p-1)x + p^2 = 0$ such that unity lies between the roots then the set of values of p is :

यदि α, β द्विघात समीकरण $(p^2 + p + 1)x^2 + (p-1)x + p^2 = 0$ के मूल हों एवं इन मूलों के मध्य इकाई आता हो तो p के मानों का समुच्चय होगा :

(A) ϕ (B) $p \in (-\infty, -1) \cup (0, \infty)$
(C) $p \in (-1, 0)$ (D) $(-1, 1)$

Ans. C

Sol. $\Rightarrow \quad af(1) < 0$
$(p^2 + p + 1) \cdot \{p^2 + p + 1 + p - 1 + p^2\} < 0$
$(p^2 + p + 1) \cdot \{2p^2 + 2p\} < 0$
$p(p + 1)(p^2 + p + 1) < 0$
$p \in (-1, 0)$
$\because \; p^2 + p + 1 > 0$



32. If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval :

यदि द्विघातीय समीकरण $x^2 - 2kx + k^2 + k - 5 = 0$ के दोनों मूल 5 से कम हैं, तो निम्न अंतराल में k स्थित है :

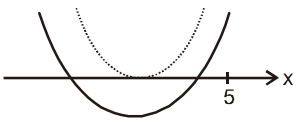
(A) (5, 6] (B) $(6, \infty)$ (C) $(-\infty, 4)$ (D) [4, 5]

Ans. C

Sol. If both roots of $ax^2 + bx + c = 0$ are less than 5, then $b^2 - 4ac \geq 0$, $f(5) > 0$ and $\frac{-b}{2a} < 5$.

Let $f(x) = x^2 - 2kx + k^2 + k - 5$ $\quad \therefore \; D \geq 0$
$\Rightarrow \quad k \leq 5$ _____(i) $\quad 4k^2 - 4(k^2 + k - 5) \geq 0$

Also, $-\frac{b}{2a} < 5 \quad \Rightarrow \quad k < 5$ _____(ii) $\quad -4k + 20 \geq 0$
and $f(5) > 0 \quad \Rightarrow \quad 25 - 10k + k^2 + k - 5 > 0$
$\Rightarrow \quad k^2 - 9k + 20 > 0 \quad \Rightarrow \quad (k-5)(k-4) > 0$
$\Rightarrow \quad k \in (-\infty, 4) \cup (5, \infty)$ _____(iii)
From equation (i), (ii) and (iii) $\Rightarrow \quad k < 4$



33. If $b > a$, then the equation $(x - a)(x - b) - 1 = 0$, has :

(A) both roots in [a, b] (B) both roots in $(-\infty, a)$
(C) both roots in $(b, +\infty)$ (D) one root in $(-\infty, a)$ and other in $(b, +\infty)$

यदि $b > a$, तो समीकरण $(x - a)(x - b) - 1 = 0$, रखती है :

(A) दोनों मूल अन्तराल $[a, b]$ में

(B) दोनों मूल $(-\infty, a)$ में

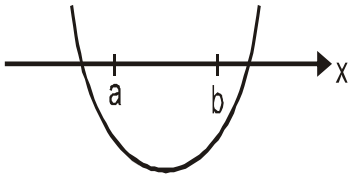
(C) दोनों मूल अन्तराल $(b, +\infty)$ में

(D) एक मूल अन्तराल $(-\infty, a)$ एवं दूसरा $(b, +\infty)$ में

Ans. D

Sol. Sol. $(x - a)(x - b) - 1 = 0$. Let $f(x) = (x - a)(x - b) - 1 \Rightarrow f(a) = -1 \Rightarrow f(b) = -1$

$\because$ the graph will be mouth opening upwards.



$\therefore$ (D) will be correct

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON ROOTS & FORMATION OF QUADRATIC EQUATION

1. Let the quadratic equation $x^2 + 3x - k = 0$ has roots a, b and $x^2 + 3x - 10 = 0$ has roots c, d such that modulus of difference of the roots of the first equation is equal to twice the modulus of the difference of the roots of the second equation. If the value of 'k' can be expressed as rational number in the lowest form as m/n then find the value of $(m + n)$.

माना द्विघात समीकरण $x^2 + 3x - k = 0$ के मूल a, b तथा $x^2 + 3x - 10 = 0$ के मूल c, d इस प्रकार है कि प्रथम समीकरण के मूलो के अन्तर का परिमाण द्वितीय समीकरण के मूलो के अन्तर के परिमाण का दुगुना है। यदि 'k' का मान m/n के रूप में व्यक्त किया जाए जहाँ m, n का HCF 1 हो तो $(m + n)$ मान क्या होगा?

Ans. 191

Sol. $x^2 + 3x - k = 0$

a, b are the roots

$$|a - b| = \left|\frac{\sqrt{D}}{a}\right| = \left|\frac{\sqrt{9 + 4k}}{1}\right| = \sqrt{9 + 4k}$$

$x^2 + 3x - 10 = 0$

c, d are the roots.

$$|c - d| = \left|\frac{\sqrt{9 + 40}}{1}\right| = \left|\sqrt{49}\right|$$

$\because \sqrt{9 + 4k} = 2 \cdot \sqrt{49}$

$9 + 4k = 4 \times 49$

$4k = 196 - 9$

$4k = 187$

$$k = \frac{187}{4} = \frac{m}{n}$$

$\Rightarrow m + n = 187 + 4 = 191$

2. A quadratic polynomial $f(x) = x^2 + ax + b$ is formed with one of its zeros being $\frac{4 + 3\sqrt{3}}{2 + \sqrt{3}}$ where a and b are integers. Also $g(x) = x^4 + 2x^3 - 10x^2 + 4x - 10$ is a biquadratic polynomial such that $g\left(\frac{4 + 3\sqrt{3}}{2 + \sqrt{3}}\right) = c\sqrt{3} + d$ where c and d are also integers. Find the values of a, b, c and d.

एक द्विघात बहुपद $f(x) = x^2 + ax + b$ इस प्रकार बनाया गया है कि इसका एक शून्य $\frac{4+3\sqrt{3}}{2+\sqrt{3}}$ है जहाँ a तथा b पूर्णांक है। साथ ही $g(x) = x^4 + 2x^3 - 10x^2 + 4x - 10$ इस प्रकार है कि $g\left(\frac{4+3\sqrt{3}}{2+\sqrt{3}}\right) = c\sqrt{3} + d$ जहाँ c तथा d भी पूर्णांक है। a, b, c एवं d का मान निकालिए।

Ans. $a = 2, b = -11, c = 4, d = -1$

Sol. $f(x) = x^2 + ax + b$

$$\alpha = \frac{4+3\sqrt{3}}{2+\sqrt{3}} = \frac{\left(4+3\sqrt{3}\right)\left(2-\sqrt{3}\right)}{\left(2+\sqrt{3}\right)\left(2-\sqrt{3}\right)} = \frac{8-4\sqrt{3}+6\sqrt{3}-9}{4-3} = 2\sqrt{3}-1$$

then β will be $\left(-2\sqrt{3}-1\right)$

sum of roots $= -a$

$2\sqrt{3}-1-2\sqrt{3}-1 = -a$

$a = 2$

Product of roots $= b$

$-\left(2\sqrt{3}-1\right)\left(2\sqrt{3}+1\right) = b$

$-(12-1) = b \Rightarrow b = -11$

$\because g\left(\frac{4+3\sqrt{3}}{2+\sqrt{3}}\right) = c\sqrt{3} + d$

$g\left(2\sqrt{3}-1\right) = c\sqrt{3} + d$, $\quad g(x) = x^4 + 2x^3 - 10x^2 + 4x - 10$

$\because x = 2\sqrt{3} - 1$

$x + 1 = 2\sqrt{3}$

$x^2 + 1 + 2x = 12$

$x^2 + 2x - 11 = 0$

Now, dividing $x^4 + 2x^3 - 10x^2 + 4x - 10$ by $x^2 + 2x - 11$, we get $x^2 + 1$ as quotient and $2x + 1$ as remainder.

$\therefore g(x) = (x^2 + 2x + 11)(x^2 + 1) + (2x + 1)$

$g(x) = 0 + (2x + 1)$

$\because g\left(2\sqrt{3}-1\right) = c\sqrt{3} + d$

$2\left(2\sqrt{3}-1\right) + 1 = c\sqrt{3} + d$

$4\sqrt{3} - 2 + 1 = c\sqrt{3} + d$

$4\sqrt{3} - 1 = c\sqrt{3} + d$

$c = 4, d = -1$

3. α, β are the roots of the equation $K(x^2 - x) + x + 5 = 0$. If K_1 & K_2 are the two values of K for which the roots α, β are connected by the relation $(\alpha/\beta) + (\beta/\alpha) = 4/5$. Find the value of $(K_1/K_2) + (K_2/K_1)$.

α, β समीकरण $K(x^2 - x) + x + 5 = 0$ के मूल है। यदि K_1 तथा K_2, K के दो मान है जिसके लिये α, β सम्बन्ध $(\alpha/\beta) + (\beta/\alpha) = 4/5$ को सन्तुष्ट करते है तो $(K_1/K_2) + (K_2/K_1)$ का मान बताइयें।

Ans. 254

Sol. $k\left(x^2 - x\right) + x + 5 = 0 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right.$

$kx^2 - kx + x + 5 = 0$

$kx^2 - x(k - 1) + 5 = 0$

$$\alpha + \beta = \frac{k-1}{k}; \alpha \cdot \beta = \frac{5}{k}$$

$$\because \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{4}{5}$$

$$\frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{4}{5}$$

$$5 \cdot \left\{ (\alpha + \beta)^2 - 2\alpha\beta \right\} = 4\alpha\beta$$

$$5 \cdot \left\{ \left(\frac{k-1}{k} \right)^2 - \frac{10}{k} \right\} = 4 \cdot \frac{5}{k}$$

$k \neq 0$

$k^2 + 1 - 2k - 10k = 4k$

$$k^2 - 16k + 1 = 0 \left\langle \begin{matrix} k_1 \\ k_2 \end{matrix} \right.$$

$k_1 + k_2 = 16$

$k_1 \cdot k_2 = 1$

Now $$\frac{k_1}{k_2} + \frac{k_2}{k_1} = \frac{k_1^2 + k_2^2}{k_1 \cdot k_2} = \frac{(k_1 + k_2)^2 - 2k_1 \cdot k_2}{k_1 \cdot k_2}$$

$$= \frac{16^2 - 2 \times 1}{1} = 254$$

CONDITION OF COMMON ROOTS

4. If the quadratic equations, $x^2 + bx + c = 0$ and $bx^2 + cx + 1 = 0$ have a common root then prove that either $b + c + 1 = 0$ or $b^2 + c^2 + 1 = bc + b + c$.

यदि द्विघात समीकरणों $x^2 + bx + c = 0$ तथा $bx^2 + cx + 1 = 0$ का एक उभयनिष्ठ मूल हो तो सिद्ध कीजिये या तो $b + c + 1 = 0$ या $b^2 + c^2 + 1 = bc + b + c$ होगा।

Ans.

Sol. $x^2 + bx + c = 0$
α, β are the roots.
$bx^2 + cx + 1 = 0$
α, γ are the roots.
$\alpha^2 + b\alpha + c = 0$
$b\alpha^2 + c\alpha + 1 = 0$

$$\frac{\alpha^2}{b-c^2} = \frac{\alpha}{bc-1} = \frac{1}{c-b^2}$$

when $\frac{\alpha}{bc-1} = \frac{1}{c-b^2}$

$$\alpha = \frac{bc-1}{c-b^2}$$

and when $\frac{\alpha^2}{b-c^2} = \frac{\alpha}{bc-1}$

$$\alpha = \frac{\left(b-c^2\right)}{\left(bc-1\right)}$$

$$\Rightarrow \frac{\left(bc-1\right)}{c-b^2} = \frac{\left(b-c^2\right)}{\left(bc-1\right)}$$

$(bc - 1)^2 = (b - c^2)(c - b^2)$
$b^2c^2 + 1 - 2bc = bc - b^3 - c^3 + b^2c^2$
$b^3 + c^3 + 1^3 = 3bc \times 1$
$\therefore (b + c + 1)\cdot(b^2 + c^2 + 1 - bc - c - b) = 0$
$\Rightarrow b + c + 1 = 0$ OR $b^2 + c^2 + 1 = bc + c + b$
Hence proved.

5. Find the value of m for which the quadratic equations $x^2 - 11x + m = 0$ & $x^2 - 14x + 2m = 0$ may have common root.

'm' का वह मान बताइये जिसके लिये द्विघात समीकरणों $x^2 - 11x + m = 0$ तथा $x^2 - 14x + 2m = 0$ के उभयनिष्ठ मूल हों।

Ans. 0, 24

Sol. $x^2 - 11x + m = 0$ ...(i)
$x^2 - 14x + 2m = 0$ ...(ii)
from (i) – (ii)
$3x - m = 0$

$$x = \frac{m}{3}$$

Now $\frac{m^2}{9}-\frac{11m}{3}+m=0$

$m^2 - 33m + 9m = 0$

$m^2 - 24m = 0$

$m = 0, 24$

6. Find the product of uncommon real roots of the two polynomials $P(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$ and $Q(x) = x^3 + 4x^2 - x - 10$.

बहुपदों $P(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$ तथा $Q(x) = x^3 + 4x^2 - x - 10$ के उन मूलो का गुणनफल ज्ञात कीजिए जो उभयनिष्ठ नही है।

Ans. 6

Sol. $P(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$

$Q(x) = x^3 + 4x^2 - x - 10$

$= x^3 + 2x^2 + 2x^2 + 4x - 5x - 10$

$= x^2 (x + 2) + 2x (x + 2) - 5 (x + 2)$

$= (x + 2) (x^2 + 2x - 5)$

$P(x) \div x^2 + 2x - 5$

We get $x^2 - 3$ as quotient and 0 as remiander,.

$\Rightarrow Q(x) = (x^2 - 3) (x^2 + 2x - 5)$

$=\left(x-\sqrt{3}\right)\left(x+\sqrt{3}\right)\left(x^2+2x-5\right)$

$\therefore$ Uncommon roots $= -2, \sqrt{3}, -\sqrt{3}$

Product $= -2\times\sqrt{3}\times\left(-\sqrt{3}\right)$

$= 6$

RANGE OF RATIONAL FUNCTIONS

7. Let α, β be real roots of the quadratic equation $x^2 - kx + k^2 + k - 5 = 0$. If m and M are respectively the minimum and maximum value of $\alpha^2 + \beta^2$, then find (m + M).

माना द्विघात समीकरण $x^2 - kx + k^2 + k - 5 = 0$ के वास्तविक मूल α, β है। यदि $\alpha^2 + \beta^2$, के न्यूनतम एवं उच्चतम मान क्रमशः m तथा M है तो (m + M) का मान ज्ञात कीजिये।

Ans. 13

Sol. $x^2 - kx + k^2 + k - 5 = 0$

α, β are the roots.

$\alpha + \beta = k$

$\alpha\beta = (k^2 + k - 5)$

$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$

$=k^2 - 2 (k^2 + k - 5)$

$= k^2 - 2k^2 - 2k + 10$

$= -k^2 - 2k + 10$

$\because$ roots are real of Quadratic Equation

$\therefore D \geq 0$

$k^2 - 4(k^2 + k - 5) \geq 0$

$k^2 - 4k^2 - 4k + 20 \geq 0$

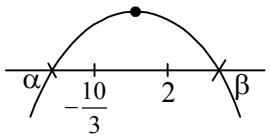
$-3k^2 - 4k + 20 \geq 0$

$3k^2 + 4k - 20 \leq 0$

$3k^2 + 10k - 6k - 20 \leq 0$

$k(3k + 10) - 2(3k + 10) \leq 0$

$(3k + 10)(k - 2) \leq 0$



$$-\frac{10}{3} \leq k \leq 2$$

from

$\alpha^2 + \beta^2 = -k^2 - 2k + 10 = f(k)$

$a < 0$,

$D = 4 + 40 = 44 > 0$

$$k = \frac{-2 + \sqrt{44}}{2} = 2{\cdot}35 \text{ or } -4{\cdot}35$$

$$\frac{-b}{2a} = \frac{2}{-2} = -1$$

$$\frac{-D}{4a} = \frac{-44}{4\times(-1)} = 11$$

$M = 11$

$$f\left(\frac{-10}{3}\right) = \frac{-100}{9} + \frac{20}{3} + 10 = \frac{-100 + 60 + 90}{9} = +\frac{50}{9}$$

$f(2) = -4 - 4 + 10 = 2 = m$

$\therefore m + M = 2 + 11 = 13$

ALWAYS POSITIVE & ALWAYS NEGATIVE BASED QUESTIONS

8. Let a, b be arbitrary real numbers. Find the smallest natural number 'b' for which the equation $x^2 + 2(a + b)x + (a - b + 8) = 0$ has unequal real roots for all $a \in R$.

माना a, b कोई वास्तविक संख्याऐं है। वह सबसे छोटी प्राकृतिक संख्या 'b' ज्ञात कीजिए जिसके लिये समीकरण $x^2 + 2(a + b)x + (a - b + 8) = 0$ के, सभी $a \in R$ के लिये, असमान वास्तविक मूल हो।

Ans. 5

Sol. $x^2 + 2(a + b)x + (a - b + 8) = 0$

$D > 0$

$4(a + b)^2 - 4(a - b + 8) > 0$

$a^2 + b^2 + 2ab - a + b - 8 > 0$

$a^2 + (2b - 1)a + (b^2 + b - 8) > 0$

$\Rightarrow D < 0$

$(2b-1)^2 - 4(b^2+b-8) < 0$
$4b^2 + 1 - 4b - 4b^2 - 4b + 32 < 0$
$-8b + 33 < 0$
$8b - 33 > 0$

$$b > \frac{33}{8}$$

$b > 4{\cdot}1$
$b = 5$

9. Find the range of values of a, such that $f(x) = \dfrac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$ is always negative.

'a' के किन मानों के लिये $f(x) = \dfrac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$ सदैव ऋणात्मक रहेगा?

Ans. $a \in \left(-\infty, -\dfrac{1}{2}\right)$

Sol. $f(x) = \dfrac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$

for $x^2 - 8x + 32 = 0$
$a > 0, D = 64 - 4 \times 32$
$= 64 - 128$
< 0
$\therefore x^2 - 8x + 32 > 0$
for $ax^2 + 2(a+1)x + 9a + 4 = 0$
$a < 0, D < 0$
$4(a+1)^2 - 4a(9a+4) < 0$
$a^2 + 1 + 2a - 9a^2 - 4a < 0$
$-8a^2 - 2a + 1 < 0$
$8a^2 + 2a - 1 > 0$
$8a^2 + 4a - 2a - 1 > 0$
$4a(2a+1) - 1(2a+1) > 0$
$(2a+1)(4a-1) > 0$

$$a \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$$

But $a < 0$

$$a \in \left(-\infty, -\frac{1}{2}\right)$$

10. We call 'p' a good number if the inequality $\frac{2x^2+2x+3}{x^2+x+1} \le p$ is satisfied for all real x. Find the smallest integral good number.

'p' एक अच्छी संख्या कहलाता है यदि असमिका $\frac{2x^2+2x+3}{x^2+x+1} \le p$ के सभी मानों के लिए सन्तुष्ट होती है। तो न्यूनतम पूर्णांक अच्छी संख्या ज्ञात कीजिए।

Ans. 4

Sol. $\frac{2x^2+2x+3}{x^2+x+1} \le p$

for $x^2 + x + 1 = 0$

$a > 0, D < 0$

$\therefore x^2 + x + 1 > 0$

$\Rightarrow 2x^2 + 2x + 3 \le p(x^2 + x + 1)$

$2x^2 - px^2 + 2x - px + 3 - p \le 0$

$(2 - p) x^2 + (2 - p)x + (3 - p) \le 0$

$a < 0, D \le 0$

$2 - p < 0$

$-p < -2$

$p > 2$

$(2 - p)^2 - 4 (2 - p) (3 - p) \le 0$

$(2 - p) [2 - p - 12 + 4p] \le 0$

$(2 - p) [3p - 10] \le 0$

$(p - 2) (3p - 10) \ge 0$

$p \le 2, \ p \ge \frac{10}{3}$

$p \ge \frac{10}{3}$

$p \ge 3 \cdot 3$

$p = 4$

11. Let $x^2 + y^2 + xy + 1 \ge a(x + y) \ \forall \ x, y \in R$. Find the possible integer(s) in the range of a.

माना $x^2 + y^2 + xy + 1 \ge a(x + y) \ \forall \ x, y \in R$ है। 'a' के परिसर में उपस्थित सभी सम्भावित पूर्णांक मान ज्ञात कीजिये।

Ans. $-1, 0, 1$

Sol. $x^2 + y^2 + xy + 1 \ge a(x + y) \quad ; \quad \forall x, y \in R$

$x^2 + xy - ax + y^2 + 1 - ay \ge 0$

$x^2 + (y - a)x + (y^2 + 1 - ay) \ge 0$

$D \le 0$

$(y - a)^2 - 4(y^2 + 1 - ay) \le 0$

$y^2 + a^2 - 2ay - 4y^2 - 4 + 4ay \le 0$

$a^2 + 2ay - 3y^2 - 4 \le 0$

$-3y^2 + 2ay + a^2 - 4 \le 0$

$3y^2 - 2ay - a^2 + 4 \ge 0$

$D \le 0$

$4a^2 - 4 \times 3(a^2 + 4) \le 0$

$a^2 + 3a^2 - 12 \le 0$

$4a^2 - 12 \le 0$

$a^2 - 3 \le 0$

$a \in \left[-\sqrt{3}, \sqrt{3}\right]$

$a = -1, 0, 1$

GENERAL EQUATION OF SECOND DEGREE IN TWO VARIABLES

12. Show that the equation $x^2 - 3xy + 2y^2 - 2x - 3y - 35 = 0$ for every real value of x, there is a real value of y and for every real value of y there is a real value of x.

सिद्ध कीजिये कि समीकरण $x^2 - 3xy + 2y^2 - 2x - 3y - 35 = 0$ में x के प्रत्येक वास्तविक मान के लिए y का एक वास्तविक मान होगा तथा y के प्रत्येक वास्तविक मान के लिए x का एक वास्तविक मान होगा।

Ans.

Sol. $x^2 - 3xy + 2y^2 - 2x - 3y - 35 = 0$

$a = 1, b = 2, h = \frac{-3}{2}, g = -1, f = \frac{-3}{2}, c = -35$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$$1\times 2\times(-35) + 2\times\left(\frac{-3}{2}\right)(-1)\times\left(\frac{-3}{2}\right) - 1\times\left(\frac{-3}{2}\right)^2 - 2\times(-1)^2 - (-35)\times\left(\frac{-3}{2}\right)^2$$

$$= -70 + 3\times\left(\frac{-3}{2}\right) - \frac{9}{4} - 2 + 35\times\frac{9}{4}$$

$$= \frac{-70}{1} - \frac{9}{2} - \frac{9}{4} - 2 + \frac{315}{4}$$

$$= \frac{-280 - 18 - 9 - 8 + 315}{4}$$

$$= \frac{-315 + 315}{4} = 0$$

Hence proved.

13. If α, β and γ are the roots of the equation $5x^3 - qx - 1 = 0, (q \in R)$ then find the value of $\frac{\alpha^2-3}{\beta\gamma}+\frac{\beta^2-3}{\gamma\alpha}+\frac{\gamma^2-3}{\alpha\beta}$.

यदि α, β एवं γ समीकरण $5x^3 - qx - 1 = 0, (q \in R)$ के मूल है, तो $\frac{\alpha^2-3}{\beta\gamma}+\frac{\beta^2-3}{\gamma\alpha}+\frac{\gamma^2-3}{\alpha\beta}$ का मान ज्ञात कीजिये।

Ans. 3

Sol. $5x^3 - qx - 1 = 0, (q \in R)$

$\alpha + \beta + \gamma = 0$

$$\alpha\beta+\beta\gamma+\gamma\alpha=\frac{-q}{5}$$

$$\alpha\beta\gamma=\frac{1}{5}$$

Now $\frac{\alpha^2-3}{\beta\gamma}+\frac{\beta^2-3}{\gamma\alpha}+\frac{\gamma^2-3}{\alpha\beta}$

$$=\frac{1}{\alpha\beta\gamma}\left[\alpha^3-3\alpha+\beta^3-3\beta+\gamma^3-3\gamma\right]$$

$$=\frac{1}{\alpha\beta\gamma}\left[\alpha^3+\beta^3+\gamma^3-3(\alpha+\beta+\gamma)\right]$$

$\because \alpha+\beta+\gamma=0 \Rightarrow \alpha^3+\beta^3+\gamma^3=3\alpha\beta\gamma$

$$=\frac{1}{\alpha\beta\gamma}\left[3\alpha\beta\gamma-3(\alpha+\beta+\gamma)\right]$$

$$=\frac{3\alpha\beta\gamma-0}{\alpha\beta\gamma}=3$$

14. Let α, β, γ are the roots of the cubic equation $x^3 + 2x^2 - x - 3 = 0$. If the absolute value of the expression $\frac{\alpha+3}{\alpha-2}+\frac{\beta+3}{\beta-2}+\frac{\gamma+3}{\gamma-2}$ can be expressed as $\frac{m}{n}$ where m and n are coprime, then find the value of (m + n).

माना α, β, γ समीकरण $x^3 + 2x^2 - x - 3 = 0$ के मूल है यदि व्यंजक $\frac{\alpha+3}{\alpha-2}+\frac{\beta+3}{\beta-2}+\frac{\gamma+3}{\gamma-2}$ का निरपेक्ष मान $\frac{m}{n}$ (जहां m तथा n सापेक्ष अभाज्य संख्या है) द्वारा प्रदर्शित किया जाता है तब (m + n) का मान ज्ञात कीजिये।

Ans. 73

Sol. Equation : $x^3 + 2x^2 - x - 3 = 0$

roots = α, β, γ

$\alpha + \beta + \gamma = -2$
$\alpha\beta + \beta\gamma + \gamma\alpha = -1$
$\alpha\beta\gamma = -3$

Let $X = \dfrac{x+3}{x-2}$

$xX - 2X = x + 3$
$xX - x = 3 + 2x$
$x(X - 1) = 3 + 2x$

Put $x = \dfrac{3+2X}{X-1}$

$$\left(\frac{3+2X}{X-1}\right)^3 + 2\left(\frac{3+2X}{X-1}\right)^2 - \left(\frac{3+2X}{X-1}\right) - 3 = 0$$

$(3 + 2X)^3 + 2(3 + 2X)^2 (X - 1) - (3 + 2X)(X - 1)^2 - 3(X - 1)^3 = 0$
$27 + 8X^3 + 18X(3 + 2X) + (2X - 2)(9 + 4X^2 + 12X) - (3 + 2X)\cdot(X^2 + 1 - 2X) - 3(X^3 - 1 - 3X(X - 1)) = 0$
$27 + 8X^3 + 54X + 36X^2 + 18X + 8X^3 + 2X^2 - 18 - 8X^2 - 24X - 3X^2 - 3 + 6X - 2X^3 - 2X + 4X^2 - 3X^3 + 3 - 9X^2 - 9X = 0$
$11X^3 + 62X^2 + 43X + 9 = 0$
$11x^3 + 62x^2 + 43x + 9 = 0$

$\dfrac{\alpha+3}{\alpha-2}, \dfrac{\beta+3}{\beta-2}, \dfrac{\gamma+3}{\gamma-2}$ are the roots

$S_1 = \dfrac{-62}{11}$

$|S_1| = \dfrac{62}{11} = \dfrac{m}{n} \Rightarrow m + n = 62 + 11 = 73$

15. Let α, β and γ are the roots of the cubic $x^3 - 3x^2 + 1 = 0$. Find a cubic whose roots are $\dfrac{\alpha}{\alpha-2}, \dfrac{\beta}{\beta-2}$ and $\dfrac{\gamma}{\gamma-2}$.

Hence or otherwise find the value of $(\alpha - 2)(\beta - 2)(\gamma - 2)$.

माना α, β एवं γ तीन घातीय समीकरण $x^3 - 3x^2 + 1 = 0$ के मूल है। वह तीन घातीय समीकरण निकालिये जिसके मूल $\dfrac{\alpha}{\alpha-2}, \dfrac{\beta}{\beta-2}$

तथा $\dfrac{\gamma}{\gamma-2}$ है एवं $(\alpha - 2)(\beta - 2)(\gamma - 2)$ का मान भी निकालिए।

Ans. $3y^3 - 9y^2 - 3y + 1 = 0$; $(\alpha - 2)(\beta - 2)(\gamma - 2) = 3$

Sol. $x^3 - 3x^2 + 1 = 0$
α, β, γ are the roots.
$x = \alpha$
cubic in y

$$y = \frac{\alpha}{\alpha - 2} = \frac{x}{x - 2}$$

$xy - 2y = x$

$x(y - 1) = 2y$

Put $x = \frac{2y}{y - 1}$

$$\left(\frac{2y}{y-1}\right)^3 - 3\left(\frac{2y}{y-1}\right)^2 + 1 = 0$$

$$\frac{8y^3}{(y-1)^3} - \frac{3 \times 4y^2}{(y-1)^2} + 1 = 0$$

$8y^3 - 12y^2 \cdot (y - 1) + (y - 1)^3 = 0$

$8y^3 - 12y^3 + 12y^2 + y^3 - 1 - 3y^2 + 3y = 0$

$9y^3 - 12y^3 + 9y^2 + 3y - 1 = 0$

$-3y^3 + 9y^2 + 3y - 1 = 0$

$3y^3 - 9y^2 - 3y + 1 = 0$

cubic in Z

$z = \alpha - 2$

$z = x - 2$

Put $x = (z + 2)$

$(z + 2)^3 - 3(z + 2)^2 + 1 = 0$

$z^3 + 8 + 6z(z + 2) - 3(z^2 + 4 + 4z) + 1 = 0$

$z^3 + 8 + 6z^2 + 12z - 3z^2 - 12 - 12z + 1 = 0$

$z^3 + 3z^2 - 3 = 0$

$\therefore$ product of roots $= (\alpha - 2)(\beta - 2)(\gamma - 2) = 3$

16. Two roots of a biquadratic $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ have their product equal to (-32). Find the value of k.

समीकरण $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ के दो मूलों का गुणनफल (-32) के बराबर है तो k का मान निकालिए।

Ans. $K = 86$

Sol. $x^4 - 18x^3 + kx^2 + 200x - 1984$

$\alpha, \beta, \gamma, \delta$ are the roots.

$\alpha + \beta + \gamma + \delta = 18$

$\Sigma\alpha\beta = k$

$\Sigma\alpha\beta\gamma = -200$

$\Sigma\alpha\beta\gamma\delta = -1984$

$\because \alpha\beta = -32$

$\Rightarrow -32 \times \gamma\delta = -1984$

$$\gamma\delta = \frac{1984}{32}$$

$\gamma\delta = 62$

$\because\ \alpha\beta\gamma + \alpha\beta\delta + \beta\gamma\delta + \alpha\gamma\delta = -200$
$-32\gamma - 32\delta + 62\beta + 62\alpha = -200$
$31\alpha + 31\beta - 16\gamma - 16\,\delta = -100$
$31(\alpha + \beta) - 16\,(\gamma + \delta) = -100$
put $\alpha + \beta = A$, $\gamma + \delta = B$
$31A - 16B = 100$ ...(i)
and $(\alpha + \beta) + (\gamma + \delta) = 18$
$A + B = 18$
$A = 18 - B$
put in (i)
$31\,(18 - B) - 16\,B = -100$
$558 - 31B - 16\,B = -100$
$-47B = -100 - 558$
$-47B = 658$

$$B = \frac{658}{47} = 14$$

$B = 14, A = 4$
$\therefore\ \alpha + \beta = 4, \gamma + \delta = 14$
$\because\ \Sigma\alpha\beta = k$
$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = k$
$\alpha(\beta + \gamma + \delta) + \beta(\gamma + \delta) + \gamma\delta = k$
$\alpha\cdot(\beta + 14) + \beta(14) + 62 = k$
$k = \alpha\beta + 14\,(\alpha + \beta) + 62$
$= -32 + 14 \times 4 + 62$
$= 30 + 56$
$= 86$

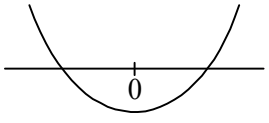
LOCATION OF ROOTS

17. Find the complete set of real values of 'a' for which both roots of the quadratic equation $(a^2 - 6a + 5)\,x^2 - \sqrt{a^2 + 2a}\;x + (6a - a^2 - 8) = 0$ lie on either side of the origin.

'a' के मानों का सम्पूर्ण समुच्चय जिसके लिए द्विघात समीकरण $(a^2 - 6a + 5)\,x^2 - \sqrt{a^2 + 2a}\;x + (6a - a^2 - 8) = 0$, के सभी मूल, मूल बिन्दु के दोनों ओर स्थित है।

Ans. $(-\infty, -2] \cup [\,0, 1) \cup (2, 4) \cup (5, \infty)$

Sol. $\left(a^2 - 6a + 5\right)x^2 - \sqrt{a^2 + 2a}\,x + \left(6a - a^2 - 8\right) = 0$
$a > 0, D > 0, a\cdot f(0) < 0$
$a^2 - 6a + 5 > 0$
$a^2 - 5a - a + 5 > 0$
$a(a - 5) - 1\,(a - 5) > 0$
$(a - 1)\,(a - 5) > 0$



$a \in (-\infty, 1) \cup (5, \infty)$

$(a^2 + 2a) - 4(a^2 - 6a + 5)\cdot(6a - a^2 - 8) > 0$
$a^2 + 2a - 4\,(6a^3 - a^4 - 8a^2 - 36a^2 + 6a^3 + 48a + 30a - 5a^2 - 40) > 0$
$a^2 + 2a - 24a^3 + 4a^4 + 32a^2 + 144a^2 - 24a^3 - 192a - 120a + 20a^2 + 160 > 0$
$4a^4 - 48a^3 + 197a^2 - 310a + 160 > 0$
$a \in R$
$a\cdot f(0) < 0$
$(a^2 - 6a + 5)\cdot(6a - a^2 - 8) < 0$
$(a - 1)\,(a - 5)\,(a - 2)\,(a - 4) > 0$

$a \in (-\infty, 1) \cup (2, 4) \cup (5, \infty)$

And $a^2 + 2a \geq 0$
$a\,(a + 2) \geq 0$
$\because$ roots are real.

$a \in (-\infty, -2] \cup [0, +\infty)$

(i) $\cap$ (ii) $\cap$ (iii)

$a \in (-\infty, -2] \cup [0, 1) \cup (2, 4) \cup (5, \infty)$

18. If α, β are the roots of the equation, $x^2 - 2x - a^2 + 1 = 0$ and γ , δ are the roots of the equation, $x^2 - 2(a + 1)x + a(a - 1) = 0$ such that $\alpha, \beta \in (\gamma, \delta)$ then find the values of 'a'.

यदि α, β समीकरण $x^2 - 2x - a^2 + 1 = 0$ के तथा γ, δ समीकरण $x^2 - 2(a + 1)x + a(a - 1) = 0$ के मूल इस प्रकार है की $\alpha, \beta \in (\gamma, \delta)$ तो 'a' के मान निकालिये।

Ans. $a \in \left(-\frac{1}{4}, 1\right)$

Sol. $x^2 - 2x - a^2 + 1 = 0$ $\quad$ $x^2 - 2(a + 1)x + a(a - 1) = 0$
α, β are the roots. $\quad$ γ, δ are the roots.

$\because \alpha, \beta \in (\gamma, \delta)$

from graph :
$a\cdot f(\alpha) < 0$, $\quad$ $a\cdot f(\beta) < 0$ $\quad$ for $f(x) = x^2 - 2(a + 1)x + a(a - 1)$
from Equation (i)

$$x = \frac{2 \pm \sqrt{4 - 4\left(-a^2 + 1\right)}}{2}$$

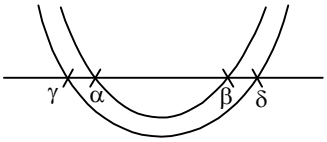
$$x = \frac{2 \pm \sqrt{4 + 4a^2 - 4}}{2}$$

$$x = \frac{2 \pm 2a}{2}$$

$x =$ $\underset{\alpha}{1 + a}$, $\underset{\beta}{1 - a}$
for 2nd equation
$a\cdot f(\alpha) < 0$
$a\cdot f(1 + a) < 0$

$1 \cdot \{(1 + a)^2 - 2(a + 1) \cdot (1 + a) + a(a - 1)\} < 0$
$\{-(a + 1)^2 + a^2 - a\} < 0$
$-a^2 - 1 - 2a + a^2 - a < 0$
$-3a - 1 < 0$
$3a + 1 > 0$

$a > \frac{-1}{3}$



and $a \cdot f(\beta) < 0$
$1 \cdot f(1 - a) < 0$
$\{(1 - a)^2 - 2(a + 1)(1 - a) + (a - 1)\} < 0$
$1 + a^2 - 2a + 2(a^2 - 1) + a^2 - a < 0$
$1 + a^2 - 2a + 2a^2 - 2 + a^2 - a < 0$
$4a^2 - 3a - 1 < 0$
$4a^2 - 4a + a - 1 < 0$
$4a(a - 1) + 1(a - 1) < 0$
$(a - 1)(4a + 1) < 0$

$-\frac{1}{4} < a < 1$

Final answer = (i) $\cap$ (ii)

$a \in \left(-\frac{1}{4}, 1\right)$

19. Find all the values of the parameter 'a' for which both roots of the quadratic equation $x^2 - ax + 2 = 0$ belong to the interval (0, 3).

प्राचल 'a' के सभी मान बताइये जिनके लिये द्विघात समीकरण $x^2 - ax + 2 = 0$ के दोनों मूल अन्तराल (0, 3) में स्थित है।

Ans. $2\sqrt{2} \le a < \frac{11}{3}$

Sol. $x^2 - ax + 2 = 0$
a, b are the roots.
(i) $D \ge 0$
$a^2 - 8 \ge 0$

$a \le -2\sqrt{2}$, $a \ge 2\sqrt{2}$

(ii) $a \cdot f(0) > 0$
$2 > 0$
(iii) $a \cdot f(3) > 0$
$9 - 3a + 2 > 0$
$3a - 11 < 0$

$a < \frac{11}{3}$

(iv) $0 < \frac{-b}{2a} < 3$

$0 < \frac{a}{2} < 3$

$0 < a < 6$

$a \in (0, 6)$

(i) $\cap$ (ii) $\cap$ (iii) $\cap$ (iv)

$a \in \left[2\sqrt{2}, \frac{11}{3}\right)$

20. At what values of 'a' do all the zeroes of the function, f(x) = $(a-2)x^2 + 2ax + a + 3$ lie on the interval (– 2, 1)?

'a' किस किन मानों के लिये फलन f(x) = $(a-2)x^2 + 2ax + a + 3$ के सभी शून्य अन्तराल (– 2, 1) में स्थित होगें?

Ans. $\left(-\infty, -\frac{1}{4}\right) \cup \{2\} \cup (5, 6]$

Sol. $f(x) = (a-2)x^2 + 2ax + (a+3)$

(i) $D \geq 0$

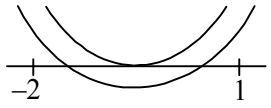
$4a^2 - 4(a-2)(a+3) \geq 0$

$a^2 - (a^2 + a - 6) \geq 0$

$a^2 - a^2 - a + 6 \geq 0$

$a - 6 \leq 0$

$a \leq 6$



(ii) $-2 < \frac{-b}{2a} < 1$

$-2 < \frac{-2a}{2(a-2)} < 1$

for $\frac{-a}{a-2} > -2$

$\frac{-a}{a-2} + 2 > 0$

$\frac{-a + 2a - 4}{a-2} > 0$

$\frac{a-4}{a-2} > 0$

$a < 2, a > 4$

for $\frac{-a}{a-2} < 1$

$\frac{-a}{a-2} - 1 < 0$

$$\frac{-a-a+2}{a-2}<0$$

$$\frac{-2a+2}{a-2}<0$$

$$\frac{2(a-1)}{a-2}>0$$

$$\frac{a-1}{a-2}>0$$

$\Rightarrow a<1, a>4$

(iii) $a\cdot f(-2)>0$

$(a-2)\{(a-2)4-4a+a+3\}>0$

$(a-2)\{4a-8-4a+a+3\}>0$

$(a-2)(a-5)>0$

$a<2, a>5$

and (iv) $a\cdot f(1)>0$

$(a-2)\{a-2+2a+a+3\}>0$

$(a-2)\{4a+1\}>0$

$$a<-\frac{1}{4}, a>2$$

(i), (ii), (iii) & (iv)

$$a\in\left(-\infty,\frac{-1}{4}\right)\cup\{2\}\cup(5,6]$$

21. Find the values of K so that the quadratic equation $x^2+2(K-1)x+K+5=0$ has atleast one positive root.

K के किन मानों के लिये द्विघात समीकरण $x^2+2(K-1)x+K+5=0$ का कम से कम एक धनात्मक मूल है।

Ans. $K\le -1$

Sol. $x^2+2(k-1)x+k+5=0$

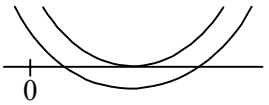
(i) $D\ge 0$

$4(k-1)^2-4(k+5)\ge 0$

$k^2+1-2k-k-5\ge 0$

$k^2-3k-4\ge 0$

$(k+1)(k-4)\ge 0$



$$\frac{-b}{2a}>0$$

$$\frac{-2(k-1)}{2}>0$$

$k<1$

$k\in(-\infty, 1)$

$f(0)>0$

$k+5>0$

$k > -5$

$k \in (-5, \infty)$

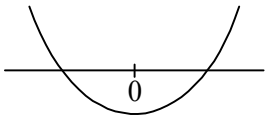
Intersection value

$k \in (-5, -1]$

(ii) $f(0) < 0$

$k + 5 < 0$

$k \in (-\infty, -5)$



(iii) $f(0) = 0$; $-\frac{b}{2a} > 0$; $D > 0$

$k = -5$; $\frac{-2(k-1)}{2} > 0$; $k < -1, k > 4$



$k - 1 < 0$

$k < 1$

Intersection value

$\Rightarrow \quad k = -5$

Final answer $k \le -1$

22. If all the solutions of the inequality $x^2 - 6ax + 5a^2 \le 0$ are also the solutions of inequality $x^2 - 14x + 40 \le 0$ then find the number of possible integral values of a.

यदि असमिका $x^2 - 6ax + 5a^2 \le 0$ के सभी हल असमिका $x^2 - 14x + 40 \le 0$ के भी हल है तो a के सम्भावित पूर्णांक मानों की संख्या बताओ।

Ans. 0

Sol. $x^2 - 14x + 40 \le 0$

$x^2 - 4x - 10x + 40 \le 0$

$x(x - 4) - 10(x - 4) \le 0$

$(x - 4)(x - 10) \le 0$

$x \in [4, 10]$

for $x^2 - 6ax + 5a^2 \le 0$

$x^2 - 5ax - ax + 5a^2 \le 0$

$x(x - 5a) - a(x - 5a) \le 0$

$(x - 5a)(x - a) \le 0$

Case–I :

if $a > 0$

$x \in [a, 5a]$

$\because \ a \ge 4, 5a \ge 10$

$a \ge 4, a \ge 2$

Not possible.

Case–II : $a < 0$

$x \in [5a, a]$

$\therefore \ x \in [4, 10]$

$\therefore$ No solution

$\therefore$ There is no integral value for a.

23. The smallest value of k, for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is_____.

k का वह न्यूनतम मान जिसके लिये समीकरण $x^2 - 8kx + 16(k^2 - k + 1) = 0$ के दोनों मूल वास्तविक एवं भिन्न है तथा कम से कम 4 के बराबर हैं, _____ है

Ans. 2

Sol. $x^2 - 8kx + 16(k^2 - k + 1) = 0;$

$D > 0$	;	$\frac{-b}{2a} > 4$;	$f(4) \geq 0$
$64k^2 - 64(k^2 - k + 1)$	;	$\frac{8k}{2} > 4$;	$16 - 32k + 16k^2 - 16k + 16 \geq 0$
$k - 1 > 0$	;	$k > 1$;	$16k^2 - 48k + 32 \geq 0$
$k > 1$	;		$k \leq 1, k \geq 2$

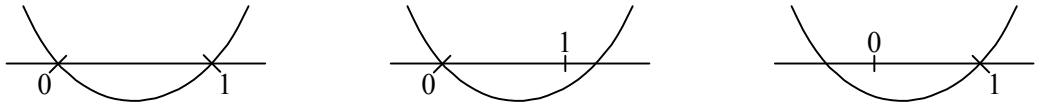
$\Rightarrow$ Minimum value of k = 2

24. Largest integral value of 'a' for which the expression $y = x^2 + ax + (a - 3)$ is non positive $\forall\ x \in [0, 1]$ is_____.

यदि व्यंजक $y = x^2 + ax + (a - 3)$ हमेशा या तो शून्य है या ऋणात्मक तो जहाँ $\forall\ x \in [0, 1]$ तो a का अधिकतम पूर्णांक में मान होगा_____.

Ans. 1

Sol.



$y = ax^2 + ax + (a - 3), \ \forall x \in [0,1]$

Let $f(x) = x^2 + ax + (a - 3)$

$\because\ f(0) \leq 0$	$f(1) \leq 0$
$a - 3 \leq 0$	$1 + a + a - 3 \leq 0$
$a \leq 3$	$2a - 2 \leq 0$
	$a \leq 1$

$\therefore\ a_{max} = 1$

25. If the equation $x^4 - ax^3 + 11x^2 - ax + 1 = 0$ has four distinct positive roots then the range of a is (m, M). Find the value of m + 2M.

यदि समीकरण $x^4 - ax^3 + 11x^2 - ax + 1 = 0$ के चार भिन्न धनात्मक मूल है तो a का परिसर (m, M) है। m + 2M का मान ज्ञात कीजिए।

Ans. 19

Sol. $x^4 - ax^3 + 11x^2 - ax + 1 = 0$

divide by x^2

$$x^2 - ax + 11 - \frac{a}{x} + \frac{1}{x^2} = 0$$

$$\left(x^2 + \frac{1}{x^2}\right) - a\left(x + \frac{1}{x}\right) + 11 = 0$$

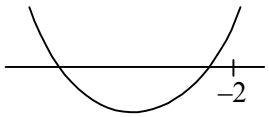
put $x + \frac{1}{x} = t$

$$x^2 + \frac{1}{x^2} = t^2 - 2$$

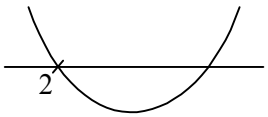
$t^2 - 2 - at + 11 = 0$

$t^2 - at + 9 = 0$

$\because x + \frac{1}{x} \le -2$, $x + \frac{1}{x} \ge 2$



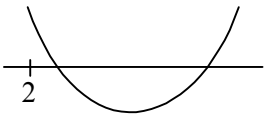
Impossible



$x + \frac{1}{x} = 2$

$x^2 - 2x + 1 = 0$

$x = 1$ not possible



$x + \frac{1}{x} = 2$

$x^2 - 2x + 1 = 0$

$(x - 1)^2 = 0$ {two equal roots}

$x = 1$ Not possible

$t^2 - at + 9 = 0$

t_1, t_2 are the roots.

(i) $D > 0$

$a^2 - 36 > 0$

$(a + 6)(a - 6) > 0$

$a \in (-\infty, -6) \cup (6, \infty)$

(ii) $\frac{-b}{2a} > 2$

$\frac{a}{2} > 2$

$a > 4$

$a \in (4, \infty)$

(iii) $a \cdot f(2) > 0$

$1 \times (4 - 2a + 9) > 0$

$2a - 13 < 0$

$a < \frac{13}{2}$

$a \in \left(-\infty, \frac{13}{2}\right)$

(i), (ii), (iii)

$a \in \left(6, \frac{13}{2}\right)$

$m = 6,\ M = \frac{13}{2}$

$\Rightarrow m + 2M = 6 + 2\cdot\frac{13}{2} = 19$

MISCELLANEOUS

26. When $y^2 + my + 2$ is divided by $(y - 1)$ then the quotient is $f(y)$ and the remainder is R_1. When $y^2 + my + 2$ is divided by $(y + 1)$ then quotient is $g(y)$ and the remainder is R_2. If $R_1 = R_2$ then find the value of m.

जब $y^2 + my + 2$ को $(y - 1)$ से भाग दिया जाता है तो भागफल $f(y)$ मिलता है तथा शेषफल R_1 है। जब $y^2 + my + 2$ का $(y + 1)$ से भाग दिया जाता है तो भागफल $g(y)$ है तथा शेषफल R_2 होगा। यदि $R_1 = R_2$ हो तो m का मान ज्ञात कीजिए।

Ans. 0

Sol. $y^2 + my + 2 = (y - 1) f(y) + R_1$ ...(i)

& $y^2 + my + 2 = (y + 1) g(y) + R_2$ ...(ii)

in Equation (i), put $y = 1$

$1 + m + 2 = R^1$

$R_1 = m + 3$

in Equation (ii), put $y = 1$

$1 - m + 2 = R_2$

$R_2 = 3 - m$

$\because\ R_1 = R_2$

$m + 3 = 3 - m$

$2m = 0$

$m = 0$

27. Let $P(x) = 4x^2 + 6x + 4$ and $Q(y) = 4y^2 - 12y + 25$. Find the unique pair of real numbers (x, y) that satisfy $P(x) \cdot Q(y) = 28$.

माना $P(x) = 4x^2 + 6x + 4$ तथा $Q(y) = 4y^2 - 12y + 25$ है। वास्तविक संख्याओं x तथा y का एकल युग्म (x, y) बताइये तो $P(x) \cdot Q(y) = 28$ को संतुष्ट करता है।

Ans. $\left(-\frac{3}{4}, \frac{3}{2}\right)$

Sol. $P(x) = 4x^2 + 6x + 4$

$\because\ a > 0,\ P(x)_{max} = \infty$

for minimum of P(x)

$\left(\frac{-b}{2a}, \frac{-D}{4a}\right) = \left(\frac{-6}{8}, 4 - \frac{36}{4\times 4}\right)$

$= \left(\frac{-3}{4}, \frac{7}{4}\right)$

$Q(x) = 4y^2 - 12y + 25$

$a > 0,\ Q(y)_{max} = \infty$

for minimum of Q(y)

$\left(\frac{-b}{2a}, \frac{-D}{4a}\right) = \left(\frac{12}{8}, 25 - \frac{144}{4\times 4}\right)$

$= \left(\frac{3}{2}, 16\right)$

$= (y, Q(y))$

$\Rightarrow P(x)\cdot P(y) = \frac{7}{4}\times 16 = 28$

$(x, y) = \left(\frac{-3}{4}, \frac{3}{2}\right)$

28. Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$.

समीकरण $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$ के वास्तविक मूलो का गुणनफल ज्ञात कीजिये।

Ans. 20

Sol. $x^2 + 18x + 30 = 2\sqrt{x + 18 + 45}$

Put $t = x^2 + 18x + 30$

$\therefore t = 2\sqrt{t + 15}$

$t^2 = 4(t + 15)$

$t^2 - 4t - 60 = 0$

$(t - 10)(t + 6) = 0$

$t = 10, -6$

$\therefore x^2 + 8x + 30 = 10$

$x^2 + 8x + 20 = 10$

$\therefore$ product of roots = 20

29. Find all real numbers x such that, $\left(x - \frac{1}{x}\right)^{\frac{1}{2}} + \left(1 - \frac{1}{x}\right)^{\frac{1}{2}} = x$.

x के वास्तविक मान ज्ञात कीजिए जिनके लिये $\left(x - \frac{1}{x}\right)^{\frac{1}{2}} + \left(1 - \frac{1}{x}\right)^{\frac{1}{2}} = x$ हो।

Ans. $x = \frac{\sqrt{5}+1}{2}$

Sol. $\left(x-\frac{1}{x}\right)^{\frac{1}{2}}+\left(1-\frac{1}{x}\right)^{\frac{1}{2}} = x$

$$x-\frac{1}{x}+1-\frac{1}{x}+2\sqrt{\left(x-\frac{1}{x}\right)\left(1-\frac{1}{x}\right)} = x^2$$

$$x-\frac{2}{x}+1+2\sqrt{x-1-\frac{1}{x}+\frac{1}{x^2}} = x^2$$

$$2\sqrt{\frac{x^3-x^2-x+1}{x^2}} = x^2-x+\frac{2}{x}-1$$

$$\frac{2\sqrt{x^3-x^2-x+1}}{x} = \frac{\left(x^3-x^2-x+2\right)}{x};x \neq 0$$

$4(x^3 - x^2 - x + 1) = (x^3 - x^2 - x + 2)^2$
put $x^2 - x^2 - x + 1 = t$
$4t = (t + 1)^2$
$t^2 + 1 + 2t - 4t = 0$
$(t - 1)^2 = 0$
$t = 1$
$\therefore\ x^3 - x^2 - x + 1 = 1$
$x(x^2 - x - 1) = 0$
$x \neq 0$
$x^2 - x - 1 = 0$

$$x = \frac{1\pm\sqrt{5}}{2}$$

$$x = \frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}$$

$$\therefore x = \frac{\left(1+\sqrt{5}\right)}{2}$$

30. Solve : $(x + 9)(x - 3)(x - 7)(x + 5) = 385$

हल करो : $(x + 9)(x - 3)(x - 7)(x + 5) = 385$

Ans. -4, 2 and $-1 \pm \sqrt{71}$

Sol. $(x + 9)(x - 3)(x - 7)(x + 5) = 385$
$(x + 9)(x - 7)(x - 3)(x + 5) = 385$

$(x^2 + 2x - 63)(x^2 + 2x - 15) = 385$
put $x^2 + 2x = t$
$(t - 63)(t - 15) = 385$
$t^2 - 15t - 63t + 945 - 385 = 0$
$t^2 - 78t + 560 = 0$
$t^2 - 70t - 8t + 560 = 0$
$(t - 70)(t - 8) = 0$
$t = 70$ and 8

for $t = 70$
$x^2 + 2x - 70 = 0$

$$x = \frac{-2 \pm \sqrt{4 + 4 \times 70}}{2}$$

$$x = -1 \pm \sqrt{71}$$

for $t = 8$
$x^2 + 2x - 8 = 0$

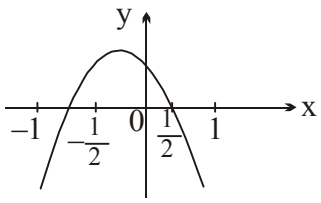
$x^2 + 4x - 2x - 8 = 0$

$x(x + 4) - 2(x + 4) = 0$
$(x + 4)(x - 2) = 0$
$x = -4, 2$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 31 to 32

Let $f(x) = ax^2 + bx + c$ be a quadratic expression and $y = f(x)$ has graph as shown in figure
यदि $f(x) = ax^2 + bx + c$ एक द्विघात व्यंजक है। एवं $y = f(x)$ का ग्राफ चित्रानुसार है।



31. Which of the following is false?

निम्न में से कौनसा विकल्प असत्य है?

(A) $ab > 0$ (B) $abc < 0$ (C) $ac < 0$ (D) $bc < 0$

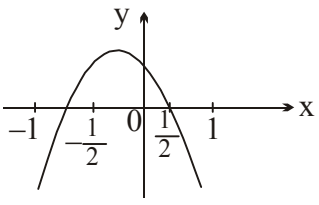
Ans. B

Sol. $f(x) = ax^2 + bx + c$
$a < 0$

$$-\frac{b}{2a} < 0$$

$$\frac{b}{2a} > 0$$

but $a < 0 \Rightarrow b < 0$ and $c > 0$
(A) Now $a \cdot b = (-)(-) = +ve$
$\Rightarrow ab > 0$



(B) abc = (–) (–) (+) = +ve
abc> 0
(C) ac < 0
= (–) (+)
= (–)
(D) bc = (–) (+) = (–)
bc < 0
∴ option B is false.

32. Which of the following is true ?

निम्न में से कौनसा विकल्प सत्य है?

(A) $a + b + c > 0$ (B) $a - b + c > 0$ (C) $a + 3b + 9c < 0$ (D) $a - 3b + 9c > 0$

Ans. D

Sol. $f(x) = ax^2 + bx + c$
(i) $f(1) < 0$
$a + b + c < 0$
(ii) $f(-1) < 0$
$a - b + c < 0$
(iii) $f\left(-\frac{1}{3}\right) > 0$

$\frac{a}{9} - \frac{b}{3} + c > 0$
$a - 3b + 9c > 0$

Paragraph for Question nos. 33 to 35

Consider a rational function $f(x) = \frac{x^2 + 3x + 1}{x^2 + x + 1}$ and a quadratic function

$g(x) = x^2 - (m + 1)x + m - 1$, where m is a parameter.

माना एक परिमेय फलन $f(x) = \frac{x^2 + 3x + 1}{x^2 + x + 1}$ है तथा एक द्विघात फलन

$g(x) = x^2 - (m + 1)x + m - 1$ है, जहाँ m प्राचल है।

33. Number of integral value(s) of 'm' so that g(x) is always positive, is :

(A) 0 (B) 1 (C) 2 (D) more than 2

g(x) सदैव धनात्मक है तो 'm' के पूर्णांक मानों की संख्या होगी :

(A) 0 (B) 1 (C) 2 (D) 2 से अधिक

Ans. A

Sol. $f(x)=\frac{x^2+3x+1}{x^2+x+1}$ and $g(x) = x^2 - (m+1)x + m - 1$

for $g'(x)$ is always +ve.

$a > 0, D < 0$

$(m+1)^2 - 4(m-1) < 0$

$m^2 + 1 + 2m - 4m + 4 < 0$

$m^2 - 2m + 1 + 4 < 0$

$(m-1)^2 + 4 < 0$

No value of m.

$\therefore$ Option A

34. Number of integral values in the range of f(x), is :

f(x) के परिसर में पूर्णांक मानों की संख्या होगी :

(A) 1 (B) 2 (C) 3 (D) more than 3

Ans. C

Sol. Let $y=\frac{x^2+3x+1}{x^2+x+1}$

Range of y

$yx^2 + yx + y - x^2 - 3x - 1 = 0$

$(y-1)x^2 + (y-3)x + (y-1) = 0$

$D \geq 0$

$(y-3)^2 - 4(y-1)^2 \geq 0$

$y^2 + 9 - 6y - 4y^2 - 4 + 8y \geq 0$

$-3y^2 + 2y + 5 \geq 0$

$3y^2 - 2y - 5 \leq 0$

$3y^2 + 3y - 5y - 5 \leq 0$

$3y(y+1) - 5(y+1) \leq 0$

$(y+1)(3y-5) \leq 0$

$-1 \leq y \leq \frac{5}{3}$

$y = -1, 0, 1$ (Integral Values)

35. If both roots of $g(x) = 0$ are greater than the smallest value of the function f(x), then 'm' lies in the interval :

यदि $g(x) = 0$ के दोनों मूल f(x) के न्यूनतम मान से अधिक है, तो 'm' किस अन्तराल में होगा :

(A) $(-\infty, -2)$ (B) $\left(-\infty, -\frac{1}{4}\right)$ (C) $(-2, \infty)$ (D) $\left(-\frac{1}{2}, \infty\right)$

Ans. D

Sol. $\because f(x) \in \left[-1, \frac{5}{3}\right]$

given that both roots of g(x) = 0 are greater than the smallest value of f(x).

$\Rightarrow D \geq 0$

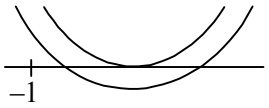
$(m+1)^2 - 4(m-1) \geq 0$

$m^2 + 1 + 2m - 4m + 4 \geq 0$

$(m-1)^2 + 4 \geq 0$

$m \in R$

$\Rightarrow \frac{-b}{2a} > -1$



$\frac{(m+1)}{2} > -1$

$m + 1 > -2$

$m > -3$

$\Rightarrow f(-1) > 0$

$1 + m + 1 + m - 1 > 0$

$2m + 1 > 0$

$m > \frac{-1}{2}$

$\Rightarrow m \in \left(-\frac{1}{2}, +\infty\right)$

D is correct.

Paragraph for Question nos. 36 to 37

For $a, b \in R - \{0\}$, let $f(x) = ax^2 + bx + a$ satisfies $f\left(x + \frac{7}{4}\right) = f\left(\frac{7}{4} - x\right) \forall x \in R$.

Also the equation $f(x) = 7x + a$ has only one real and distinct solution.

$a, b \in R - \{0\}$ के लिए, माना $f(x) = ax^2 + bx + a$, $f\left(x + \frac{7}{4}\right) = f\left(\frac{7}{4} - x\right) \forall x \in R$ को संतुष्ट करता है तथा समीकरण $f(x) = 7x + a$ का केवल एक वास्तविक तथा भिन्न हल है।

36. The value of $(a + b)$ is equal to :

$(a + b)$ का मान बराबर है :

(A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. For $a, b \in R - \{0\}$, let $f(x) = ax^2 + bx + a$ satisfies $f\left(x + \frac{7}{4}\right) = f\left(\frac{7}{4} - x\right) \forall x \in R$.

Also the equation $f(x) = 7x + a$ has only one real and distinct solution.

$\Rightarrow f\left(x+\frac{7}{4}\right)=f\left(\frac{7}{4}-x\right)$

$a\left(x+\frac{7}{4}\right)^2+b\left(x+\frac{7}{4}\right)+a=a\left(\frac{7}{4}-x\right)^2+b\left(\frac{7}{4}-x\right)+a$

$a\left(x^2+\frac{49}{16}+\frac{7x}{2}\right)+bx+\frac{7b}{4}=a\left(\frac{49}{16}+x^2-\frac{7}{2}x\right)+\frac{7b}{4}-bx$

$ax^2+\frac{49}{16}a+\frac{7ax}{2}+bx=ax^2+\frac{49}{16}a-\frac{7ax}{2}-bx$

$7ax = -2bx$

$7a + 2b = 0$

And $f(x) = 7x + a$

$ax^2 + bx + a = 7x + a$

$ax^2 + (b - 7)x = 0$

$\because$ one root is real and distinct.

$D = 0$

$\therefore (b - 7)^2 + 4 \times a \times 0 = 0$

$b = 7$

$\Rightarrow 7a + 2 \times 7 = 0$

$7a = -14$

$a = -2$

$a + b = -2 + 7 = 5$

37. The minimum value of f(x) in $\left[0, \frac{3}{2}\right]$ is equal to :

अन्तराल $\left[0, \frac{3}{2}\right]$ में f(x) का न्यूनतम मान है :

(A) $\frac{-33}{8}$ (B) 0 (C) 4 (D) – 2

Ans. D

Sol. $\because f(x) = ax^2 + bx + a$

$= -2x^2 + 7x - 2$

$V=\left(-\frac{b}{2a},\frac{D}{4a}\right)\equiv\left(\frac{-7}{2(-2)},c-\frac{b^2}{4a}\right)\equiv\left(\frac{7}{4},-2+\frac{49}{8}\right)\equiv\left(\frac{7}{4},\frac{33}{8}\right)$

$f(0) = 0 + 0 - 2,$ $f\left(\frac{3}{2}\right)=-2\left(\frac{9}{4}\right)+7\left(\frac{3}{2}\right)-2$

$= -2$ $=-\frac{9}{2}+\frac{21}{2}-2$

$$= \frac{-9+21-4}{2} = \frac{8}{2} = 4$$

Min value of f(x) in $\left[0, \frac{3}{2}\right] = -2$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

38. The value(s) of 'p' for which the equation $ax^2 - px + ab = 0$ and $x^2 - ax - bx + ab = 0$ may have a common root, given a, b are non zero real numbers, is :

p के वह मान जिसके लिये समीकरण $ax^2 - px + ab = 0$ तथा $x^2 - ax - bx + ab = 0$ एक उभयनिष्ठ मूल रखती है। दिया गया है a, b अशून्य वास्तविक संख्या है :

(A) $a + b^2$ (B) $a^2 + b$ (C) $a(1 + b)$ (D) $b(1 + a)$

Ans. BC

Sol. Equation $ax^2 - px + ab = 0$ ...(i)

$x^2 - ax - bx + ab = 0$

from equation, $x = a, b$

$\because$ both equations may have a common root.

for $x = a$

$a^3 - ap + ab = 0$

$a(a^2 - p + b) = 0$

$a \neq 0$, $p = a^2 + b$

for $x = b$

$ab^2 - pb + ab = 0$

$b(ab - p + a) = 0$

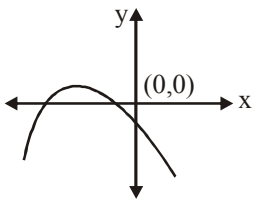
$b \neq 0$

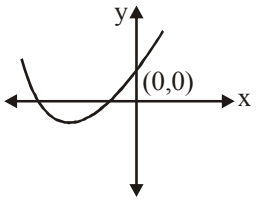
$p = a(1 + b)$

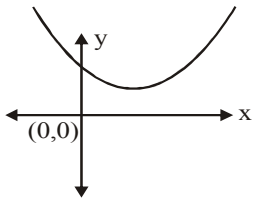
Option B, C are correct.

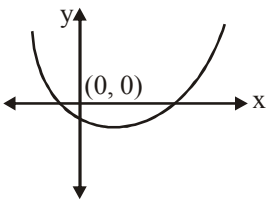
39. For which of the following graphs of quadratic expression $y = ax^2 + bx + c$, then product abc is positive?

द्विघात व्यंजक $y = ax^2 + bx + c$ के निम्न में से कौनसे आरेखों के लिए गुणन abc धनात्मक है?

(A)


(B)


(C)


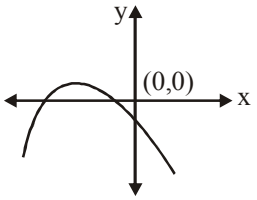
(D)


Ans. BD

Sol. (A) $a < 0$

$$-\frac{b}{2a} < 0$$

$$\frac{b}{2a} > 0$$



but $a < 0$

$b < 0$

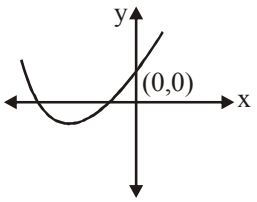
and $c < 0$

$\therefore a \cdot b \cdot c = (-ve)(-ve)(-ve) = -ve$

(B) $a > 0$

$$-\frac{b}{2a} < 0$$

$$\frac{b}{2a} > 0$$



but $a > 0 \Rightarrow b > 0$

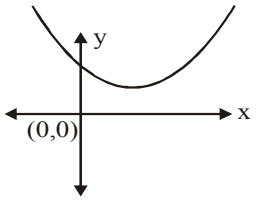
and $c > 0$

Now $abc = (+)(+)(+) = +ve$

(C) $a > 0$

$$-\frac{b}{2a} > 0$$

$$\frac{b}{2a} < 0$$



but $a > 0 \Rightarrow b < 0$

$c > 0$

Now $abc = (+)(-)(+)$

$= -ve$

(D) $a > 0$

$$\frac{-b}{2a} > 0$$

$$\frac{b}{2a} < 0$$

but $a > 0$ then $b < 0$

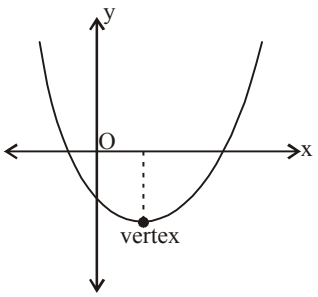
and $c < 0$

Now $abc = (+)(-)(-)$

$= +ve$

40. Graph of $y = ax^2 + bx + c = 0$ is given adjacently. What conclusions can be drawn from this graph :

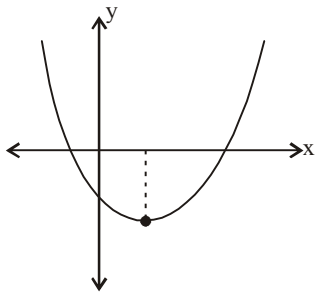
फलन $y = ax^2 + bx + c = 0$ का रेखाचित्र दर्शाया गया है, तब :



(A) $a > 0$ (B) $b < 0$ (C) $c < 0$ (D) $b^2 - 4ac > 0$

Ans. ABCD

Sol. $a > 0$



$$-\frac{b}{2a} > 0$$

$$\frac{b}{2a} < 0$$

but $a > 0$ then $b < 0$

and $c < 0$, $D > 0$

$b^2 - 4ac > 0$

All options are correct.

41. The values of a for which $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$ does not have a real solution is :

a के मान, जिसके लिए समीकरण $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$ का कोई वास्तविक हल नहीं है, होगा :

(A) –10 (B) 12 (C) 5 (D) –30

Ans. BCD

Sol. $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$

$\frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x+4)} + \frac{a}{30} = 0$, $x \neq 1, 2$

$30(x - 3) + a(x + 4) = 0$

$30x - 90 + ax + 4a = 0$

$x(30 + a) = 90 - 4a$

$x = \frac{90 - 4a}{30 + a}$

for $a = -30$, No solution

for $x = 1$

$$1 = \frac{90 - 4a}{30 + a}$$

$30 + a = 90 - 4a$

$5a = 60$

$a = 12$

for $x = 2$

$$2 = \frac{90 - 4a}{30 + a}$$

$60 + 2a = 90 - 4a$

$6a = 30$

$a = 5$

for $a = 5, 12, -30$

No solution.

42. If α, β are the roots of $4x^2 - 16x + c = 0, c > 0$ such that $1 < \alpha < 2 < \beta < 3$, then value of 'c' can be :

यदि α, β समीकरण $4x^2 - 16x + c = 0, c > 0$ के मूल है इस प्रकार कि $1 < \alpha < 2 < \beta < 3$, तो 'c' का मान हो सकता है :

(A) 11 (B) 12 (C) 13 (D) 14

Ans. CD

Sol. $4x^2 - 16x + c = 0, c > 0$

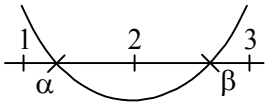
$\alpha < 2 < \beta < 3$

$f(1) > 0,$ $f(3) > 0$

$4 - 16 + c > 0,$ $36 - 48 + c > 0$

$c > 12$ $-12 + c > 0$

$c > 12$



$f(2) < 0 \quad \Rightarrow 16 - 32 + c < 0$

$-16 + c < 0$

$c < 16$

$\because c \in (12,16)$

Options C and D are correct.

43. If the quadratic polynomial $P(x) = (p - 3)x^2 - 2px + 3p - 6$ ranges from $[0, \infty)$ for every $x \in R$, then the value of p can be :

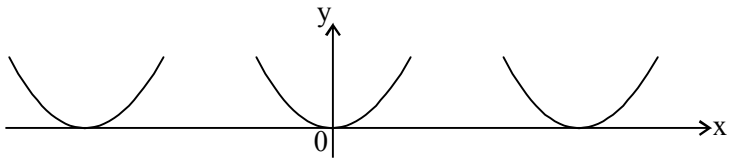
यदि द्विघात बहुपद $P(x) = (p - 3)x^2 - 2px + 3p - 6$ का परिसर $[0, \infty)$ $x \in R$ के लिए है, तो p का मान निम्न में से कौनसा हो सकता है :

(A) $\frac{3}{2}$ (B) 4 (C) 6 (D) 7

Ans. CD

Sol. $P(x) = (p - 3)x^2 - 2px + 3p - 6$

$\because P(x) \in [0,\infty)$



$\Rightarrow a > 0$

$P - 3 > 0 \qquad P > 3$

$\Rightarrow D \le 0$

$4p^2 - 4(p-3)(3p-6) \le 0$

$p^2 - (3p^2 - 6p - 9p + 18) \le 0$

$p^2 - 3p^2 + 15p - 18 \le 0$

$-2p^2 + 15p - 18 \le 0$

$2p^2 - 15p + 18 \ge 0$

$2p^2 - 12p - 3p + 18 \ge 0$

$2p(p-6) - 3(p-6) \ge 0$

$(p-6)(2p-3) \ge 0$

$p \ge 6, \; p \le \frac{3}{2}$

$\Rightarrow p \ge 6$

$\therefore$ Correct option C, D.

44. If $f(x) = 2x^2 + kx + 50$, $k \ge 20$ and magnitude of difference of the roots of the equation $f(x) = 0$ is 8, then which of the following can be true :

(A) minimum value of $f(x)$ is -32 (B) minimum value of $f(x)$ is -64

(C) graph of $f(x)$ cuts positive y-axis (D) graph of $f(x)$ cuts negative y-axis

यदि $f(x) = 2x^2 + kx + 50$, $k \ge 20$ तथा समीकरण $f(x) = 0$ के मूलों के अन्तर का मापांक 8 हो, तो निम्न में से कौनसा कथन सत्य हो सकता है :

(A) $f(x)$ का न्यूनतम मान -32 है (B) $f(x)$ का न्यूनतम मान -64 है।

(C) $f(x)$ का आरेख धनात्मक y-अक्ष को काटता है (D) $f(x)$ का आरेख ऋणात्मक y-अक्ष पर काटता है

Ans. AC

Sol. $f(x) = 2x^2 + kx + 50$, $k \ge 20$

$$|\alpha - \beta| = \left|\frac{\sqrt{D}}{a}\right|$$

$$\left|\frac{\sqrt{D}}{a}\right| = 8$$

$\sqrt{D} = 8 \times 2$

$D = 64 \times 4 = 256$

$$\frac{-D}{4a} = \frac{-256}{4 \times 2} = -32$$

$f(0) = 50 = +ve$

$\therefore$ A, C are correct.

45. Let α and β ($\alpha > \beta$) be roots of the quadratic equation $ax^2 + bx + c = 0$. If $0 < a < c$ and $a + c = b$, then which of the following statement(s) is/are correct?

(A) Larger root is 1 (B) Larger root is – 1 (C) Smaller root is $\frac{c}{a}$ (D) b is greater than zero

माना α तथा β ($\alpha > \beta$) वर्ग समीकरण $ax^2 + bx + c = 0$ के मूल है। यदि $0 < a < c$ तथा $a + c = b$, तो निम्न में से कौनसा कथन सत्य है/हैं?

(A) बड़ा मूल 1 है (B) बड़ा मूल – 1 है (C) छोटा मूल $\frac{c}{a}$ है (D) b शून्य से बड़ा है

Ans. BD

Sol. $ax^2 + bx + c = 0 \begin{cases} \alpha \\ \beta \end{cases}$

Give $\alpha < \beta$, $0 < a < c$, $b = a + c$

$ax^2 + ax + cx + c = 0$

$ax(x + 1) + c(x + 1) = 0$

$(x + 1)(ax + c) = 0$

$x = -1, -\frac{c}{a}$

$\because a > 0, c > 0 \Rightarrow b > 0$

$\because c > a \Rightarrow \frac{c}{a} > 1$

$-\frac{c}{a} < -1$

$\therefore$ B, D are correct.

MATCH THE COLUMN TYPE QUESTION(S)

46. Consider the quadratic trinomial $f(x) = 2x^2 - 10px + 7p - 1$, where p is a parameter. Find the range of p in the following conditions given in column-I :

Column-I	**Column-II**
(A) If both roots of $f(x) = 0$ are confined in $(-1, 1)$ then	(P) $\left(\frac{2}{5}, \infty\right)$
(B) Exactly one root of $f(x) = 0$ lies in $(-1, 1)$	(Q) ϕ
(C) Both roots of $f(x) = 0$ are greater than 1	(R) $\left(-\frac{1}{17}, \frac{1}{3}\right)$
(D) One root of $f(x) = 0$ is greater than 1 and other root of $f(x) = 0$ is less than – 1	(S) $\left(-\infty, -\frac{1}{17}\right] \cup \left[\frac{1}{3}, \infty\right)$

(T) $\left(-\infty,-\frac{1}{17}\right)\cup\left(\frac{1}{3},\infty\right)$

द्विघात त्रिपद $f(x) = 2x^2 - 10px + 7p - 1$, जहाँ p एक प्राचल है। दी गई शर्तों के अनुसार p के परिसर का मिलान कीजिए :

Column-I	**Column-II**
(A) यदि $f(x) = 0$ के दोनों मूल अन्तराल $(-1, 1)$ में स्थित हो तो	(P) $\left(\frac{2}{5}, \infty\right)$
(B) $f(x) = 0$ का एक मूल अन्तराल $(-1, 1)$ में स्थित हो	(Q) ϕ
(C) $f(x) = 0$ के दोनों मूल एक से अधिक हो	(R) $\left(-\frac{1}{17},\frac{1}{3}\right)$
(D) $f(x) = 0$ का एक मूल एक से अधिक तथा अन्य मूल -1 से छोटा हो तो	(S) $\left(-\infty,-\frac{1}{17}\right]\cup\left[\frac{1}{3},\infty\right)$
	(T) $\left(-\infty,-\frac{1}{17}\right)\cup\left(\frac{1}{3},\infty\right)$

Ans. (A) R (B) S (C) Q (D) Q

Sol. $f(x) = 2x^2 - 10px + 7p - 1$

(A) If both roots of $f(x) = 0$ confined in $(-1,1)$

(i) $D \geq 0$

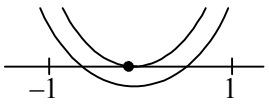
$100p^2 - 8(7p - 1) \geq 0$

$4[25p^2 - 14p + 2] \geq 0$

$25p^2 - 14p + 2 \geq 0$

$\because D < 0, P \in R$

(ii) $-1 < \frac{-b}{2a} < 1$



$-1 < \frac{10p}{4} < 1$

$-\frac{2}{5} < p < \frac{2}{5}$

$p \in \left(-\frac{2}{5},\frac{2}{5}\right)$

(iii) $a \cdot f(1) > 0$

$2 \cdot \{2 + 10p + 7p - 1\} > 0$

$17p + 1 > 0$

$p > -\frac{1}{17}$

(iv) $a \cdot f(1) > 0$

$2 \cdot \{2 - 10p + 7p - 1\} > 0$

$-3p + 1 > 0$

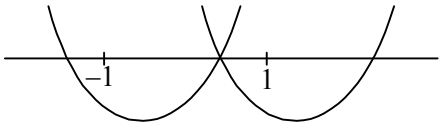
$3p - 1 < 0$

$p < \frac{1}{3}$

Now (i) $\cap$ (ii) $\cap$ (iii) $\cap$ (iv)

$p\left(-\frac{1}{17}, \frac{1}{3}\right)$

$A \to R$



(B) Exactly one root of f(x) = 0 lies in (–1, 1)

Case–I :

$D > 0$

$p \in R$

$f(-1) < 0$

$2 + 10p + 7p - 1 < 0$

$17p + 1 < 0$

$p < -\frac{1}{17} \Rightarrow p \in \left(-\frac{1}{17}, \infty\right)$

Case–II :

$D > 0 \Rightarrow p \in R$

$p(1) < 0$

$2 - 10p + 7p - 1 < 0$

$1 - 3p < 0$

$p > \frac{1}{3} \Rightarrow p \in \left(\frac{1}{3}, \infty\right)$

$(i) \cup (ii)$

$p \in \left(-\infty, -\frac{1}{17}\right) \cup \left(\frac{1}{3}, \infty\right)$

for $p = -\frac{1}{17} \Rightarrow$

$f(x) = 2x^2 - 10x\left(-\frac{1}{17}\right) + 7\left(-\frac{1}{17}\right) - 1$

$= 2x^2 + \frac{10}{17}x - \frac{7}{17} - 1$

$= 2x^2 + \frac{10}{17}x - \frac{24}{17}$

$= \frac{1}{17}\left[34x^2 + 10x - 24\right]$

$= \frac{2}{17}\left[17x^2 + 5x - 12\right]$

$$= \frac{2}{17}\left[17x^2 + 17x - 12x - 12\right]$$

$$= \frac{2}{17}\left[17x(x+1) - 12(x+1)\right]$$

$$= \frac{2}{17}(x+1)(17x-12)$$

for $p = \frac{1}{3}$

$$f(x) = 2x^2 - 10x\cdot\left(\frac{1}{3}\right) + 7\left(\frac{1}{3}\right) - 1$$

$$= 2x^2 - \frac{10x}{3} + \frac{4}{3}$$

$$= \frac{1}{3}\left[6x^2 - 10x + 4\right] = \frac{2}{3}\left[3x^2 - 5x + 2\right]$$

$$= \frac{2}{3}\left[3x^2 - 3x - 2x + 2\right] = \frac{2}{3}\left[3x(x-1) - 2(x-1)\right]$$

$$= \frac{2}{3}(x-1)(3x-2)$$

$\because p = -\frac{1}{17}$ & $\frac{1}{3}$ satisfying the condition

$\therefore p = -\frac{1}{17}, \frac{1}{3}$ includes $p \in \left(-\infty, -\frac{1}{17}\right] \cup \left[\frac{1}{3}, \infty\right)$

$B \rightarrow S$

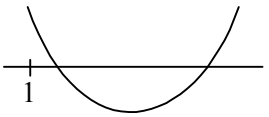
(C) Both roots of f(x) = 0 are greater than 1

(i) $D > 0 \Rightarrow p \in R$

(ii) $f(1) > 0$

$2 - 10p + 7p - 1 > 0$

$1 - 3p > 0$



$$p < \frac{1}{3} \Rightarrow p \in \left(-\infty, \frac{1}{3}\right)$$

(iii) $-\frac{b}{2a} > 1$

$$\frac{10p}{4} > 1$$

$$p > \frac{2}{5}$$

$$p \in \left(\frac{2}{5}, \infty\right)$$

$p \in \phi$

$C \rightarrow Q$

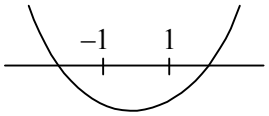
(D) One root of f(x) = 0 is greater than 1 and other root of f(x) = 0 is less than –1.

(i) $f(-1) < 0$

$2 + 10p + 7p - 1 < 0$

$17p + 1 < 0$

$p < -\frac{1}{17}$



(ii) $f(1) < 0$

$2 - 10p + 7p - 1 < 0$

$1 - 3p < 0$

$p > \frac{1}{3}$

(i) $\cap$ (ii)

$p \in \phi$

$D \rightarrow Q$

47. The roots of the quadratic equation, $x^2 - 5x + 6 = 0$ is α and β. Then find the following :

Column-I	**Column-II**
(A) $\alpha - \beta$	(P) 1
(B) $\alpha^2 + \beta^2$	(Q) – 13
(C) $\frac{\alpha^3 + \beta^3}{35}$	(R) – 1
(D) $\frac{\alpha^4 + \beta^4 - 62}{\alpha^3 + \beta^3}$	(S) 13

समीकरण $x^2 - 5x + 6 = 0$ के मूल α तथा β है। तब निम्न का मान प्राप्त करे :

Column-I	**Column-II**
(A) $\alpha - \beta$	(P) 1
(B) $\alpha^2 + \beta^2$	(Q) – 13
(C) $\frac{\alpha^3 + \beta^3}{35}$	(R) – 1
(D) $\frac{\alpha^4 + \beta^4 - 62}{\alpha^3 + \beta^3}$	(S) 13

Ans. (A) PR (B) S (C) P (D) P

Sol. $x^2 - 5x + 6 = 0 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right.$

$\alpha + \beta = 5$
$\alpha\beta = 6$

(A) $|\alpha - \beta| = \left|\frac{\sqrt{D}}{a}\right| = \left|\frac{\sqrt{25-24}}{1}\right| = 1 \Rightarrow (\alpha - \beta) = \pm 1$

$A \rightarrow P, R$
(B) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 25 - 12 = 13$
$B \rightarrow S$

(C) $\frac{\alpha^3 + \beta^3}{35} = \frac{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)}{35} = \frac{125 - 3\times 6\times 5}{35} = \frac{125-90}{35}$

$= \frac{35}{35} = 1$
$C \rightarrow P$

(D) $\frac{\alpha^4 + \beta^4 - 62}{\alpha^3 + \beta^3} = \frac{\left(\alpha^2+\beta^2\right)^2 - 2\alpha^2\beta^2 - 62}{35} = \frac{(13)^2 - 2\times 36 - 62}{35}$

$= \frac{169 - 72 - 62}{35} = \frac{169-134}{35} = \frac{35}{35} = 1$
$D \rightarrow P$

48. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$: **[JEE-2007]**

Column-I	**Column-II**
(A) If $-1 < x < 1$, then f(x) satisfies	(P) $0 < f(x) < 1$
(B) If $1 < x < 2$, the f(x) satisfies	(Q) $f(x) < 0$
(C) If $3 < x < 5$, then f(x) satisfies	(R) $f(x) > 0$
(D) If $x > 5$, then f(x) satisfies	(S) $f(x) < 1$

माना $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$: **[JEE-2007]**

Column-I	**Column-II**
(A) If $-1 < x < 1$, then f(x) satisfies	(P) $0 < f(x) < 1$
(B) If $1 < x < 2$, the f(x) satisfies	(Q) $f(x) < 0$
(C) If $3 < x < 5$, then f(x) satisfies	(R) $f(x) > 0$
(D) If $x > 5$, then f(x) satisfies	(S) $f(x) < 1$

Ans. (A) PRS (B) QS (C) QS (D) PRS

Sol. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6} = \frac{(x-5)(x-1)}{(x-2)(x-3)}$

Let $y = \dfrac{x^2 - 6x + 5}{x^2 - 5x + 6}$

(i) $f(x) < 0$

$$\frac{(x-5)(x-1)}{(x-2)(x-3)} < 0$$

$$x \in (1,2) \cup (3,5)$$

(ii) $f(x) < 1$

$$\frac{x^2 - 6x + 5}{x^2 - 5x + 6} - 1 < 0$$

$$\frac{x^2 - 6x + 5 - x^2 + 5x - 6}{(x-2)(x-3)} < 0$$

$$-\frac{(x+1)}{(x-2)(x-3)} < 0$$

$$\frac{(x+1)}{(x-2)(x-3)} > 0$$

$$x \in (-1,2) \cup (3,+\infty)$$

(iii) $f(x) > 0$

$x \in (-\infty,1) \cup (2, 3) \cup (5, \infty)$

(iv) $0 < f(x) < 1$

(ii) $\cap$ (iii)

$x \in (-1, 1) \cup (5, \infty)$

Exercise J-M

1. Let α, β be two roots of the equation $x^2 + (20)^{1/4} x + (5)^{1/2} = 0$. Then $\alpha^8 + \beta^8$ is equal to : **[JEE(Main)July2021]**

माना α, β समीकरण $x^2 + (20)^{1/4} x + (5)^{1/2} = 0$ के दो मूल हैं। तो $\alpha^8 + \beta^8$ बराबर है : **[JEE(Main)July2021]**

(A) 100 (B) 10 (C) 160 (D) 50

Ans. D

Sol. $x^2 + (20)^{1/4} + (5)^{1/2} = 0$

$\alpha + \beta = -(20)^{1/4}$

$\alpha \cdot \beta = 5^{1/2}$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$\alpha^2 + \beta^2 = 20^{1/2} - 2.5^{1/2} = 0$

$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2 = -10$

$\alpha^8 + \beta^8 = (\alpha^4 + \beta^4)^2 - 2\alpha^4\beta^4 = 50$

2. If α, β are roots of the equation $x^2 + 5(\sqrt{2})x + 10 = 0$, $\alpha > \beta$ and $P_n = \alpha^n - \beta^n$ for each positive integer n, then the value of $\left(\frac{P_{17}P_{20} + 5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19} + 5\sqrt{2}P_{18}^2}\right)$ is equal to _____. **[JEE(Main)July2021]**

यदि समीकरण $x^2 + 5(\sqrt{2})x + 10 = 0$, के α तथा β, $\alpha > \beta$ दो मूल हैं तथा $P_n = \alpha^n - \beta^n$, (प्रत्येक धन पूर्णांक n के लिए) है, तो $\left(\frac{P_{17}P_{20} + 5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19} + 5\sqrt{2}P_{18}^2}\right)$ का मान है _____. **[JEE(Main)July2021]**

Ans. 1

Sol. $x^2 + 5\sqrt{2}x + 10 = 0 <_{\beta}^{\alpha}$

$\alpha^2 + 5\sqrt{2}\alpha = -10$ & $P_n = \alpha^n - \beta^n$

$\beta^2 + 5\sqrt{2}\beta = -10$

$$\Rightarrow \frac{P_{17}}{P_{18}}\left[\frac{P_{20} + 5\sqrt{2}P_{19}}{P_{19} + 5\sqrt{2}P_{18}}\right]$$

$$\Rightarrow \frac{P_{17}}{P_{18}}\left[\frac{\alpha^{20} - \beta^{20} + 5\sqrt{2}(\alpha^{19} - \beta^{19})}{\alpha^{19} - \beta^{19} + 5\sqrt{2}(\alpha^{18} - \beta^{18})}\right]$$

$$\Rightarrow \frac{P_{17}}{P_{18}}\left[\frac{\alpha^{18}(\alpha^2 + 5\sqrt{2}\alpha) - \beta^{18}(\beta^2 + 5\sqrt{2}\beta)}{\alpha^{17}(\alpha^2 + 5\sqrt{2}\alpha) - \beta^{17}(\beta^2 + 5\sqrt{2}\beta)}\right]$$

$$\Rightarrow \frac{P_{17}}{P_{18}}\left[\frac{(-10)(\alpha^{18}-\beta^{18})}{(-10)(\alpha^{17}-\beta^{17})}\right]$$

$$\Rightarrow \frac{P_{17}}{P_{18}} \times \frac{P_{18}}{P_{17}} = 1$$

3. If $a + b + c = 1$, $ab + bc + ca = 2$ and $abc = 3$, then the value of $a^4 + b^4 + c^4$ is equal to _____.

[JEE(Main)July2021]

यदि $a + b + c = 1$, $ab + bc + ca = 2$ तथा $bc = 3$ हैं, तो $a^4 + b^4 + c^4$ बराबर है _____

[JEE(Main)July2021]

Ans. 13

Sol. $a + b + c = 1 \quad ab + bc + ca = 2 \quad abc = 3$

$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$1 = a^2 + b^2 + c^2 + 4$

$a^2 + b^2 + c^2 = -3$

$\Rightarrow (a^2 + b^2 + c^2)^2 = a^4 + b^4 + c^4 + 2(a^2b^2 + b^2c^2 + c^2a^2)$

$9 - 2(a^2b^2 + b^2c^2 + c^2a^2) = a^4 + b^4 + c^4$

$\Rightarrow (ab + bc + ca)^2 = a^2b^2 + b^2c^2 + c^2a^2 + 2(ab^2c + abc^2 + a^2bc)$

$4 = a^2b^2 + b^2c^2 + c^2a^2 + 2abc(b + c + a)$

$4 = a^2b^2 + b^2c^2 + c^2a^2 + 6$

$-2 = a^2b^2 + b^2c^2 + c^2a^2$

$\Rightarrow a^4 + b^4 + c^4 = 9 - 2(-2)$

$a^4 + b^4 + c^4 = 13$

4. The number of real roots of the equation $e^{4x} - e^{3x} - 4e^{2x} - e^x + 1$ is equal to_____.

[JEE(Main)July2021]

समीकरण $e^{4x} - e^{3x} - 4e^{2x} - e^x + 1 = 0$ के वास्तविक मूलों की संख्या है _____

[JEE(Main)July2021]

Ans. 2

Sol. put $e^x = t$

$t^4 - t^3 - 4t^2 - t + 1 = 0$

$t^4 + t^3 - 2t^3 - 2t^2 - 2t^2 - 2t + t + 1 = 0$

$t^3(t + 1) - 2t^2(t + 1) - 2t(t + 1) + 1(t + 1) = 0$

$(t + 1)\cdot(t^3 - 2t^2 - 2t + 1) = 0$

$(t + 1)[t^3 + t^2 - 3t^2 - 3t + t + 1] = 0$

$(t + 1)^2\cdot(t^2 - 3t + 1) = 0$

$t = -1$ (rejected)

$t^2 - 3t + 1 = 0$

$$t = \frac{3 \pm \sqrt{9-4}}{2} = \frac{3 \pm \sqrt{5}}{2}$$

$e^x = \frac{3+\sqrt{5}}{2}$ or $e^x = \frac{3-\sqrt{5}}{2}$

$x = \ln\left(\frac{3+\sqrt{5}}{2}\right)$ or $\left(\frac{3-\sqrt{5}}{2}\right)$

5. If $x^2 + 9y^2 - 4x + 3 = 0$, x, $y \in R$, then x and y respectively lie in the intervals : **[JEE(Main)Aug.2021]**

यदि $x^2 + 9y^2 - 4x + 3 = 0$, x, $y \in R$ है, तो x तथा y क्रमशः निम्न में से किस अंतराल में है?

[JEE(Main)Aug.2021]

(A) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

(B) [1, 3] and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

(C) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and [1, 3]

(D) [1, 3] and [1, 3]

Ans. B

Sol.

$x^2 - 4x + (9y^2 + 3) = 0$	;	$9y^2 + (x^2 - 4x + 3) = 0$
$x \in R$	;	$y \in R$
$D \geq 0$	;	$D \geq 0$
$16 - 4\cdot(9y^2 + 3) \geq 0$	;	$0 - 4\cdot 9\cdot(x^2 - 4x + 3) \geq 0$
$4 - 9y^2 - 3 \geq 0$	;	$(x - 1)(x - 3) \leq 0$
$1 - 9y^2 \geq 0$	;	$x \in [1, 3]$

$y \in \left[-\frac{1}{3}, \frac{1}{3}\right]$

6. The set of all values of $k > -1$, for which the equation
$(3x^2 + 4x+3)^2 - (k+1)(3x^2 + 4x +3)(3x^2 + 4x+2) + k(3x^2 + 4x +2)^2 = 0$ has real roots, is :

[JEE(Main)Aug.2021]

$k > -1$ के सभी मानों, जिनके लिए समीकरण
$(3x^2 + 4x+3)^2 - (k+1)(3x^2 + 4x +3)(3x^2 + 4x+2) + k(3x^2 + 4x +2)^2 = 0$
के वास्तविक मूल हैं, का समुच्चय है : **[JEE(Main)Aug.2021]**

(A) [2, 3]

(B) $\left(1, \frac{5}{2}\right)$

(C) $\left[\frac{1}{2}, \frac{3}{2}\right) - \{1\}$

(D) $\left[-\frac{1}{2}, 1\right)$

Ans. B

Sol. Given equation can be expressed

$$\left(\frac{3x^2+4x+3}{3x^2+4x+2}\right)^2-(k+1)\left(\frac{3x^2+4x+3}{3x^2+4x+2}\right)+k=0$$

$t^2-(k+1)t+k=0$; Let $t=\left(\frac{3x^2+4x+3}{3x^2+4x+2}\right)$

$t^2-kt-t+k=0$

$t(t-k)-1(t-k)=0$

$(t-1)(t-k)=0$

$t\neq 1$

for $t=k$

$$\frac{3x^2+4x+3}{3x^2+4x+2}=k$$

$3x^2+4x+3=3x^2k+4xk+2k$

$3x^2(k-1)+4x(k-1)+2k-3=0$

$x\in R$; $k\neq 1$

$D\geq 0$

$16(k-1)^2-k(k-1)(2k-3)\geq 0$

$$k\in\left(1,\frac{5}{2}\right]$$

7. The sum of all integral values of $k(k\neq 0)$ for which the equation $\frac{2}{x-1}-\frac{1}{x-2}=\frac{2}{k}$ in x has no real roots, is _____. **[JEE(Main)Aug.2021]**

$k(k\neq 0)$ के सभी पूर्णांक मानों, जिनके लिए x समीकरण $\frac{2}{x-1}-\frac{1}{x-2}=\frac{2}{k}$ का कोई वास्तविक मूल नहीं है, का योग है _____। **[JEE(Main)Aug.2021]**

Ans. 66

Sol. $\frac{2}{x-1}-\frac{1}{x-2}=\frac{2}{k}$; $k\neq 0$

$$\frac{2(x-2)-(x-1)}{(x^2-3x+2)}=\frac{2}{k}$$

$2x^2-6x+4=k(2x-4-x+1)$

$2x^2-6x+4=kx-3k$

$2x^2-x(k+6)+(4+3k)=0$

$D<0$

$(k+6)^2-4\cdot 2\cdot(4+3k)<0$

$k^2 + 36 + 12k - 32 - 24k < 0$

$k^2 - 12k + 4 < 0$

$$k = \frac{12 \pm \sqrt{144-16}}{2} = \frac{12 \pm \sqrt{128}}{2}$$

$$k = 6 \pm 4\sqrt{2}$$

$$k \in \left(6-4\sqrt{2}, 6+4\sqrt{2}\right)$$

$k \in (0.344, 11.656)$

$k \in \{1, 2, 3, ...11\}$

Sum = 66

8. Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to _____. **[JEE(Main)Aug.2021]**

माना $\lambda \neq 0$, R में है। यदि समीकरण $x^2 - x + 2\lambda = 0$ के मूल α तथा β हैं और समीकरण $3x^2 - 10x + 27\lambda = 0$ के मूल α तथा γ हैं, तो $\frac{\beta\gamma}{\lambda}$ बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 18

Sol. $\lambda \neq 0$

$$x^2 - x + 2\lambda = 0 \begin{cases} \alpha \\ \beta \end{cases} \quad ; \qquad 3x^2 - 10x + 27\lambda = 0 \begin{cases} \alpha \\ \gamma \end{cases}$$

$\alpha\beta = 2\lambda$ ...(i) $\qquad \alpha\gamma = \frac{27\lambda}{3} = 9\lambda$ ...(ii)

By (i) × (ii)

$\alpha^2\beta\gamma = 18\lambda^2$ ...(iii)

$\because$ both equations have one common root.

$\therefore \alpha^2 - \alpha + 2\lambda = 0$

$3\alpha^2 - 10\alpha + 27\lambda = 0$

$$\frac{\alpha^2}{-27\lambda + 20\lambda} = \frac{\alpha}{6-27} = \frac{1}{-10+3}$$

$$\frac{\alpha^2}{-7\lambda} = \frac{\alpha}{-21} = \frac{1}{-7}$$

$\Rightarrow \alpha^2 = \lambda$

from equation (iii)

$\lambda \cdot \beta\gamma = 18\lambda^2$

$$\frac{\beta\gamma}{\lambda} = 18$$

9. If the sum of the squares of the reciprocals of the roots α and β of the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to : **[JEE(Main)June2022]**

यदि समीकरण $3x^2 + \lambda x - 1 = 0$ के मूल α तथा β के व्युत्क्रमों के वर्गों का योगफल 15 है, तो $6(\alpha^3 + \beta^3)^2$ बराबर है : **[JEE(Main)June2022]**

(1) 18 (2) 24 (3) 36 (4) 96

Ans. Official Answer NTA (2)

Sol. $3x^2 + \lambda x - 1 = 0$

$$\alpha + \beta = \frac{-\lambda}{3}, \qquad \alpha\beta = \frac{-1}{3}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = 15$$

$$\frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} = 15$$

$$\frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2} = 15$$

$\Rightarrow \lambda^2 = 9 \qquad \Rightarrow \lambda = \pm 3$

$$6\left(\alpha^3 + \beta^3\right)^2 = 6\left(\left(\alpha + \beta\right)^3 - 3\alpha\beta\left(\alpha + \beta\right)\right)^2$$

$= 6(4) = 24$

10. The sum of all the real roots of the equation $(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$ is : **[JEE(Main)June2022]**

समीकरण $(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$ के सभी वास्तविक मूलों का योग है : **[JEE(Main)June2022]**

(1) $\log_e 3$ (2) $-\log_e 3$ (3) $\log_e 6$ (4) $-\log_e 6$

Ans. Official Answer NTA (2)

Sol. $(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$

$e^{2x} - 4 = 0 \qquad 6e^{2x} - 5e^x + 1 = 0$

$e^{2x} = 4 \qquad e^x = \frac{1}{2}$

$e^{2x} = 4 \qquad e^x = \frac{1}{3}$

Now if we consider roots as x_1, x_2, x_3 then

$$e^{x_1} = \frac{1}{2}$$

$$e^{x_2} = \frac{1}{3}$$

$e^{2x_3} = 4$

take logrithm on base 'e' and add

$$x_1 + x_2 + x_3 = \ell n\left(\frac{1}{2}\right) + \ell n\left(\frac{1}{3}\right) + \frac{1}{2}\ell n(4)$$

$$x_1 + x_2 + x_3 = \ell n\left(\frac{1}{2}.\frac{1}{3}.2\right)$$

$$x_1 + x_2 + x_3 = \ell n\left(\frac{1}{3}\right)$$

$= -\ell n3$

11. Let a, b $\in$ R be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root α. If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to : **[JEE(Main)June2022]**

माना a, b $\in$ R इस प्रकार है कि समीकरण $ax^2 - 2bx + 15 = 0$ का एक पुनरावृत्त मूल (repeated root) α है। यदि समीकरण $x^2 - 2bx + 21 = 0$ के मूल α तथा β हैं, तो $\alpha^2 + \beta^2$ बराबर है : **[JEE(Main)June2022]**

(1) 37 (2) 58 (3) 68 (4) 92

Ans. Official Answer NTA (2)

Sol. $ax^2 - 2bx + 15 = a(x - \alpha)^2$

$x^2 - 2bx + 21 = (x - \alpha)(x - \beta)$

$$2\alpha = \frac{2b}{a},\ \alpha^2 = \frac{15}{a}$$

$\alpha + \beta = 2b,\ \alpha\beta = 21$

again, $a\alpha^2 - 2b\alpha + 15 = 0$

$\alpha^2 - 2b\alpha + 21 = 0$

$(a - 1)\alpha^2 - 6 = 0$

$$\alpha^2 = \frac{6}{a-1} = \frac{15}{a} \quad \Rightarrow a = \frac{5}{3}$$

$$\alpha^2 = \frac{15}{5/3} = 9$$

$$\beta^2 = \frac{(21)^2}{\alpha^2} = 49$$

$\alpha^2 + \beta^2 = 58$

12. Let f(x) be a quadratic polynomial such that f(–2) + f(3) = 0. If one of the roots of f(x) = 0 is –1, then the sum of the roots of f(x) = 0 is equal to : **[JEE(Main)June2022]**

माना f(x) एक द्विघातीय बहुपद है जिसके लिए f(–2) + f(3) = 0 है। यदि f(x) = 0 का एक मूल –1 है, तो f(x) = 0 के मूलों का योगफल बराबर है : **[JEE(Main)June2022]**

(1) $\frac{11}{3}$ (2) $\frac{7}{3}$ (3) $\frac{13}{3}$ (4) $\frac{14}{3}$

Ans. Official Answer NTA (1)

Sol. Let $f(x) = a(x-\alpha)(x-\beta)$ .....(1)

for $f(x) = 0$; $x = -1$. [$\because$ one root is -1]

$\Rightarrow f(x) = 0 = a(x+1)(x-\beta)$ .....(2)

Now, $f(-2) + f(3) = 0$

$$a\left[(-2+1).(-2-\beta)\right] + a\left[(3+1)(3-\beta)\right] = 0$$

$$a\left[2+\beta+12-4\beta\right] = 0$$

$$\Rightarrow 14 - 3\beta = 0 \qquad \Rightarrow \beta = \frac{14}{3}$$

$\alpha = -$ (given), $\qquad \beta = \frac{14}{3}$

$$\text{sum of roots} = \frac{14}{3} - 1 = \frac{11}{3}$$

13. Let α and β be the roots of the equation $x^2 + (2i-1) = 0$. Then, the value of $|\alpha^8 + \beta^8|$ is equal to :

[JEE(Main)June2022]

माना समीकरण $x^2 + (2i-1) = 0$ के मूल α तथा β हैं। तो $|\alpha^8 + \beta^8|$ का मान बराबर है :

[JEE(Main)June2022]

(1) 50 (2) 250 (3) 1250 (4) 1500

Ans. Official Answer NTA (1)

Sol. α, β are the roots of equation

$x^2 + 2i - 1 = 0$

$$\alpha = +\sqrt{-2i+1} \qquad \alpha^2 = 1-2i$$

$$|\alpha|^2 = \sqrt{5} \quad \alpha^4 = 5 \quad \alpha^8 = 25$$

Similarly $\beta^8 = 25$

$$\alpha^8 + \beta^8 = 50$$

14. For a natural number n, let $\alpha_n = 19^n - 12^n$. Then, the value of $\frac{31\alpha_9 - \alpha_{10}}{57\alpha_8}$ is _____.

[JEE(Main)June2022]

एक धन पूर्णांक n के लिए, माना $\alpha_n = 19^n - 12^n$ है। तो $\frac{31\alpha_9 - \alpha_{10}}{57\alpha_8}$ का मान है _____

[JEE(Main)June2022]

Ans. Official Answer NTA (4)

Sol. $a_n = 19^n - 12^n$

$$\frac{31\alpha_9-\alpha_{10}}{57\alpha_8}=\frac{31\left(19^9-12^9\right)-\left(19^{10}-12^{10}\right)}{57\alpha_8}$$

$$=\frac{19^9(31-19)-12^9(31-12)}{57\alpha_8}$$

$$=\frac{19^9\cdot12-12^{19}\cdot19}{57\alpha_8}$$

$$=\frac{12\cdot19\left(19^8-12^8\right)}{57\alpha_8}=4$$

15. The sum of cubes of all the roots of the equation $x^4-3x^3-2x^2+3x+1=0$ is _____.

[JEE(Main)June2022]

समीकरण $x^4-3x^3-2x^2+3x+1=0$ के सभी मूलों के घनों का योग है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (36)

Sol. $x^4-3x^3-2x^2+3x+1=0$

$x=0$ is not the root of this equation so divide it by x^2

$$x^2-3x-2+\frac{3}{x}+\frac{1}{x^2}=0$$

$$x^2+\frac{1}{x^2}-2-3\left(x-\frac{1}{x}\right)=0$$

$$\left(x-\frac{1}{x}\right)^2-3\left(x-\frac{1}{x}\right)=0$$

$x-\frac{1}{x}=0,$ $\qquad$ $x-\frac{1}{x}=3$

$x^2-1=0$ $\qquad$ $x^2-3x-1=0$

$x=\pm1$ $\qquad$ $\gamma+\delta=3$

$\alpha=1,\ \beta=-1$ $\qquad$ $\gamma\delta=-1$

$\alpha^3+\beta^3+\gamma^3+\delta^3$

$1-1+(\gamma+\delta)\left((\gamma+\delta)^2-3\gamma\delta\right)$

$0+3(9-3(-1))$

$=0+3(12)=36$

16. Let p and q be two real numbers such that $p+q=3$ and $p^4+q^4=369$. Then $\left(\frac{1}{p}+\frac{1}{q}\right)^{-2}$ is equal to ____.

[JEE(Main)June2022]

माना दो वास्तविक संख्याओं p तथा q के लिए $p+q=3$ तथा $p^4+q^4=369$ हैं। तो $\left(\frac{1}{p}+\frac{1}{q}\right)^{-2}$ बराबर है _____

Ans. Official Answer NTA (4)

Sol. $p + q = 3$ $\quad\quad$ $p^4 + q^4 = 369$

$$\left(\frac{1}{p}+\frac{1}{q}\right)^{-2} = \left(\frac{p+q}{pq}\right)^{-2} = \left(\frac{pq}{p+q}\right)^{2}$$

$$= \frac{(pq)^2}{9}$$

$$(p)^2 + \left(q^2\right)^2 = \left(p^2+q^2\right)^2 - 2(pq)^2$$

$$369 = \left((p+q)^2 - (2pq)\right)^2 - 2(pq)^2$$

$$369 = \left(9 - (2pq)\right)^2 - 2(pq)^2$$

$pq = t$

$$369 = (9-2t)^2 - 2t^2$$

$$369 = 81 + 4t^2 - 36t - 2t^2$$

$2t^2 - 36t - 288 = 0$

$t^2 - 18t - 144 = 0$

$t^2 - 24t + 6t - 144 = 0$

$t(t - 24) + 6(t - 24) = 0$

$t = 24$ $\quad$ or $\quad$ $t = -6$

$$\left(\frac{1}{p}+\frac{1}{q}\right)^{-2} = \frac{(pq)^2}{9} = \frac{36}{9} = 4$$

or $\frac{576}{9} = 64$

17. If the sum of all the roots the equation $e^{2x} - 11e^{x} - 45e^{-x} + \frac{81}{2} = 0$ is $\log_e p$, then p is equal to _____.

[JEE(Main)June2022]

यदि समीकरण $e^{2x} - 11e^{x} - 45e^{-x} + \frac{81}{2} = 0$ के सभी हलों का योगफल $\log_e p$ है, तो p बराबर है _____

[JEE(Main)June2022]

Ans. Official Answer NTA (45)

Sol. $$e^{2x} - 11e^{x} - 45e^{-x} + \frac{81}{2} = 0$$

Let $e^x = t$

$$t^2 - 11t - 45 \times \frac{1}{t} + \frac{81}{2} = 0$$

$2t^3 - 22t^2 + 81t - 90 = 0 \begin{cases} t_1 \\ t_2 \\ t_3 \end{cases}$

$t_1t_2t_3 = \frac{90}{2} = 45$

$e^{x_1}.e^{x_2}.e^{x_3} = 45$

$e^{x_1+x_2+x_3} = 45$

$x_1 + x_2 + x_3 = \log_e 45$

So p = 45

18. Let α, β be the roots of the equation $x^2 - 4\lambda x + 5 = 0$ and α, γ be the roots of the equation $x^2 - \left(3\sqrt{2} + 2\sqrt{3}\right)x + 7 + 3\lambda\sqrt{3} = 0$, $\lambda > 0$. If $\beta + \gamma = 3\sqrt{2}$, then $(\alpha + 2\beta + \gamma)^2$ is equal to _____.

[JEE(Main)June2022]

माना समीकरण $x^2 - 4\lambda x + 5 = 0$ के मूल α, β हैं तो समीकरण $x^2 - \left(3\sqrt{2} + 2\sqrt{3}\right)x + 7 + 3\lambda\sqrt{3} = 0$ के मूल α, γ हैं, $\lambda > 0$ है। यदि $\beta + \gamma = 3\sqrt{2}$ है, तो $(\alpha + 2\beta + \gamma)^2$ बराबर है _____. **[JEE(Main)June2022]**

Ans. Official Answer NTA (98)

Sol. $\alpha + \beta = 4\lambda$ _____(1)

$\alpha + \gamma = 3\sqrt{2} + 2\sqrt{3}$ _____(2)

$\beta + \gamma = 3\sqrt{2}$ _____(3)

(1) + (2) + (3)

$\alpha + \beta + \gamma = \frac{4\lambda + 6\sqrt{2} + 2\sqrt{3}}{2}$

$= 2\lambda + 3\sqrt{2} + \sqrt{3}$

$\Rightarrow \gamma = 3\sqrt{2} + \sqrt{3} - 2\lambda$

$\beta = 2\lambda - \sqrt{3}$

$\alpha = 2\lambda + \sqrt{3}$

$\alpha + 2\beta + \gamma = 4\lambda + 3\sqrt{2}$

$\alpha\beta = 5$

$\left(2\lambda + \sqrt{3}\right)\left(2\lambda - \sqrt{3}\right) = 5$

$4\lambda^2 - 3 = 5$

$\lambda^2 = 2$

$\lambda = \sqrt{2}$

$\alpha + 2\beta + \gamma = 4\sqrt{2} + 3\sqrt{2}$

$= \left(7\sqrt{2}\right)^2$

$= 98$

19. The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$ is : **[JEE(Main)June2022]**

समीकरण $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$ के वास्तविक हलों की संख्या है : **[JEE(Main)June2022]**

Ans. Official Answer NTA (2)

Sol. $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$

Let $f(x) = e^{2x}\left(e^{2x} + \frac{1}{e^{2x}} + 4\left(e^{x} + \frac{1}{e^{x}}\right) - 58\right)$

Let $e^{x} + \frac{1}{e^{x}} = t$

Let $h(t) = t^2 + 4t - 58 = 0$

$t = \frac{-4 \pm \sqrt{16 + 4.58}}{2}$

$\frac{-4 \pm 2\sqrt{62}}{2}$

$t_1 = -2 - 2\sqrt{62}$

$t_2 = -2 + 2\sqrt{62}$ (not possbile)

$t \geq 2$

$e^{x} + \frac{1}{e^{x}} = -2 + 2\sqrt{62}$

$e^{2x} - \left(-2 + 2\sqrt{62}\right)e^{x} + 1 = 0$

$\left(-2 + 2\sqrt{62}\right) - 4$

$4 + 4.62 - 8\sqrt{62} - 4$

$248 - 8\sqrt{62} > 0$

both roots are positive

2 real roots

20. The minimum value of the sum of the square of the roots of $x^2 + (3 - a)x + 1 = 2a$ is :

[JEE(Main)July2022]

समीकरण $x^2 + (3 - a)x + 1 = 2a$ के मूलों के वर्गों के योगफल का निम्नतम मान है :

[JEE(Main)July2022]

(1) 4 (2) 5 (3) 6 (4) 8

Ans. Official Answer NTA (3)

Sol. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

let $f(a) = (3 - a)^2 - 2(1 - 2a)$

$f(a) = a^2 - 2a + 7$

$f(a) = (a - 1)^2 + 6$

$f(a))_{min.} = 6$

21. If α, β are the roots of the equation $x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_5 3}}\right)x + 3\left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0$, then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, is : **[JEE(Main)July2022]**

यदि समीकरण $x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_5 3}}\right)x + 3\left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0$ के मूल α, β हैं, तो वह समीकरण, जिसके मूल $\alpha + \frac{1}{\beta}$ तथा $\beta + \frac{1}{\alpha}$ हैं, है : **[JEE(Main)July2022]**

(1) $3x^2 - 20x - 12 = 0$ (2) $3x^2 - 10x - 4 = 0$

(3) $3x^2 - 10x + 2 = 0$ (4) $3x^2 - 20x + 16 = 0$

Ans. Official Answer NTA (2)

Sol. Bonus because 'x' is missing the correct will be,

$$x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_5 3}}\right)x + 3\left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0$$

$$3^{\sqrt{\log_3 5}} = 3^{\sqrt{\log_3 5}\cdot\sqrt{\log_3 5}\cdot\sqrt{\log_5 3}} = 3^{\log_3 5\cdot\sqrt{\log_5 3}}$$

$$= \left(3^{\log_3 5}\right)^{\sqrt{\log_5 3}} = 5^{\sqrt{\log_5 3}}$$

$$3^{\sqrt[3]{\log_3 5}} = 3^{\log_3 5\cdot\sqrt[3]{(\log_5 3)^2}} = \left(3^{\log_3 5}\right)^{(\log_3 5)^{2/3}}$$

$$= 5^{(\log_5 3)^{2/3}}$$

So, equation is $x_2 - 5x - 3 = 0$ and roots are α & β $\{\alpha + \beta = 5; \alpha\beta = -3\}$

New roots are $\alpha + \frac{1}{\beta}$ & $\beta + \frac{1}{\alpha}$

i.e., $\frac{\alpha\beta + 1}{\beta}$ & $\frac{\alpha\beta + 1}{\alpha}$ i.e., $\frac{-2}{\beta}$ & $\frac{-2}{\alpha}$

Let $\frac{-2}{\alpha} = t \Rightarrow \alpha = \frac{-2}{t}$

As $\alpha^2 - 5\alpha - 3 = 0$

$\Rightarrow \frac{4}{t^2} + \frac{10}{t} - 3 = 0$

$\Rightarrow 4 + 10t - 3t^2 = 0$

$\Rightarrow 3t^2 - 10t - 4 = 0$

i.e., $3x^2 - 10x - 4 = 0$

22. Let α, β be the roots of the equation $x^2 - \sqrt{2}x + \sqrt{6} = 0$ and $\frac{1}{\alpha^2} + 1, \frac{1}{\beta^2} + 1$ be the roots of the equation $x^2 + ax + b = 0$. Then the roots of the equation $x^2 - (a + b - 2)x + (a + b + 2) = 0$ are :

[JEE(Main)July2022]

(1) non-real complex numbers (2) real and both negative

(3) real and both positive (4) real and exactly one of them is positive

माना समीकरण $x^2 - \sqrt{2}x + \sqrt{6} = 0$ के मूल α, β हैं तथा समीकरण $x^2 + ax + b = 0$ के मूल $\frac{1}{\alpha^2} + 1, \frac{1}{\beta^2} + 1$ हैं। तो समीकरण $x^2 - (a + b - 2)x + (a + b + 2) = 0$ के मूल : **[JEE(Main)July2022]**

(1) अवास्तविक सम्मिश्र संख्याऐं है (2) वास्तविक तथा दोनों ऋणात्मक हैं

(3) वास्तविक तथा दोनों धनात्मक हैं (4) वास्तविक तथा उनकें से ठीक एक धनात्मक है

Ans. Official Answer NTA (2)

Sol. $a = \frac{-1}{\alpha^2} - \frac{1}{\beta^2} - 2$

$b = \frac{1}{\alpha^2} + \frac{1}{\beta^2} + 1 + \frac{1}{\alpha^2\beta^2}$

$a + b = \frac{1}{(\alpha\beta)^2} - 1 = \frac{1}{6} - 1 = -\frac{5}{6}$

$x^2 - \left(-\frac{5}{6} - 2\right)x + \left(2 - \frac{5}{6}\right) = 0$

$6x^2 + 17x + 7 = 0$

$x = -\frac{7}{3}, x = -\frac{1}{2}$ are the roots

Both roots are real and negative.

23. If for some p, q, r $\in$ R, not all have same sign, one of the roots of the equation $(p^2 + q^2)x^2 - 2q(p + r)x + q^2 + r^2 = 0$ is also a root of the equation $x^2 + 2x - 8 = 0$, then $\frac{q^2 + r^2}{p^2}$ is equal to _____.

[JEE(Main)July2022]

यदि p, q, r $\in$ R, सभी धनात्मक या सभी ऋणात्मक नहीं है, के लिए समीकरण $(p^2 + q^2)x^2 - 2q(p + r)x + q^2 + r^2 = 0$ का एक मूल समीकरण $x^2 + 2x - 8 = 0$ का भी एक मूल है, तो $\frac{q^2 + r^2}{p^2}$ बराबर है _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (272)

Sol. $(px - q)^2 + (qx - r)^2 = 0$

$$\Rightarrow x = \frac{q}{p} = \frac{r}{q} = -4$$

$$\Rightarrow \frac{q^2 + r^2}{p^2} = 272$$

24. The sum of all real value of x for which $\frac{3x^2 - 9x + 17}{x^2 + 3x + 10} = \frac{5x^2 - 7x + 19}{3x^2 + 5x + 12}$ is equal to _____.

[JEE(Main)July2022]

x के सभी मानों, जिसके लिए $\frac{3x^2 - 9x + 17}{x^2 + 3x + 10} = \frac{5x^2 - 7x + 19}{3x^2 + 5x + 12}$ है, का योग बराबर है _____

[JEE(Main)July2022]

Ans. Official Answer NTA (6)

Sol. $$\frac{3x^2 - 9x + 17}{x^2 + 3x + 10} = \frac{5x^2 - 7x + 19}{3x^2 + 5x + 12}$$

$$\frac{x^2 + 3x + 10 + 2x^2 - 12x + 7}{x^2 + 3x + 10} = \frac{3x^2 + 5x + 12 + 2x^2 - 12x + 7}{3x^2 + 5x + 12}$$

$$1 + \frac{2x^2 - 12x + 7}{x^2 + 3x + 10} = 1 + \frac{2x^2 - 12x + 7}{3x^2 + 5x + 12}$$

$$\left(2x^2 - 12x + 7\right)\left(\frac{1}{x^2 + 3x + 10} - \frac{1}{3x^2 + 5x + 12}\right) = 0$$

$2x^2 - 12x + 7 = 0$ OR $3x^2 + 5x + 12 = x^2 + 3x + 10$

$x = \frac{12 \pm \sqrt{D}}{4}$ $\qquad$ $2x^2 + 2x + 2 = 0$

$\qquad\qquad\qquad\qquad$ $x^2 + x + 1 = 0$

Sum or Roots = 6 $\qquad$ No solution.

25. Let α, β ($\alpha > \beta$) be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n$, $n \in N$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to _____. **[JEE(Main)July2022]**

माना $x^2 - x - 4 = 0$ के मूल α, β ($\alpha > \beta$) हैं। यदि $P_n = \alpha^n - \beta^n$, $n \in N$ है, तो $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ बराबर है_____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (16)

Sol. $P_n = \alpha^n - \beta^n$ $\qquad$ $x^2 - x - 4 = 0$

$$\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}} \quad .....(1)$$

As $P_n - P_{n-1} = \left(\alpha^n - \beta^n\right) - \left(\alpha^{n-1} - \beta^{n-1}\right)$

$= \alpha^{n-2}\left(\alpha^2 - \alpha\right) - \beta^{n-2}\left(\beta^2 - \beta\right)$

$= 4\left(\alpha^{n-2} - \beta^{n-2}\right)$

$P_n - P_{n-1} = 4P_{n-2}$

Hence Expression (1)

$$\frac{P_{16}\left(P_{15} - P_{14}\right) - P_{15}\left(P_{15} - P_{14}\right)}{P_{13}P_{14}}$$

$$= \frac{\left(P_{15} - P_{14}\right)\left(P_{16} - P_{15}\right)}{P_{13}P_{14}} = \frac{\left(4P_{13}\right)\left(4P_{14}\right)}{P_{13}P_{14}} = 16$$

Exercise J - A

1. A value of b for which equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have one root in common is : **[JEE(Adv)-2011]**

b के किस मान के लिये समीकरणों $x^2 + bx - 1 = 0$ तथा $x^2 + x + b = 0$ का एक मूल उभयनिष्ठ होगा : **[JEE(Adv)-2011]**

(A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

Ans. B

Sol. $x^2 + bx - 1 = 0$

$x^2 + x + b = 0$

$\frac{x^2}{b^2+1} = \frac{x}{-1-b} = \frac{1}{1-b} \Rightarrow x = \frac{b^2+1}{-(b+1)} = \frac{-(b+1)}{1-b} \Rightarrow (b^2+1)(1-b) = (b+1)^2$

$\Rightarrow \quad b^2 - b^3 + 1 - b = b^2 + 2b + 1 \Rightarrow b^3 + 3b = 0 \Rightarrow b = 0 ; b^2 = -3 \Rightarrow b = 0, \pm\sqrt{3}\,i$

2. The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation $p(p(x)) = 0$ has : **[JEE(Adv)-2014]**

(A) only purely imaginary roots (B) all real roots

(C) two real and two purely imaginary roots (D) neither real nor purely imaginary roots

समीकरण $p(x) = 0$, सभी गुणांक वास्तविक है, के शुद्ध काल्पनिक मूल है तो समीकरण $p(p(x)) = 0$ के : **[JEE(Adv)-2014]**

(A) केवल शुद्ध काल्पनिक मूल होगें (B) सभी मूल वास्तविक होगे

(C) दो वास्तविक तथा दो काल्पनिक मूल होगे (D) ना तो वास्तविक ना ही शुद्ध काल्पनिक मूल होगे

Ans. D

Sol. $p(x)$ will be of the form $ax^2 + c$. Since it has purely imaginary roots only.

Since $p(x)$ is zero at imaginary values while $ax^2 + c$ takes real value only at real 'x', no root is real.

Also $p(p(x)) = 0 \Rightarrow p(x)$ is purely imaginary $\Rightarrow ax^2 + c =$ purely imaginary

Hence x can not be purely imaginary since x^2 will be negative in that case and $ax^2 + c$ will be real.

Thus .(D) is correct.

3. Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is (are) a subset (s) of S? **[JEE(Adv)-2015]**

माना कि S उन सभी शून्येतर (non-zero) वास्तविक संख्याओं α का समुच्चय (set) है जिनके लिए द्विघाती समीकरण $\alpha x^2 - x + \alpha = 0$ के दो विभिन्न वास्तविक मूल x_1 और x_2 असमिका $|x_1 - x_2| < 1$ को सन्तुष्ट करते हैं। निम्नलिखित अन्तरालों में से कौनसा (से) समुच्चय S के उपसमुच्चय है (हैं)? **[JEE(Adv)-2015]**

(A) $\left(-\frac{1}{2},-\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}},0\right)$ (C) $\left(0,\frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}},\frac{1}{2}\right)$

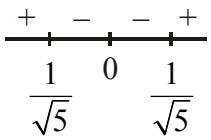
Ans. AD

Sol. $(x_1+x_2)^2-4x_1x_2<1$

$$\frac{1}{\alpha^2}-4<1$$

$$\Rightarrow \quad 5-\frac{1}{\alpha^2}>0$$

$$\frac{5\alpha^2-1}{\alpha^2}>0$$



$$\alpha\in\left(-\infty,-\frac{1}{\sqrt{5}}\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right) \qquad \text{_____(1)}$$

$D > 0$

$1 - 4\alpha^2 > 0$

$$\alpha\in\left(-\frac{1}{2},\frac{1}{2}\right) \qquad \text{_____(2)}$$

(1) and (2)

$$\alpha\in\left(-\frac{1}{2},\frac{1}{\sqrt{5}}\right)\cup\left(\frac{1}{\sqrt{5}},\frac{1}{\sqrt{2}}\right)$$

Paragraph for Question nos. 4 to 5

Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$. For $n = 0,1, 2,.....$, let $a_n = p\alpha^n + q\beta^n$.

FACT : if a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$. **[JEE(Adv)-2017]**

माना कि p, q पूर्णांक है एवं α, β समीकरण $x^2 - x - 1 = 0$ के मूल हैं, जहाँ $\alpha \neq \beta$ है। $n = 0,1, 2,.....$, के लिये माना कि $a_n = p\alpha^n + q\beta^n$.

तथ्यः यदि a तथा b परिमेय संख्यायें हैं एवं $a + b\sqrt{5} = 0$ है, तब $a = 0 = b$. **[JEE(Adv)-2017]**

4. If $a_4 = 28$, then $p + 2q =$

यदि $a_4 = 28$ है, तब $p + 2q =$

(A) 12 (B) 21 (C) 14 (D) 7

Ans. A

Sol. $a_{n+2} = a_{n+1} + a_n$

$a_4 = a_3 + a_2 = 3a_1 + 2a_0 = 3p\alpha + 3q\beta + 2(p + q)$

As $\alpha = \frac{1+\sqrt{5}}{2}$, $\beta = \frac{1-\sqrt{5}}{2}$, we get

$$a_4 = 3p\left(\frac{1+\sqrt{5}}{2}\right) + 3q\left(\frac{1-\sqrt{5}}{2}\right) + 2p + 2q = 28$$

$$\left(\frac{3p}{2} + \frac{3q}{2} + 2p + 2q - 28\right) = 0 \text{.....(i)}$$

and $\frac{3p}{2} - \frac{3q}{2} = 0$.....(ii)

$\Rightarrow$ $p = q$ (from (ii))

$\Rightarrow$ $7p = 28$ (from (i) and (ii))

$\Rightarrow$ $p = 4$

$\Rightarrow$ $q = 4$

$\Rightarrow$ $p + 2q = 12$

5. $a_{12} =$

(A) $a_{11} + 2a_{10}$ (B) $a_{11} + a_{10}$ (C) $a_{11} - a_{10}$ (D) $2a_{11} + a_{10}$

Ans. B

Sol. As α and β are roots of equation $x^2 - x - 1 = 0$, we get :

$\alpha^2 - \alpha - 1 = 0 \quad \Rightarrow \quad \alpha^2 = \alpha + 1$

$\beta^2 - \beta - 1 = 0 \quad \Rightarrow \quad \beta^2 = \beta + 1$

$\therefore \quad a_{11} + a_{10} = p\alpha^{11} + q\beta^{11} + p\alpha^{10} + q\beta^{10}$

$= p\alpha^{10}(\alpha + 1) + q\beta^{10}(\beta + 1)$

$= p\alpha^{10} \times \alpha^2 + q\beta^{10} \times \beta^2$

$= p\alpha^{12} + q\beta^{12}$

$= a_{12}$

6. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is : **[JEE(Adv)-2020]**

मान लीजिये कि a, b द्विघातीय बहुपद $x^2 + 20x - 2020$ के भिन्न वास्तविक मूलों को दर्शाते हैं, एवं मान लीजिये कि c, d द्विघातीय बहुपद $x^2 - 20x + 2020$ के भिन्न समिश्र मूलों को दर्शाते हैं। तब $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ का मान है : **[JEE(Adv)-2020]**

(A) 0 (B) 8000 (C) 8080 (D) 16000

Ans. D

Sol. $x^2 + 20x - 2020 = 0$ a, b are its roots

$x^2 - 20x + 2020 = 0$ c,d are its roots

$a + b = -20$ $ab = -2020$

$c + d = 20$ $cd = 2020$

$E = ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$

$= a^2c - ac^2 + a^2d - ad^2 + b^2c + b^2d - bd^2$

$= a^2(c + d) + b^2(c + d) - c^2(a + b) - d^2(a + b)$

$= (a^2 + b^2)(c + d) - (a + b)(c^2 + d^2)$

$= ((a + b)^2 - 2ab)(c + d) - (a + b)((c + d)^2 - 2cd)$

$= (400 + 4040)(20) - (-20)(400 - 4040)$

$= 88800 - 72800 = 16000$

7. Let R^2 denote $R \times R$. Let $S = \{(a, b, c) : a, b, c \in R$ and $ax^2 + 2bxy + cy^2 > 0$ for all $(x, y) \in R^2 - \{(0,0)\}\}$. Then which of the following statements is(are) TRUE ? **[JEE(Adv.)2024]**

(A) $\left(2, \frac{7}{2}, 6\right) \in S$

(B) If $\left(3, b, \frac{1}{12}\right) \in S$, then $|2b| < 1$.

(C) For any given $(a, b, c) \in S$, the system of linear equations
$ax + by = 1$
$bx + cy = -1$
has a unique solution.

(D) For any given $(a, b, c) \in S$, the system of linear equations
$(a + 1)x + by = 0$
$bx + (c + 1)y = 0$
has a unique solution.

माना कि R^2, $R \times R$ को दर्शाता है। माना कि $S = \{(a, b, c) : a, b, c \in R$ और सभी $(x, y) \in R^2 - \{(0,0)\}$ के लिए $ax^2 + 2bxy + cy^2 > 0\}$ है। तब निम्नलिखित कथनों में से कौनसा(से) सत्य है(हैं) ? **[JEE(Adv.)2024]**

(A) $\left(2, \frac{7}{2}, 6\right) \in S$

(B) यदि $\left(3, b, \frac{1}{12}\right) \in S$, तब $|2b| < 1$ है।

(C) किसी दिये गये $(a, b, c) \in S$ के लिए, रैखिक समीकरणों के निकाय

$ax + by = 1$

$bx + cy = -1$

का एक अद्वितीय हल है।

(D) किसी दिये गए $(a, b, c) \in S$ के लिए, रैखिक समीकरणों के निकाय

$(a + 1)x + by = 0$

$bx + (c + 1)y = 0$

का एक अद्वितीय हल है।

Ans. BCD

Sol. $ax^2 + 2bxy + cy^2 > 0$

$\Rightarrow a\left(\frac{x}{y}\right)^2 + 2b\left(\frac{x}{y}\right) + c > 0$ $\quad$ $c\left(\frac{y}{x}\right)^2 + 2b\left(\frac{y}{x}\right) + a > 0$

$\Rightarrow a > 0$ and $D < 0$ $\quad$ $c > 0$ and $D < 0$

$b^2 < ac$ and $a > 0$ and $c > 0$

(A) $\frac{49}{4} < 12$ $\quad$ incorrect

(B) $b^2 < \frac{3}{12} \Rightarrow 4b^2 < 1 \quad \Rightarrow |2b| < 1$ $\quad$ correct

(C) for unique solution $\frac{a}{b} \neq \frac{b}{c} \Rightarrow b^2 \neq ac$ $\quad$ correct

(D) $\frac{a+1}{b} \neq \frac{b}{c+1} \Rightarrow \underbrace{b^2 - ac}_{(-)ve} \neq \underbrace{a + c + 1}_{(+)ve}$ $\quad$ correct

8. Let $f(x) = x^4 + ax^3 + bx^2 + c$ be a polynomial with real coefficients such that $f(1) = -9$. Suppose that $i\sqrt{3}$ is a root of the equation $4x^3 + 3ax^2 + 2bx = 0$, where $i = \sqrt{-1}$. If α_1, α_2, α_3 and α_4 are all the roots of the equation $f(x) = 0$, then $|\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 + |\alpha_4|^2$ is equal to _____. **[JEE(Adv.)2024]**

माना कि $f(x) = x^4 + ax^3 + bx^2 + c$ वास्तविक गुणांकों वाला एक ऐसा बहुपद है कि $f(1) = -9$ है। मान लीजिये कि $i\sqrt{3}$, समीकरण $4x^3 + 3ax^2 + 2bx = 0$ का एक मूल है, जहाँ $i = \sqrt{-1}$ है। यदि α_1, α_2, α_3 और α_4, समीकरण $f(x) = 0$ के सभी मूल हैं, तब $|\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 + |\alpha_4|^2$ का मान _____ है। **[JEE(Adv.)2024]**

Ans. 20

Sol. $f(1) = -9$

$a + b + c = -10$

$4x^3 + 3ax^2 + 2bx = 0$

$x\left(4x^2 + 3ax + 2b\right) = 0$

$\Rightarrow \left(4x^2 + 3ax + 2b\right) = 0 \begin{matrix} \nearrow i\sqrt{3} \\ \searrow -i\sqrt{3} \end{matrix}$

$a = 0,\ \frac{2b}{4} = 3 \Rightarrow b = 6$

$c = -10 - a - b = -16$

$f(x) = x^4 + 6x^2 - 16 = 0$

$(x^2 + 8)\ (x^2 - 2) = 0$

$x = \pm\sqrt{2}\ ,\ \pm 2\sqrt{2}i$

$\sum|\alpha|^2 = 20$

9. Let $a = 3\sqrt{2}$ and $b = \frac{1}{5^{\frac{1}{6}}\sqrt{6}}$. If x, y ∈ R such that $3x + 2y = \log_a (18)^{\frac{5}{4}}$ and $2x - y = \log_b\left(\sqrt{1080}\right)$, then $4x + 5y$ is equal to _____. **[JEE(Adv.)2024]**

माना कि $a = 3\sqrt{2}$ और $b = \frac{1}{5^{\frac{1}{6}}\sqrt{6}}$ हैं। यदि x, y ∈ R इस प्रकार हैं कि $3x + 2y = \log_a (18)^{\frac{5}{4}}$ और $2x - y = \log_b\left(\sqrt{1080}\right)$, तब $4x + 5y$ बराबर _____ है। **[JEE(Adv.)2024]**

Ans. 8

Sol. $\log_{3\sqrt{2}}\left(3\sqrt{2}\right)^{\frac{5}{2}} = \frac{5}{2} = 3x + 2y \quad \Rightarrow 3x + 2y = \frac{5}{2} \quad _____(1)$

$2x - y = \log_{\left(5^{\frac{1}{6}}\sqrt{6}\right)^{-1}} \sqrt{5^1 6^3} = -3 \quad \Rightarrow 4x - 2y = -6 \quad _____(2)$

From equ. (1) and (2), $x = -\frac{1}{2}$, $y = 2$

$4x + 5y = -2 + 10 = 8$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CHOICE TYPE QUESTIONS

1. If one root of the quadratic equation $ax^2 + bx + c = 0$ be n^{th} root of the other then $(ac^n)^{1/n+1} + (a^nc)^{1/n+1} =$

यदि द्विघात समीकरण $ax^2 + bx + c = 0$ का एक मूल दूसरे कि n^{th} घात हो, $(ac^n)^{1/n+1} + (a^nc)^{1/n+1} =$

(A) $b^{1/n+1}$ (B) $b^{1/n}$ (C) b (D) –b

Ans. D

Sol. $\alpha.\alpha^n = \frac{c}{a} \quad \Rightarrow \quad \alpha = \left(\frac{c}{a}\right)^{\frac{1}{n+1}} \quad \Rightarrow \quad \alpha + \alpha^n = -\frac{b}{a} \quad \Rightarrow \quad \left(\frac{c}{a}\right)^{\frac{1}{n+1}} + \left(\frac{c}{a}\right)^{\frac{n}{n+1}} = -\frac{b}{a}$

$a^{1-\frac{1}{n+1}}.c^{\frac{1}{n+1}} + c^{\frac{n}{n+1}}.a^{1-\frac{n}{n+1}} + b = 0 \quad \Rightarrow \quad a^{\frac{n}{n+1}}.c^{\frac{1}{n+1}} + a^{\frac{1}{n+1}}.c^{\frac{n}{n+1}} + b = 0$

$\left(a^n c\right)^{\frac{1}{n+1}} + \left(a c^n\right)^{\frac{1}{n+1}} + b = 0$ Proved.

2. If α, β are the roots of the equation $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$, then $a S_{n+1} + c S_{n-1} =$

यदि α, β समीकरण $ax^2 + bx + c = 0$ के मूल हो तथा $S_n = \alpha^n + \beta^n$ तथा $a S_{n+1} + c S_{n-1} =$

(a) $b S_n$ (B) $b^2 S_n$ (C) $2bS_n$ (D) $-bS_n$

Ans. D

Sol. $ax^2 + bx + c = 0 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right.; \quad S_n = \alpha^n + \beta^n$

$a\alpha^2 + b\alpha + c = 0$

$\Rightarrow a\alpha^2 + c = -b\alpha$

$a\beta^2 + b\beta + c = 0$

$a\beta^2 + c = -b\beta$

Now $aS_{n+1} + c \cdot S_{n-1}$

$= a\left(\alpha^{n+1} + \beta^{n+1}\right) + c\cdot\left(\alpha^{n-1} + \beta^{n-1}\right)$

$= a\alpha^{n+1} + c\cdot\alpha^{n-1} + a\beta^{n+1} + c\cdot\beta^{n-1}$

$= \alpha^n\left(a\alpha + \frac{c}{\alpha}\right) + \beta^n\left(a\beta + \frac{c}{\beta}\right)$

$= \frac{\alpha^n}{\alpha}\left(a\alpha^2 + c\right) + \frac{\beta^n}{\beta}\left(a\beta^2 + c\right)$

$$= \frac{\alpha^n}{\alpha}(b\alpha) + \frac{\beta^n}{\beta}(-b\beta)$$

$$= -b\left(\alpha^n + \beta^n\right) = -bS_n$$

3. If the equation $x^2 + 2(k+1)x + 9k - 5 = 0$ has only negative roots, then :

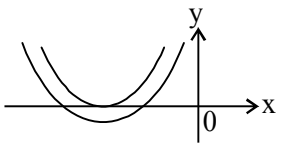
यदि समीकरण $x^2 + 2(k+1)x + 9k - 5 = 0$ के केवल ऋणात्मक मूल हों, तो :

(A) $k \le 0$ (B) $k \ge 0$ (C) $k \ge 6$ (D) $k \le 6$

Ans. C

Sol. Equation $x^2 + 2(k+1)x + (9k-5) = 0$

$\because$ Roots only negative



(i) $\therefore D \ge 0$

$4(k+1)^2 - 4(9k-5) \ge 0$

$k^2 + 1 + 2k - 9k + 5 \ge 0$

$k^2 - 7k + 6 \ge 0$

$(k-6)(k-1) \ge 0$

$k \le 1, k \ge 6$

(ii) $-\frac{b}{2a} < 0$

$k \in (-1, \infty)$

(iii) $f(0) > 0$

$k \in \left(\frac{5}{9}, \infty\right)$

(i) $\cap$ (ii) $\cap$ (iii)

$k \in [6, \infty)$

4. The set of values of 'a' for which $f(x) = ax^2 + 2x(1-a) - 4$ is negative for exactly three integral values of x, is :

'a' के मानों का समुच्चय जिसके लिये $f(x) = ax^2 + 2x(1-a) - 4$ के ठीक तीन पूर्णांक मानों के लिये ऋणात्मक है, होगा :

(A) $(0, 2)$ (B) $(0, 1]$ (C) $[1, 2)$ (D) $[2, \infty)$

Ans. C

Sol. $f(x) = ax^2 + 2x(1-a) - 4$

$$x = \frac{-2(1-a) \pm \sqrt{4(1-a)^2 + 16a}}{2a}$$

$$x = \frac{+2(a-1) \pm 2\sqrt{1+a^2-2a+4a}}{2a}$$

$$x = \frac{(a-1) \pm \sqrt{(1+a)^2}}{a}$$

$$x = \frac{a-1+1+a}{a} = \frac{2a}{a} = 2,$$

and $x = \frac{a-1-1-a}{a}$

$$x = -\frac{2}{a}$$

$$-\frac{2}{a} < -1, \frac{-2}{a} \geq -2$$

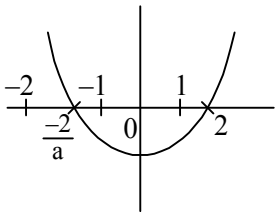
$$\frac{-2}{a} + 1 < 0, -\frac{2}{a} + 2 \geq 0$$

$$\frac{a-2}{a} < 0, \frac{2a-2}{a} \geq 0$$

$$a < 2, \frac{2(a-1)}{a} \geq 0$$

$$a \geq 1$$

$$\therefore a \in [1, 2)$$



5. The number of real solutions of the equation $(15 + 4\sqrt{14})^t + (15 - 4\sqrt{14})^t = 30$ are, where $t = x^2 - 2|x|$:

समीकरण $(15 + 4\sqrt{14})^t + (15 - 4\sqrt{14})^t = 30$ के वास्तविक हलों की संख्या क्या होगी, जहाँ $t = x^2 - 2|x|$:

(A) 0 (B) 2 (C) 4 (D) 6

Ans. C

Sol. $\left(15 + 4\sqrt{14}\right)^t + \left(15 - 4\sqrt{14}\right)^t = 30$

Let $u = 15 + 4\sqrt{14}$

$$\frac{1}{u} = 15 - 4\sqrt{14}$$

$$u^t + \frac{1}{u^t} = 30$$

put $u^t = p$

$$p+\frac{1}{p}=30$$

$p^2 - 30p + 1 = 0$

$$p=\frac{30\pm\sqrt{900-4}}{2}$$

$$p=\frac{30\pm\sqrt{896}}{2}$$

$$p=\frac{30\pm 8\sqrt{14}}{2}$$

$$p=15\pm 4\sqrt{14}$$

$$p=15+4\sqrt{14},15-4\sqrt{14}$$

$u^t = 15+4\sqrt{14}$, $u^t = 15-4\sqrt{14}$

$$\left(15+4\sqrt{14}\right)^t=\left(15+4\sqrt{14}\right)^1$$

$t = 1$

And

$$\left(15+4\sqrt{14}\right)^t=15-4\sqrt{14}$$

$$\left(15+4\sqrt{14}\right)^{t+1}=\left(15-4\sqrt{14}\right).\left(15+4\sqrt{14}\right)$$

$$\left(15+4\sqrt{14}\right)^{t+1}=1$$

$t + 1 = 0$

$t = -1$

for $t = 1$

$x^2 - 2|x| = 1$

for $x \geq 0$	;	for $x < 0$
$x^2 - 2x - 1 = 0$	;	$x^2 + 2x - 1 = 0$
$x=\frac{+1\pm\sqrt{2}}{1}$	;	$x = -1+\sqrt{2}, -1-\sqrt{2}$

$x = +1+\sqrt{2}, +1-\sqrt{2}$

for $t = -1$

$x^2 - 2|x| = -1$

$x^2 - 2|x| + 1 = 0$

for $\|x\| < 0$	;	for $x \geq 0$
$x^2 + 2x + 1 = 0$	;	$x^2 - 2x + 1 = 0$
$(x + 1)^2 = 0$	;	$(x - 1)^2 = 0$
$x = -1$	;	$x = 1$

Total no. of solutions = 4

6. If α, β are the roots of $ax^2 + bx + c = 0, (a \neq 0)$, and $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0, (A \neq 0)$ for some constant δ, then the value of $\frac{b^2 - 4ac}{a^2}$ is :

(A) $\frac{B^2 - 4AC}{A^2}$ (B) $\frac{B^2 + 4AC}{A^2}$ (C) $\frac{A^2 - 4BC}{A^2}$ (D) None of these

यदि α, β समीकरण $ax^2 + bx + c = 0, (a \neq 0)$ के मूल व किसी अचर राशि δ के लिए $\alpha + \delta, \beta + \delta$ समीकरण $Ax^2 + Bx + C = 0, (A \neq 0)$ के मूल हों, तो $\frac{b^2 - 4ac}{a^2}$ का मान होगा :

(A) $\frac{B^2 - 4AC}{A^2}$ (B) $\frac{B^2 + 4AC}{A^2}$ (C) $\frac{A^2 - 4BC}{A^2}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $ax^2 + bx + c = 0 \begin{matrix}\nearrow \alpha \\ \searrow \beta\end{matrix} \Rightarrow \alpha + \beta = -\frac{b}{a}, \ \alpha\beta = \frac{c}{a} \Rightarrow Ax^2 + Bx + C = 0 \begin{matrix}\nearrow \alpha+\delta \\ \searrow \beta+\delta\end{matrix}$

$(\alpha + \delta) + (\beta + \delta) = -\frac{B}{A}, \quad (\alpha + \delta)(\beta + \delta) = \frac{C}{A} \quad \because \quad |(\alpha + \delta) - (\beta + \delta)| = |(\alpha - \beta)|$

$\Rightarrow \quad \sqrt{\frac{B^2}{A^2} - \frac{4C}{A}} = \sqrt{\frac{b^2}{a^2} - \frac{4c}{a}} \Rightarrow \frac{b^2 - 4ac}{a^2} = \frac{B^2 - 4AC}{A^2}$ Hence proved

7. Given a, b, c are non negative real numbers and if $a^2 + b^2 + c^2 = 1$, then the value of $a + b + c$ is :

दिया गया है कि a, b, c अऋणात्मक वास्तविक संख्यायें है तथा यदि $a^2 + b^2 + c^2 = 1$ है तो $a + b + c$ का मान होगा :

(A) ≥ 3 (B) ≥ 2 (C) $\leq \sqrt{2}$ (D) $\leq \sqrt{3}$

Ans. D

Sol. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2\sum ab = 1 + 2\sum ab$

put $a^2 + b^2 + c^2 \geq ab + bc + ca$

or $\sum ab \leq 1$

$\therefore \quad (a + b + c)^2 \leq 1 + 2$

$\therefore \quad (a + b + c) \leq \sqrt{3}$

Alternatively : using RMS $\geq$ AM in a, b, c

$\sqrt{\frac{a^2 + b^2 + c^2}{3}} \geq \frac{a + b + c}{3} \quad \Rightarrow \quad \sqrt{3} \geq (a + b + c)^2 \quad \Rightarrow \quad a + b + c \leq \sqrt{3}$

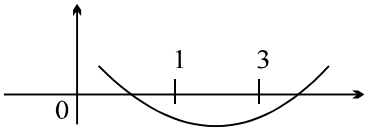
8. The set of values of 'a' for which the inequality, $(x - 3a)(x - a - 3) < 0$ is satisfied for all $x \in [1, 3]$ is :

'a' के मानों का समुच्चय जिनके लिये असमिका $(x - 3a)(x - a - 3) < 0$ सभी $x \in [1, 3]$ के लिये संतुष्ट होती है, क्या होगा :

(A) $(1/3, 3)$ (B) $(0, 1/3)$ (C) $(-2, 0)$ (D) $(-2, 3)$

Ans. B

Sol. Equation is $x^2 - (4a + 3)x + 3a(a + 3)$



$f(1) < 0$ and $f(3) < 0$

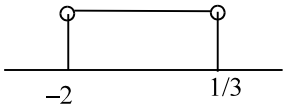
$(1-3a)(1 - a - 3) < 0 \Rightarrow 1 - a - 3 - 3a + 3a^2 + 9a < 0$

$\Rightarrow \quad 3a^2 + 5a - 2 < 0$

$3a^2 + 6a - a - 2 < 0$

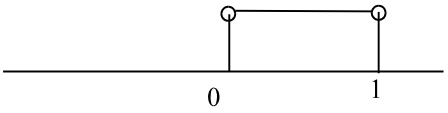
$3a(a + 2) - (a + 2) < 0$

$(a + 2)(3a - 1) < 0 \Rightarrow$



again $(3-3a)(-a) < 0 \Rightarrow (a - 1)a < 0$

$\Rightarrow \quad 0 < a < 1$



Hence $a \in \left(0, \frac{1}{3}\right)$

9. If α and β are the roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$, then the equation $x^2 - 4qx + 2q^2 - r = 0$ has always $(p, q, r, s \in R)$:

(A) two real roots (B) two positive roots

(C) two negative roots (D) one positive and one negative root

यदि समीकरण $x^2 + px + q = 0$ के मूल α तथा β है तथा समीकरण $x^2 - rx + s = 0$ के मूल α^4, β^4 है तो समीकरण $x^2 - 4qx + 2q^2 - r = 0$ के सदैव $(p, q, r, s \in R)$:

(A) दो वास्तविक मूल होगें (B) दो धनात्मक मूल होगें

(C) दो ऋणात्मक मूल होगें (D) एक धनात्मक तथा एक ऋणात्मक मूल होगा

Ans. A

Sol. $x^2 + px + q = 0 \begin{cases} \alpha \\ \beta \end{cases}$; $\alpha^4 + \beta^4 = \left(\alpha^2 + \beta^2\right)^2 - 2\alpha^2\beta^2$

$\alpha + \beta = -p$, $\alpha\beta = q$; $r = \left[(\alpha + \beta)^2 - 2\alpha\beta\right]^2 - 2\alpha^2\beta^2$

And $x^2 - rs + q = 0$; $r = [p^2 - 2q]^2 - 2q^2$

$\alpha^4 + \beta^4 = r = +ve$
Now the equation : $x^2 - 4qx + 2q^2 - r = 0$
$D = 16q^2 - 4(2q^2 - r)$
$= 16q^2 - 8q^2 + 4r$

$= 8q^2 + 4r = 8q^2 + 4\left[\left(p^2 - 2q\right)^2 - 2q^2\right]$

$= 8q^2 + 4(p^2 - 2q)^2 - 8q^2$
$= 4(p^2 - 2q)^2$
$= +ve$
$\therefore$ Equation have two real roots.

10. Number of integral values of 'a' for which the equation $x^2 - (a + 1)x + a - 1 = 0$, has integral roots, is equal to :

'a' के पूर्णांक मानों की संख्या समीकरण $x^2 - (a + 1)x + a - 1 = 0$ के लिये पूर्णांक मूल रखती है, बराबर है :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. A

Sol. $x^2 - (a + 1)x + (a - 1) = 0$
$D =$ per. sq^r.
$D = P^2$
Let $P = 2$
$(a + 1)^2 - 4(a - 1) = 4$
$a^2 + 1 + 2a - 4a + 4 = 4$
$(a^2 + 1 - 2a) = 0$
$(a - 1)^2 = 0$
$a = 1$
There is only one case.
$\therefore$ Integral value of $a = 1$.

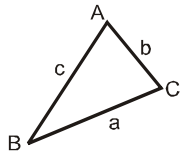
11. Let a, b, c be the sides of a triangle. No two of them are equal and $\lambda \in R$. If the roots of the equation $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real, then :

माना a, b, c त्रिभुज की भुजाऐं है, जिनमें से कोई भी दो समान नही है एवं $\lambda \in R$ है। यदि समीकरण $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ के मूल वास्तविक है तो :

(A) $\lambda < \frac{4}{3}$ (B) $\lambda > \frac{5}{3}$ (C) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$ (D) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$

Ans. A

Sol. We, know that $a + b > c$, $b + c > a$ and $c + a > b$ $\Rightarrow$ $c - a < b$, $a - b < c$, $b - c < a$
squaring on both sides and adding $(c - a)^2 + (a - b)^2 + (b - c)^2 < a^2 + b^2 + c^2$
$a^2 + b^2 + c^2 - 2(ab + bc + ca) < 0 \Rightarrow (a + b + c)^2 - 4(ab + bc + ca) < 0$



$\Rightarrow$ $\frac{(a + b + c)^2}{ab + bc + ca} < 4$ _____(i)

Now roots of equation $x^2 + 2(a + b + c)\, x + 3\lambda\,(ab + bc + ca) = 0$ are real, then $D \geq 0$

$\Rightarrow \quad 4\,(a + b + c)^2 - 4.\,3\lambda\,(ab + bc + ca) \geq 0 \quad \Rightarrow \quad \dfrac{(a+b+c)^2}{ab+bc+ca} \geq 3\lambda$

So $\quad 3\lambda \leq \dfrac{(a+b+c)^2}{ab+bc+ca} < 4 \Rightarrow \quad \lambda < \dfrac{4}{3}$

12. If α, β are the roots of equation $x^2 + x - 2 = 0$, then the value of $\dfrac{\alpha\beta^4\,(\beta+1)^4 + \beta\alpha^4\,(\alpha+1)^4}{\alpha^2+\beta^2+\alpha+\beta}$ is equal to :

यदि α, β समीकरण $x^2 + x - 2 = 0$ के मूल है, तो $\dfrac{\alpha\beta^4\,(\beta+1)^4 + \beta\alpha^4\,(\alpha+1)^4}{\alpha^2+\beta^2+\alpha+\beta}$ का मान है :

(A) 2 (B) -2 (C) 4 (D) -4

Ans. D

Sol. $x^2 + x - 2 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right.$ $\quad \alpha + \beta = -1,\ \alpha\beta = -2,\ \alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 1 + 4 = 5$

$\alpha^2 + \alpha - 2 = 0 \qquad \beta^2 + \beta - 2 = 0$

$\alpha\,(\alpha + 1) = 2 \qquad \beta\,(\beta + 1) = 2$

$\alpha + 1 = \dfrac{2}{\alpha} \qquad \beta + 1 = \dfrac{2}{\beta}$

Now, $\dfrac{\alpha\beta^4(\beta+1)^4 + \beta\alpha^4(\alpha+1)^4}{\alpha^2+\beta^2+\alpha+\beta}$

$= \dfrac{\alpha\beta^4 \cdot \dfrac{2^4}{\beta^4} + \beta\alpha^4 \cdot \dfrac{2^4}{\alpha^4}}{5-1}$

$= \dfrac{16(\alpha+\beta)}{4} = 4 \times (-1) = -4$

13. If α, β, γ are the roots of the cubic $x^3 - 2x + 3 = 0$ then the value of $\dfrac{1}{\alpha^3+\beta^3+6} + \dfrac{1}{\beta^3+\gamma^3+6} + \dfrac{1}{\gamma^3+\alpha^3+6}$ equals :

यदि α, β, γ घनीय समीकरण $x^3 - 2x + 3 = 0$ के मूल है, तो $\dfrac{1}{\alpha^3+\beta^3+6} + \dfrac{1}{\beta^3+\gamma^3+6} + \dfrac{1}{\gamma^3+\alpha^3+6}$ का मान है :

(A) $\dfrac{1}{3}$ (B) $\dfrac{-1}{3}$ (C) $\dfrac{1}{2}$ (D) $\dfrac{-1}{2}$

Ans. B

Sol. $x^3 - 2x + 3 = 0 \begin{cases} \alpha \\ \beta \\ \gamma \end{cases}$

$\alpha + \beta + \gamma = 0$ $\qquad\qquad$ $\alpha\beta + \beta\gamma + \gamma\alpha = -2$

$\alpha + \beta = -\gamma, \quad \beta + \gamma = -\alpha, \quad \gamma + \alpha = -\beta$ $\qquad$ $\alpha\beta\gamma = -3$

$\Rightarrow \alpha^3 - 2\alpha + 3 = 0$

$\alpha^3 + 3 = 2\alpha$

Similarly

$\beta^3 + 3 = 2\beta, \quad \gamma^3 + 3 = 2\gamma$

Now $\qquad \dfrac{1}{\alpha^3+\beta^3+6} + \dfrac{1}{\beta^3+\gamma^3+6} + \dfrac{1}{\gamma^3+\alpha^3+6}$

$$= \frac{1}{(\alpha^3+3)+(\beta^3+3)} + \frac{1}{(\beta^3+3)+(\gamma^3+3)} + \frac{1}{(\gamma^3+3)+(\alpha^3+3)}$$

$$= \frac{1}{(2\alpha+2\beta)} + \frac{1}{(2\beta+2\gamma)} + \frac{1}{(2\gamma+2\alpha)}$$

$$= \frac{1}{2(\alpha+\beta)} + \frac{1}{2(\beta+\gamma)} + \frac{1}{2(\gamma+\alpha)}$$

$$= \frac{1}{-2\gamma} + \frac{1}{-2\alpha} + \frac{1}{-2\beta}$$

$$= -\frac{1}{2}\left[\frac{\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma}\right]$$

$$= -\frac{1}{2} \times \frac{-2}{-3} = -\frac{1}{3}$$

14. If α, β be roots of $x^2 + px + 1 = 0$ and γ, δ are the roots of $x^2 + qx + 1 = 0$ then $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta) =$

(A) $p^2 + q^2$ (B) $p^2 - q^2$ (C) $q^2 - p^2$ (D) None of these

यदि α, β समीकरण $x^2 + px + 1 = 0$ तथा γ, δ समीकरण $x^2 + qx + 1 = 0$ के मूल हों, तो $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta) =$

(A) $p^2 + q^2$ (B) $p^2 - q^2$ (C) $q^2 - p^2$ (D) इनमें से कोई नहीं

Ans. C

Sol. $x^2 + px + 1 = 0 \begin{cases} \alpha \\ \beta \end{cases}$; $x^2 + qx + 1 = 0 \begin{cases} \gamma \\ \delta \end{cases}$

$\alpha + \beta = -p, \ \alpha\beta = 1$; $\gamma + \delta = -q, \ \gamma\delta = 1$

$$(\alpha-\gamma)(\beta-\gamma)(\alpha+\delta)(\beta+\delta)$$

$$=\left(\alpha\beta-\alpha\gamma-\beta\gamma+\gamma^2\right)\cdot\left(\alpha\beta+\alpha\cdot\delta+\beta\delta+\delta^2\right)$$

$$=\left(1-\gamma(\alpha+\beta)+\gamma^2\right)\cdot\left(1+\delta(\alpha+\beta)+\delta^2\right)$$

$$=\left(1+p\gamma+\gamma^2\right)\cdot\left(1-p\delta+\delta^2\right)$$

$$=1-p\delta+\delta^2+p\gamma-p^2\gamma\delta+p\gamma\delta^2+\gamma^2-p\gamma^2\delta+\gamma^2\delta^2$$

$$=1-p\delta+\delta^2+p\gamma-p^2+p\delta+\gamma^2-p\gamma+1$$

$$=2-p^2+\gamma^2+\delta^2$$

$$=2-p^2+(\gamma+\delta)^2-2\gamma\delta$$

$=2 - p^2 + (-q)^2 - 2 \times 1$

$= 2 - p^2 + q^2 - 2 = q^2 - p^2$

15. If a, b, c are real numbers satisfying the condition $a + b + c = 0$ then the roots of the quadratic equation $3ax^2 + 5bx + 7c = 0$ are :

(A) positive (B) negative (C) real and distinct (D) imaginary

यदि a, b, c वास्तविक संख्यायें शर्ते $a + b + c = 0$ को संतुष्ट करती है, तो द्विघात समीकरण $3ax^2 + 5bx + 7c = 0$ के मूल :

(A) धनात्मक (B) ऋणात्मक (C) वास्तविक एवं भिन्न (D) काल्पनिक

Ans. C

Sol. If $a + b + c = 0$

$b = -(a + c)$

$b^2 = (a^2 + c^2 + 2ac)$

And Quadratic Equatio : $3ax^2 + 5bx + 7c = 0$

$D = 25b^2 - 84ac$

$D = 25(a^2 + c^2 + 2ac) - 84ac$

$D = 25a^2 + 25c^2 + 50ac - 84ac$

$D = 25a^2 + 25c^2 - 34ac$

$\because \dfrac{25a^2+25c^2}{2} \geq \left(25a^2 \times 25c^2\right)^{\frac{1}{2}}$

$25a^2 + 25c^2 \geq 50ac$

$25a^2 + 25c^2 - 50ac \geq 0$

$\therefore 25a^2 + 25c^2 - 34ac \geq 0$

$D \geq 0$

$\therefore$ Roots are real and distinct.

16. Let a, b, c be three real numbers such that $a + b + c = 0$ and $a^2 + b^2 + c^2 = 2$. Then the value of $(a^4 + b^4 + c^4)$ is equal to :

तीन वास्तविक संख्याये a, b, c इस प्रकार है कि $a + b + c = 0$ तथा $a^2 + b^2 + c^2 = 2$ है। तो $(a^4 + b^4 + c^4)$ का मान बराबर होगा:

(A) 2 (B) 5 (C) 6 (D) 8

Ans. A

Sol. Given $a + b + c = 0$

and $a^2 + b^2 + c^2 = 2$

$(a^2 + b^2 + c^2)^2 = 4$

$a^4 + b^4 + c^4 + 2[a^2b^2 + b^2c^2 + c^2a^2] = 4$ ...(i)

$(a + b + c)^2 = a^2 + b^2 + c^2 + 2\Sigma ab$

$0 = 2 + 2\Sigma ab$

$\Sigma ab = -1$

$(ab + bc + ca)^2 = a^2b^2 + b^2c^2 + c^2a^2 + 2(ab^2c + abc^2 + a^2bc)$

$(-1)^2 = a^2b^2 + b^2c^2 + c^2a^2 + 2abc(a + b + c)$

$\Rightarrow a^2b^2 + b^2c^2 + c^2a^2 = 1$

$\therefore$ from (i)

$a^4 + b^4 + c^4 + 2 \times 1 = 4$

$a^4 + b^4 + c^4 = 2$

17. The number of solution of the equation $e^{2x} + e^x + e^{-2x} + e^{-x} = 3(e^{-2x} + e^x)$ is :

(A) 0 (B) 2 (C) 1 (D) more than 2

समीकरण $e^{2x} + e^x + e^{-2x} + e^{-x} = 3(e^{-2x} + e^x)$ के हलों की संख्या क्या है :

(A) 0 (B) 2 (C) 1 (D) 2 से अधिक

Ans. C

Sol. $e^{2x} + e^x + e^{-2x} + e^{-x} = 3(e^{-2x} + e^x)$

put $t = e^x$

$$t^2 + t + \frac{1}{t^2} + \frac{1}{t} = 3\left(\frac{1}{t^2} + t\right)$$

$t^4 + t^3 + 1 + t = 3(1 + t^3)$; $t \neq 0$

$t^4 + t^3 - 3t^3 + t + 1 - 3 = 0$

$t^4 - 2t^3 + t - 2 = 0$

$t^3 (t - 2) + 1 (t - 2) = 0$

$(t - 2) (t^3 + 1) = 0$

$(t - 2) (t + 1) (t^2 - t + 1) = 0$

$t = 2, -1$ $\downarrow$

$e^x = 2$ $(D < 0)$

$x = \ln 2$

Only one solution

18. Complete set of values of 'a' such that $y = \frac{x^2 - x}{1 - ax}$ $(x \in R)$ attains all real values is :

यदि $y=\frac{x^2-x}{1-ax}$ $(x \in R)$ का मान सभी वास्तविक संख्याओं को प्राप्त होता a का मान होगा :

(A) $[1, 4]$ (B) $(0, 4]$ (C) $(0, 1]$ (D) $(1, \infty)$

Ans. D

Sol. $y=\frac{x^2-x}{1-ax}$

$y=\frac{x(x-1)}{(1-ax)}, a \neq 1$

$y - axy = x^2 - x$
$x^2 + x - axy - y = 0$
$x^2 + (1 - ay)x - y = 0$
$D \geq 0$
$(1 - ay)^2 + 4y \geq 0$
$1 + a^2y^2 - 2ay + 4y \geq 0$
$a^2y^2 + (4 - 2a)\, y + 1 \geq 0$
$D \leq 0$
$(4 - 2a)^2 - 4a^2 \leq 0$
$16 + 4a^2 - 16a - 4a^2 \leq 0$
$16(a - 1) \geq 0$
$a \geq 1$
$a \in (1, \infty)$ $\because a \neq 1$

19. If the roots of the equation $bx^2 + cx + a = 0$ be imaginary, then for all real values of x, the expression $3b^2x^2 + 6bcx + 2c^2$ is : **[AIEEE-2009]**

(A) Greater than 4 ab (B) Less than 4ab (C) Greater than –4ab (D) Less than – 4ab

यदि समीकरण $bx^2 + cx + a = 0$ के मूल काल्पनिक हो तो x के सभी वास्तविक मानों के लिए व्यंजक $3b^2x^2 + 6bcx + 2c^2$: **[AIEEE-2009]**

(A) 4ab से अधिक है (B) 4ab से कम है (C) –4ab से अधिक है (D) – 4ab से कम है

Ans. C

Sol. $bx^2 + cx + a = 0$
Roots are imaginary $c^2 - 4ab < 0$
$f(x) = 3b^2x^2 + 6bcx + 2c^2$
$D = 36b^2c^2 - 24b^2c^2 = 12b^2c^2$
$\because \quad 3b^2 > 0$

$\therefore \quad f(x) \geq \left(-\frac{D}{4a}\right)$

$f(x) = - c^2$
Now $c^2 - 4ab < 0$
$c^2 < 4ab$

$-c^2 > -4ab$

$\therefore f(x) > -4ab.$

20. Let for $a \neq a_1 \neq 0$, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$. If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is : **[AIEEE-2011]**

माना $a \neq a_1 \neq 0$ के लिए, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ तथा $p(x) = f(x) - g(x)$. यदि केवल $x = -1$ के लिए $p(x) = 0$ हो, तथा $p(-2) = 2$ है, तो $p(2)$ का मान है : **[AIEEE-2011]**

(A) 18 (B) 3 (C) 9 (D) 6

Ans. A

Sol. $P(x) = 0 \quad \Rightarrow \quad f(x) = g(x) \quad \Rightarrow \quad ax^2 + bx + c = a_1x^2 + b_1x + C,$

$\Rightarrow \quad (a - a_1)x^2 + (b - b_1)x + (c - c_1) = 0.$

It has only one solution $x = -1$

$\Rightarrow \quad b - b_1 = a - a_1 + c - c_1$ _____(1)

vertex $(-1, 0) \Rightarrow \quad \frac{b - b_1}{2(a - a_1)} = -1 \quad \Rightarrow \quad b - b_1 = 2(a - a_1)$ _____(2)

$\Rightarrow \quad f(-2) - g(-2) = 2 \quad \Rightarrow \quad 4a - 2b + c - 4a_1 + 2b_1 - c_1 = 2$

$\Rightarrow \quad 4(a - a_1) - 2(b - b_1) + (c - c_1) = 2$ .... (3)

by (1), (2) and (3) $(a - a_1) = (c - c_1) = \frac{1}{2}(b - b_1) = 2$

Now $P(2) = f(2) - g(2) = 4(a - a_1) + 2(b - b_1) + (c - c_1) = 8 + 8 + 2 = 18$

21. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has : **[AIEEE-2012]**

(A) exactly four real roots (B) Infinite number of real roots

(C) No real roots (D) Exactly one real root

समीकरण $e^{\sin x} - e^{-\sin x} - 4 = 0$ के : **[AIEEE-2012]**

(A) मात्र चार वास्तविक मूल (B) अनन्त वास्तविक मूल हैं

(C) कोई वास्तविक मूल नहीं है (D) मात्र एक वास्तविक मूल हैं

Ans. C

Sol. Let $e^{\sin x} = t \quad \Rightarrow \quad t^2 - 4t - 1 = 0 \quad \Rightarrow \quad t = \frac{4 \pm \sqrt{16 + 4}}{2} \quad \Rightarrow \quad t = e^{\sin x} = 2 \pm \sqrt{5}$

$\Rightarrow \quad e^{\sin x} = 2 - \sqrt{5}, e^{\sin x} = 2 + \sqrt{5} \Rightarrow \quad e^{\sin x} = 2 - \sqrt{5} < 0$, (rejected)

$\Rightarrow \quad \sin x = \ln(2 + \sqrt{5}) > 1$ (rejected)

hence no solution

22. If α and β are roots of the equation $px^2 + qx + r = 0$, $p \neq 0$. If p, q , r are in AP and $\frac{1}{\alpha}+\frac{1}{\beta}=4$ then value of $|\alpha-\beta|$ is: [JEE(Main)-2014]

यदि समीकरण $px^2 + qx + r = 0$, $p \neq 0$ के मूल α तथा β है तथा p, q , r स.श्रे. में है, $\frac{1}{\alpha}+\frac{1}{\beta}=4$ तो $|\alpha-\beta|$ का मान होगा : **[JEE(Main)-2014]**

(A) $\frac{\sqrt{34}}{9}$ (B) $\frac{2\sqrt{13}}{9}$ (C) $\frac{\sqrt{61}}{9}$ (D) $\frac{2\sqrt{17}}{9}$

Ans. B

Sol. $px^2 + qx + r = 0 \begin{matrix}\alpha \\ \beta\end{matrix}$; p, q, r $\rightarrow$ A.P. ; $2q = p + r$

$$\frac{1}{\alpha}+\frac{1}{\beta}=4 \quad \Rightarrow \quad \frac{\alpha+\beta}{\alpha\beta}=4 \quad \Rightarrow \quad \frac{-q}{r}=4$$

$q = -4r$ _____(i)

$\therefore$ $-8r = p + r$

$p = -9r$ _____(ii)

$$|\alpha-\beta| = \sqrt{(\alpha+\beta)^2-4\alpha\beta} = \sqrt{\frac{q^2}{p^2}-\frac{4r}{p}}$$ by (i) और and (ii) से

$$= \frac{\sqrt{q^2-4pr}}{|p|} = \frac{\sqrt{16r^2+36r^2}}{|-9r|} = \frac{2\sqrt{13}}{9}$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

23. Which of the following statement(s) is/are True?

(A) The equation $\sqrt{x+1}+\sqrt{x-1}=1$ has no real solution

(B) If $0 < p < \pi$ then the quadratic equation, $(\cos p - 1)x^2 + \cos px + \sin p = 0$ has real roots

(C) If $2a + b + c = 0$ $(c \neq 0)$ then the quadratic equation, $ax^2 + bx + c = 0$ has no root in (0, 2)

(D) If x and y are positive real numbers and m, n are any positive integers then; $\frac{x^n \cdot y^m}{\left(1+x^{2n}\right)\left(1+y^{2m}\right)} > \frac{1}{4}$

निम्नलिखित कथनों में से कौनसा कथन सत्य है?

(A) समीकरण $\sqrt{x+1}+\sqrt{x-1}=1$ का कोई वास्तविक हल नही है

(B) यदि $0 < p < \pi$ है तो द्विघात समीकरण $(\cos p - 1)\, x^2 + \cos px + \sin p = 0$ के वास्तविक मूल है

(C) यदि $2a + b + c = 0$ $(c \neq 0)$ तो द्विघात समीकरण $ax^2 + bx + c = 0$ के अन्तराल (0, 2) में कोई वास्तविक मूल नही है

(D) यदि x तथा y दो धनात्मक संख्याये है तथा m, n दो धनात्मक पूर्णांक है तो; $\frac{x^n \cdot y^m}{\left(1+x^{2n}\right)\left(1+y^{2m}\right)} > \frac{1}{4}$

Ans. AB

Sol. (A) Equation $\sqrt{x+1}+\sqrt{x-1}=1$

$x \geq -1, x \geq 1$

Now $\sqrt{x+1}+\sqrt{x-1}=1$

$x + 1 + x - 1 + 2\sqrt{x^2-1} = 1$

$2\sqrt{x^2-1} = (1-2x)$

$4(x^2 - 1) = 1 + 4x^2 - 4x$

$4x^2 - 4 = 1 + 4x^2 - 4x$

$4x = 5$

$x = \frac{5}{4}$

at $\sqrt{1+\frac{5}{4}}+\sqrt{\frac{5}{4}-1}$

$=\frac{3}{2}+\frac{1}{2}=\frac{4}{2}=2 \neq 1$

$\therefore$ There is no real solution.

(B) if $0 < p < \pi$

$(\cos p - 1)x^2 + \cos px + \sin p = 0$

$D = \cos^2 p - 4\sin p(\cos p - 1)$

$D = \cos^2 p - 4 \sin p \cos p + 4 \sin p$

$D = \cos^2 p + 4 \sin p (1 - \cos p)$

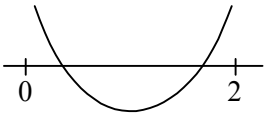
$D = \cos^2 p + 4\sin p \cdot 2\sin^2 \frac{p}{2}$

$D = \cos^2 p + 8\sin p \cdot \sin^2 \frac{p}{2}$

+ve +ve

$\because D \geq 0$

(C) if $2a + b + c = 0, (c \neq 0)$



Quadaratic Equation $ax^2 + bx + c = 0$

Let $f(x) = ax^2 + bx + c$

$f(0) = 0 + 0 + c = c$

$f(2) = 4a + 2b + c$

$= 2 (2a + b) + c$

$= 2(-c) + c$

$= -2c + c$

$= - c$

here f(0), f(2) will be in same sign

(D) $\dfrac{x^n \cdot y^m}{\left(1+x^{2n}\right)\left(1+y^{2m}\right)} > \dfrac{1}{4}$

$4x^n y^m > 1 + x^{2n} + y^{2m} + x^{2n} \cdot y^{2m}$

$x^{2n} + y^{2m} + 1 + x^{2n} \cdot y^{2m} - 4x^n y^m < 0$

$x^{2n} + y^{2m} - 2x^n \cdot y^m + 1 + x^{2n} \cdot y^{2m} - 2x^n y^m < 0$

$(x^n - y^m)^2 + (1 - x^n \cdot y^m)^2 < 0$

+ve +ve ↑

24. Let $(a-1)(x^2+\sqrt{3}x+1)^2 - (a+1)(x^4-x^2+1) \le 0 \ \forall \ x \in R$, then which of the following is/are correct?

(A) $a \in \left[-\dfrac{1}{\sqrt{3}}, \dfrac{4}{\sqrt{3}}\right]$ (B) Largest possible value of a is $\sqrt{3}$

(C) Number of possible integral values of a is 3 (D) Sum of all possible integral values of a is '0'

माना $(a-1)(x^2+\sqrt{3}x+1)^2 - (a+1)(x^4-x^2+1) \le 0 \ \forall \ x \in R$, तो निम्न में से कौन सत्य है?

(A) $a \in \left[-\dfrac{1}{\sqrt{3}}, \dfrac{4}{\sqrt{3}}\right]$ (B) a का अधिकतम सम्भावित मान $\sqrt{3}$ है

(C) a के सम्भावित पूर्णांक मानों की संख्या 3 है (D) a के समी सम्भावित पूर्णांक मानो का योग शून्य है

Ans. CD

Sol. $(a-1)\left(x^2+\sqrt{3}x+1\right)^2 - (a+1)\left(x^4-x^2+1\right) \le 0$

$(a-1)\left(x^2+\sqrt{3}x+1\right)^2 - (a+1)\left(x^2+\sqrt{3}x+1\right)\left(x^2-\sqrt{3}x+1\right) \le 0$

$\because x^4 - x^2 + 1 = \left(x^2+\sqrt{3}x+1\right)\left(x^2-\sqrt{3}x+1\right)$

$\Rightarrow \left(x^2+\sqrt{3}x+1\right)\left[(a-1)\left(x^2+\sqrt{3}x+1\right) - (a+1)\left(x^2-\sqrt{3}x+1\right)\right] \le 0$

↓

$D < 0$, always positive (removed)

$\Rightarrow a\left(x^2+\sqrt{3}x+1\right) - a\left(x^2-\sqrt{3}x+1\right) - \left(x^2+\sqrt{3}x+1\right) - \left(x^2-\sqrt{3}x+1\right) \le 0$

$a\left(x^2+\sqrt{3}x-x^2+\sqrt{3}x-1\right) - \left(x^2+\sqrt{3x}+1+x^2-\sqrt{3}x+1\right) \le 0$

$2\sqrt{3}ax - 2x^2 - 2 \le 0$

$x^2 - \sqrt{3}ax + 1 \ge 0$

$\because$ $a > 0$ then $D \le 0$ (should be)

$\therefore$ $3a^2 - 4 \le 0$

$\left(\sqrt{3}a+2\right)\left(\sqrt{3}a-2\right) \le 0$

$a \in \left[-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right]$

$a = -1, 0, 1$

and sum of integral value $= -1 + 0 + 1 = 0$

C, D options are right.

25. Let $f(x) = x^2 + ax + b$. If the maximum and the minimum values of $f(x)$ are 3 and 2 respectively for $0 \le x \le 2$, then the possible ordered pair(s) of (a, b) is/are :

माना $f(x) = x^2 + ax + b$ है। यदि $f(x)$ का, $0 \le x \le 2$ के लिये, अधिकतम एवं न्यूनतम मान क्रमशः 3 एवं 2 है, तो (a, b) के संभव क्रमित युग्म/युग्मों की संख्या है :

(A) (–2, 3) (B) (– 3/2, 2) (C) (– 5/2, 3) (D) (– 5/2, 2)

Ans. A

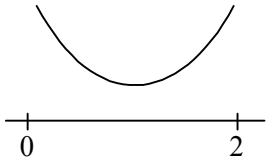
Sol. $f(x) = x^2 + ax + b$

for $x \in [0, 2]$

$f(x) \in [2, 3]$

$f_{max}(0) = b = 3$

$f_{min} = 2$



$-\frac{D}{4a} = 2$

$\frac{4ac - b^2}{4a} = 2$

$\frac{4 \times 1 \times b - a^2}{4} = 2$

$12 - a^2 = 8$

$a^2 = 4$

$a = \pm 2$

$-\frac{b}{2a} = \frac{-a}{2} \Rightarrow$ for $a = 2$

$\frac{2}{2} = -1$ (wrong)

for $a = -2$

$\frac{+2}{2} = 1$ (right)

$a = -2$

$(a, b) = (-2, 3)$

26. If the expression $y = \frac{x - k}{x^2 - 4x + k}$ can take all real values for permissible real values of x, then the possible integral value of k can be :

यदि व्यंजक $y = \frac{x-k}{x^2-4x+k}$ x के अनुमेय (permissible) वास्तविक मानों के लिये सभी वास्तविक मान ले सकता है, तो k का संभव पूर्णांक मान हो सकता है :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. BC

Sol. If the expression $y = \frac{x-k}{x^2-4x+k}$ can take all real values for permissible real values of x

$\Rightarrow y = \frac{x-k}{x^2-4x+k}$

$yx^2 - 4xy + ky - x + k = 0$

$yx^2 - x(4y + 1) + ky + k = 0$

$y \neq 0, \quad D \geq 0$

$(4y + 1)^2 - 4y(ky + k) \geq 0$

$16y^2 + 1 + 8y - 4ky^2 - 4ky \geq 0$

$(16 - 4k)y^2 + (8 - 4k)y + 1 \geq 0$

for this

(i) $a > 0$

$16 - 4k > 0$

$4k < 16$

$k < 4$

(ii) $D < 0$

$(8 - 4k)^2 - 4(16 - 4k) < 0$

$16(2 - k)^2 - 16(4 - k) < 0$

$4 + k^2 - 4k - 4 + k < 0$

$k^2 - 3k < 0$

$k(k - 3) < 0$

$k \in (0, 3)$

(i) $\cap$ (ii)

$k \in (0, 3)$

$k = 1, 2$

B, C options are right.

Exercise C - II

1. Let p and q be the two roots of the equation, $mx^2 + x(2 - m) + 3 = 0$. Let m_1, m_2 be the two values of m satisfying $\frac{p}{q} + \frac{q}{p} = \frac{2}{3}$. Determine the numerical value of $\frac{m_1}{m_2^2} + \frac{m_2}{m_1^2}$.

यदि p तथा q समीकरण $mx^2 + x(2 - m) + 3 = 0$ के दो मूल है। यदि m_1, m_2, m के दो मान है जो $\frac{p}{q} + \frac{q}{p} = \frac{2}{3}$ को संतुष्ट करते है। $\frac{m_1}{m_2^2} + \frac{m_2}{m_1^2}$ का आंशिक मान ज्ञात कीजिए।

Ans. 99

Sol. $mx^2 + x(2-m) + 3 = 0 \left\langle \begin{matrix} p \\ q \end{matrix} \right.$

$p + q = \frac{m-2}{m}, pq = \frac{3}{m}$

$\Rightarrow p^2 + q^2 = (p + q)^2 - 2pq$

$\Rightarrow p^2 + q^2 = \frac{(m-2)^2}{m^2} - \frac{6}{m}$

$= \frac{(m-2)^2 - 6m}{m^2}$

$= \frac{m^2 + 4 - 10m}{m^2}$

$\Rightarrow \frac{p}{q} + \frac{q}{p} = \frac{2}{3}$

$\frac{p^2 + q^2}{pq} = \frac{2}{3}$

$3 \cdot (p^2 + q^2) = 2pq$

$3 \cdot \frac{(m^2 + 4 - 10m)}{m^2} = 2 \cdot \frac{3}{m}$

$m^2 + 4 - 10m = 2m$

$m^2 - 12m + 4 = 0 \left\langle \begin{matrix} m_1 \\ m_2 \end{matrix} \right.$

$m_1 + m_2 = 12, m_1 \cdot m_2 = 4$

$\Rightarrow m_1^3 + m_2^3 = (m_1 + m_2)^3 - 3m_1m_2(m_1 + m_2)$

$= (12)^3 - 3 \times 4 \times 12$

$= 12^3 - 12^2$
$= 12^2 (12 - 1)$
$= 144 \times 11$

$\Rightarrow \frac{m_1}{m_2^2} + \frac{m_2}{m_1^2}$

$= \frac{m_1^3 + m_2^3}{m_1^2 \cdot m_2^2}$

$= \frac{144 \times 11}{4 \times 4}$
$= 99$

2. Solve for x the following equation : $\log_{(2x+3)}(6x^2 + 23x + 21) = 4 - \log_{(3x+7)}(4x^2 + 12x + 9)$

x के मान ज्ञात करने के लिए निम्न समीकरण को हल करें : $\log_{(2x+3)}(6x^2 + 23x + 21) = 4 - \log_{(3x+7)}(4x^2 + 12x + 9)$

Ans. $x = -\frac{1}{4}$

Sol. $\log_{(2x+3)}(6x^2 + 23x + 21) = 4 - \log_{(3x+7)}(4x^2 + 12x + 9)$
$\log_{(2x+3)} (2x + 3) \cdot (3x + 7) = 4 - \log_{(3x+7)} \cdot (2x + 3)^2$
$1 + \log_{(2x+3)}(3x + 7) = 4 - 2\log_{(3x+7)} (2x + 3)$

put $t = \log_{(2x+3)} (3x + 7)$ $\Rightarrow \frac{1}{t} = \log_{(3x+7)} (2x + 3)$

$t + \frac{2}{t} - 3 = 0$
$t^2 - 3t + 2 = 0$
$t^2 - 2t - t + 2 = 0$
$t(t - 2) - 1(t - 2) = 0$
$(t - 1) (t - 2) = 0$
$t = 1, 2$
for $t = 1$
$\log_{(2x+3)} (3x + 7) = 1$
$2x + 3 = 3x + 7$
$x = -4$
for $t = 2$
$\log_{(2x+3)} (3x + 7) = 2$
$(2x + 3)^2 = 3x + 7$
$4x^2 + 9 + 12x - 3x - 7 = 0$
$4x^2 + 9x + 2 = 0$
$4x^2 + 8x + x + 2 = 0$
$4x(x + 2) + 1 (x + 2) = 0$
$(x + 2) (4x + 1) = 0$

$x = -2, -\frac{1}{4}$

at $x = -4$

$2x + 3 = -5 < 0$ (wrong)

$3x + 7 = -5 < 0$ (wrong)

at $x = -2$

$2x + 3 = -1 < 0$ (wrong)

$3x + 7 = 1 < 0$

at $x = -\frac{1}{4}$

$2x + 3 = \frac{2}{-4} + 3 = \frac{5}{2} > 0$

$3x + 7 = 7 - \frac{3}{2} = \frac{11}{2} > 0$

$\therefore x = -\frac{1}{4}$

3. Find the values of 'a' for which $-3 < [(x^2+ax-2)/(x^2+x+1)] < 2$ is valid for all real x.

'a' का मान ज्ञात कीजिए, यदि असमिका $-3 < [(x^2+ax-2)/(x^2+x+1)] < 2$, x के वास्तविक मानों के लिए सत्य है।

Ans. $-2 < a < 1$

Sol. $-3 < \frac{x^2+ax-2}{x^2+x+1} < 2$

when		and when
$\frac{x^2+ax-2}{x^2+x+1} > -3$		$\frac{x^2+ax-2}{x^2+x+1} < 2$
$4x^2 + x\cdot(a+3) + 1 > 0$		$x^2 + (2-a)x + 4 > 0$
$D < 0$		$D < 0$
$(a+3)^2 - 16 < 0$		$(a-2)^2 - 16 < 0$
$(a+3+4)\cdot(a+3-4) < 0$		$(a-2+4)(a-2-4) < 0$
$(a+7)(a-1) < 0$		$(a+2)(a-6) < 0$
$a \in (-7, 1)$	$\cap$	$a \in (-2, 6)$
$\Rightarrow a \in (-2, 1)$	or	$-2 < a < 1$

4. Find the minimum value of $\frac{\left(x+\frac{1}{x}\right)^6 - \left(x^6+\frac{1}{x^6}\right) - 2}{\left(x+\frac{1}{x}\right)^3 + x^3 + \frac{1}{x^3}}$ for $x > 0$.

धनात्मक x के लिए $\frac{\left(x+\frac{1}{x}\right)^6-\left(x^6+\frac{1}{x^6}\right)-2}{\left(x+\frac{1}{x}\right)^3+x^3+\frac{1}{x^3}}$ का न्यूनतम मान बताइये।

Ans. $Y_{Min}=6$

Sol. $\frac{\left(x+\frac{1}{x}\right)^6-\left(x^6+\frac{1}{x^6}\right)-2}{\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)}$

$=\frac{\left(x+\frac{1}{x}\right)^6-\left\{\left(x^3\right)^2+\left(\frac{1}{x^3}\right)^2\right\}-2}{\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)}$

$=\frac{\left(x+\frac{1}{x}\right)^6-\left\{\left(x^3+\frac{1}{x^3}\right)^2-2\right\}-2}{\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)}$

$=\frac{\left(\left(x+\frac{1}{x}\right)^3\right)^2-\left(x^3+\frac{1}{x^3}\right)^2+2-2}{\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)}$

$=\frac{\left[\left(x+\frac{1}{x}\right)^3+x^3+\frac{1}{x^3}\right]\cdot\left[\left(x+\frac{1}{x}\right)^3-\left(x^3+\frac{1}{x^3}\right)\right]}{\left[\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)\right]}$

$=x^3+\frac{1}{x^3}+3\cdot\left(x+\frac{1}{x}\right)-x^3-\frac{1}{x^3}$

$=3\left(x+\frac{1}{x}\right)$

$\because x+\frac{1}{x}\geq 2$

$\Rightarrow$ Miimum value = 3×2
$= 6$

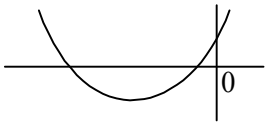
5. Find all numbers p for each of which the least value of the quadratic trinomial $4x^2 - 4px + p^2 - 2p + 2$ on the interval $0 \le x \le 2$ is equal to 3.

'p' के किन मानों के लिए द्विघात $4x^2 - 4px + p^2 - 2p + 2$ का अन्तराल $0 \le x \le 2$ में न्यूनतम मान 3 है।

Ans. $p = 1 - \sqrt{2}$ or $5 + \sqrt{10}$

Sol. $4x^2 - 4px + p^2 - 2p + 2 = 0$

Case–I :



$$-\frac{b}{2a} < 0$$

$$\frac{4p}{8} < 0$$

$p < 0$

$f(0) = 3$

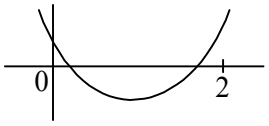
$p^2 - 2p + 2 - 3 = 0$

$p^2 - 2p - 1 = 0$

$$p = 1 + \sqrt{2}, 1 - \sqrt{2}$$

$$\Rightarrow p = 1 - \sqrt{2}$$

Case–II :

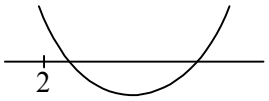


$$0 < -\frac{b}{2a} < 2$$

$$0 < \frac{p}{2} < 2$$

$0 < p < 4$

Case–III :



$$-\frac{b}{2a} > 2$$

$$\frac{p}{2} > 2$$

$f(2) = 3$

$16 - 8p + p2 - 2p + 2 = 3$

$p^2 - 10p + 15 = 0$

$$p = 5 + \sqrt{10}, 5 - \sqrt{10}$$

$$\therefore p = 1 - \sqrt{2}, 5 + \sqrt{10}$$

6. Find the values of 'b' for which the equation $2\log_{\frac{1}{25}}(bx+28) = -\log_5(12-4x-x^2)$ has only one solution.

'b' के किन मानों के लिए समीकरण $2\log_{\frac{1}{25}}(bx+28) = -\log_5(12-4x-x^2)$ का केवल एक हल होगा।

Ans. $(-\infty,-14)\cup\{4\}\cup\left[\frac{14}{3},\infty\right)$

Sol. $2\log_{\frac{1}{25}}(bx+28) = -\log_5(12-4x-x^2)$

$-\log_5(bx+28) = -\log_5(12-4x-x^2)$

$\Rightarrow x^2 + (b+4)x + 16 = 0$

for only one solution

$D = 0$

$(b+4)^2 = 64$

$b = 4, -12$

at $b = 4$

$x^2 + 8x + 16 = 0$

$x = -4$

$\Rightarrow$ at $b = 4$, $x = -4$

$bx + 28 = 4\cdot(-4) + 28 = 12 > 0$; $12 - 4x - x^2 = 12 + 16 - 16 = 12 > 0$

at $b = -12$

$x^2 - 8x + 16 = 0$

$x = 4$

at $b = -12$, $x = 4$

$bx + 28 = -48 + 28 = -20 < 0$ (rejected)

$\Rightarrow bx + 28 > 0$ And $12 - 4x - x^2 > 0$

$x > \frac{-28}{b}$ $x^2 + 4x - 12 < 0$

$x > \frac{-28}{4}$ $(x+6)(x-2) < 0$

$x > -7$ $x \in (-6, 2)$

Itersection value

$x \in (-6, 2) \leftarrow$ domain of log

$f(-6)\cdot f(2) < 0$

$(28 - 6b)\cdot(28 + 2b) < 0$

$b \in (-\infty,-14)\cup\left(\frac{14}{3},\infty\right)$

at $b = -14$

$x = 8, 2$ (not lies in domain)

at $b = \frac{14}{3}$ (x lies in domain)

$$\Rightarrow b \in (-\infty, -14) \cup \{4\} \cup \left[\frac{14}{3}, \infty\right)$$

7. Solve the system of equations :

$\log_a x \ \log_a(xyz) = 48,\ \log_a y \ \log_a(xyz) = 12,\ \log_a z \ \log_a(xyz) = 84$ where $a > 0, a \neq 1$

समीकरणों के निकाय को हल करें :

$\log_a x \ \log_a(xyz) = 48,\ \log_a y \ \log_a(xyz) = 12,\ \log_a z \ \log_a(xyz) = 84$ जहाँ $a > 0, a \neq 1$

Ans. (a^4, a, a^7) or $\left(\frac{1}{a^4}, \frac{1}{a}, \frac{1}{a^7}\right)$

Sol. $\log_a x \cdot \log_a(xyz) = 48$ ...(1)

$\log_a y \cdot \log_a(xyz) = 12$ ...(2)

$\log_a z \cdot \log_a(xyz) = 84$ ...(3)

By (3) ÷ (1)

$$\log_x z = \frac{7}{4}$$

$$z = x^{\frac{7}{4}}$$

$$x = z^{\frac{4}{7}}$$

By (3) ÷ (2)

$$\log_y z = \frac{84}{12}$$

$$z = y^7$$

$$y = z^{\frac{1}{7}}$$

from equation (1)

$\log_a x \cdot \log_a(xyz) = 48$

$$\log_a z^{\frac{4}{7}} \cdot \log_a\left(z^{\frac{4}{7}} \cdot z^{\frac{1}{7}} \cdot z\right) = 48$$

$$\frac{4}{7}\cdot\frac{12}{7}\cdot(\log_a z)^2 = 48$$

$(\log_a z)^2 = (7)^2$

$\log_a z = \pm 7$

$z = a^7, a^{-7}$

$$\Rightarrow x = \left(a^{\frac{4}{7}}\right)^7 = a^4 \text{ or } a^{-4}$$

$y = z^{\frac{1}{7}} = \left(a^7\right)^{\frac{1}{7}} = a \text{ or } a^{-1}$

$z = a^7 \text{ or } a^{-7}$

(a^4, a, a^7) or (a^{-4}, a^{-1}, a^{-7})

8. Let $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$. Find $y \in N$.

माना कि $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$ यदि $y \in N$ तो y का मान ज्ञात कीजिए।

Ans. $y = 6$

Sol. Let $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$

$$y = \sqrt{\log_2 3 \cdot \left(\log_2 2^2 \cdot 3\right) \cdot \left(\log_2 2^4 \cdot 3\right) \cdot \left(\log_2 2^6 \cdot 3\right) + 16} - (\log_2 2^2 \cdot 3) \cdot (\log_2 2^4 \cdot 3) + 10$$

$$y = \sqrt{\log_2 3 \cdot (2 + \log_2 3) \cdot (4 + \log_2 3) \cdot (6 + \log_2 3) + 16} - (2 + \log_2 3)(4 + \log_2 3) + 10$$

put $\log_2 3 = x$

$$y = \sqrt{x(x+2)(x+4)(x+6) + 16} - (x+2)(x+4) + 10$$

$$y = \sqrt{\left(x^2 + 6x\right)\left(x^2 + 6x + 8\right) + 16} - \left(x^2 + 6x + 8\right) + 10$$

put $t = x^2 + 6x$

$$y = \sqrt{t(t+8) + 16} - (t+8) + 10$$

$y = t + 4 - t - 8 + 10$

$y = 6$

9. Find 'a' for which the equation $x^2 + (a - b)x + (1 - a - b) = 0$ has two distinct and unequal roots for $\forall b \in R$?

'a' के वह मान ज्ञात कीजिए जिसके लिए समीकरण $x^2 + (a - b)x + (1 - a - b) = 0$ के दो भिन्न–भिन्न तथा असमान मूल हों, जहाँ $\forall b \in R$?

Ans. $a > 1$

Sol. $x^2 + (a - b)x + (1 - a - b) = 0$

$D > 0$

$(a - b)^2 - 4(1 - a - b) > 0$

$a^2 + b^2 - 2ab - 4 + 4a + 4b > 0$

$b^2 + b \cdot (4 - 2a) + (a^2 + 4a - 4) > 0$

$D < 0$ for $\forall b \in R$

$(4 - 2a)^2 - 4 \cdot 1 \cdot (a^2 + 4a - 4) < 0$

$16 + 4a^2 - 16a - 4a^2 - 16a + 16 < 0$

$32 - 32a < 0$

$a > 1$

10. Let a, b, c be real numbers, $a \neq 0$. If α is root of $a^2x^2 + bx + c = 0$. β is the root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$, then show that the equation $a^2x^2 + 2bx + 2c = 0$ has a root γ that always satisfies $\alpha < \gamma < \beta$.

माना a, b, c वास्तविक संख्याऐं है, जहाँ $a \neq 0$ यदि α समीकरण $a^2x^2 + bx + c = 0$ का एक मूल है तथा β समीकरण $a^2x^2 - bx - c = 0$ का एक मूल जहाँ $0 < \alpha < \beta$, तब प्रदर्शित कीजिए कि समीकरण $a^2x^2 + 2bx + 2c = 0$ का मूल γ है जो संबंध $\alpha < \gamma < \beta$ को हमेशा संतुष्ट करता है।

Ans.

Sol. $\because$ α is root of $a^2x^2 + bx + c = 0$

$a^2\alpha^2 + b\alpha + c = 0$

$b\alpha + c = -a^2\alpha^2$ ...(1)

β is root of $a^2x^2 - bx - c = 0$

$a^2\beta^2 - b\beta - c = 0$

$a^2\beta^2 = b\beta + c$ ...(2)

Let $f(x) = a^2x^2 + 2bx + 2c$

$f(\alpha) = a^2\alpha^2 + 2(b\alpha + c)$

$= a^2\alpha^2 - 2a^2\alpha^2$

$= -a^2\alpha^2 < 0$

$f(\beta) = a^2\beta^2 + 2b\beta + 2c$

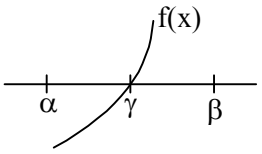
$= a^2\beta^2 + 2(b\beta + c)$

$= 3\alpha^2\beta^2 > 0$

and given that

γ is root of $a^2x^2 + 2bx + 2c = 0$

$\therefore$ $\alpha < \gamma < \beta$



11. If α, β are the roots of the equation, $(\tan^2 135°)x^2 - (\text{cosec}10° - \sqrt{3}\sec 10°)x + \tan^2 240° = 0$ then find the quadratic equation whose roots are $(2\alpha + \beta)$ and $(\alpha + 2\beta)$.

यदि α, β समीकरण $(\tan^2 135°)x^2 - (\text{cosec}10° - \sqrt{3}\sec 10°)x + \tan^2 240° = 0$ के मूल है तो वह द्विघात समीकरण ज्ञात कीजिए जिसके मूल $(2\alpha + \beta)$ तथा $(\alpha + 2\beta)$ है।

Ans. $x^2 - 12x + 35 = 0$

Sol. $\left(\tan^2 135^\circ\right)x^2 - \left(\text{cosec}10^\circ - \sqrt{3}\sec 10^\circ\right)x + \tan^2 240^\circ = 0 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right.$

$$\alpha + \beta = \frac{\text{cosec}10^\circ - \sqrt{3}\sec 10^\circ}{\tan^2 135^\circ} = 4$$

a new quadratic equation whose roots are : $(2\alpha + \beta)$ and $(\alpha + 2\beta)$

$= (\alpha + \beta + \alpha)$ ad $(\alpha + \beta + \beta)$

$= (4 + \alpha)$ and $(4 + \beta)$

quad. in y : $y = 4 + \alpha$

$y = 4 + x$

put $x = y - 4$

$\left(\tan^2 135^\circ\right)\cdot(y-4)^2 - \left(\operatorname{cosec} 10^\circ - \sqrt{3}\sec 10^\circ\right)\cdot(y-4) + \tan^2 240^\circ = 0$

$(y-4)^2 - 4(y-4) + 3 = 0$

$y^2 + 16 - 8y - 4y + 16 + 3 = 0$

$y^2 - 12y + 35 = 0$

12. Let $x = \frac{1}{3}$ or $x = -15$ satisfies the equation, $\log_8(kx^2 + wx + f) = 2$. If k, w and f are relatively prime positive integers then find the value of $k + w + f$.

माना $x = \frac{1}{3}$ या $x = -15$ समीकरण $\log_8(kx^2 + wx + f) = 2$ को संतुष्ट करते है। यदि k, w तथा f सापेक्षतः अभाज्य धनात्मक संख्याऐं हो तो $k + w + f$ का मान ज्ञात कीजिए।

Ans. 96

Sol. Let $x = \frac{1}{3}$ or $x = -15$ satisfies the equation

$\log_8(kx^2 + wx + f) = 2$

$kx^2 + wx + f = 64$

$$kx^2 + wx + f - 64 = 0 \left\langle \begin{matrix} \frac{1}{3} \\ -15 \end{matrix} \right.$$

$$\text{sum of roots} \Rightarrow \frac{1}{3} - 15 = -\frac{w}{k}$$

$$-\frac{44}{3} = -\frac{w}{k}$$

$3w = 44k$

$$w = \frac{44k}{3}$$

$$\text{Product of roots} \Rightarrow \frac{1}{3}\times(-15) = \frac{f-64}{k}$$

$-5k = f - 64$

$f = 64 - 5k$

for +ve f; $k = 12$

and given that k, w and f are relatively prime positive integers (whose HCF = 1)

for $w = \frac{44k}{3}$

k can be; 3,6,9,12

for $k = 3$, $w = 44\times\frac{3}{3} = 44$, $f = 64 - 15 = 9$ (Common factor = 1)

for k = 6, $w = 44 \times \frac{6}{3} = 88$, (Common factor = 1,2)

for k = 9, $w = 44 \times \frac{9}{3} = 132$ (Common factor = 1,3)

for k = 12, $w = 44 \times \frac{12}{3} = 176$ (Common factor = 1,4)

$\therefore$ k = 3, w = 44, f = 49

$\Rightarrow$ k + w + f = 3 + 44 + 49 = 96

13. If roots of the equation $x^2 - 10cx - 11d = 0$ are a, b and those of $x^2 - 10ax - 11b = 0$ are c, d, then find the value of a + b + c + d. (a, b, c and d are distinct numbers)

यदि समीकरण $x^2 - 10cx - 11d = 0$ के मूल a, b है तथा $x^2 - 10ax - 11b = 0$ के मूल c, d है तो a + b + c + d का मान बताइये। (a, b, c व d परस्पर भिन्न संख्याऐं है)

Ans. 1210

Sol. $x^2 - 10cx - 11d = 0 \begin{cases} a \\ b \end{cases}$ ...(1) $x^2 - 10ax - 11b = 0 \begin{cases} c \\ d \end{cases}$ ...(2)

a + b = 10c c + d = 10a

$\Rightarrow$ a + b – c – d = 10c – 10a

b – d = 10c – 10a + c – a

(b – d) = 11(c – a)

$\Rightarrow$ a + b + c + d = 10 (c + a)

from equation (1)

$a^2 - 10ca - 11d = 0$ ...(3)

from equation (2)

$c^2 - 10ca - 11b = 0$ ...(4)

from (3) – (4)

$a^2 - c^2 + 11 (b - d) = 0$

(a – c) (a + c) = –11(b – d)

–(c – a) (c + a) = $-11 \cdot 11(c - a)$

c + a = 121

$\Rightarrow$ a + b + c + d = 10 × 121 = 1210

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 14 to 16

Consider the cubic equation

$x^3 - (1 + \cos\theta + \sin\theta)x^2 + (\cos\theta \sin\theta + \cos\theta + \sin\theta)x - \sin\theta \cdot \cos\theta = 0$

whose roots are x_1, x_2 and x_3.

तीन घातीय समीकरण

$x^3 - (1 + \cos\theta + \sin\theta)x^2 + (\cos\theta \sin\theta + \cos\theta + \sin\theta)x - \sin\theta \cdot \cos\theta = 0$

के मूल x_1, x_2 तथा x_3 है।

14. The value of $x_1^2 + x_2^2 + x_3^2$ equals :

$x_1^2 + x_2^2 + x_3^2$ का मान बराबर है :

(A) $2\cos\theta$ (B) $\sin\theta(\sin\theta + \cos\theta)$ (C) 1 (D) 2

Ans. D

Sol. $S_1 = x_1 + x_2 + x_3 = 1 + \cos\theta + \sin\theta$
$S_2 = \Sigma x_1 \cdot x_2 = (\cos\theta \sin\theta + \cos\theta + \sin\theta)$
$S_3 = x_1 \cdot x_2 \cdot x_3 = \sin\theta \cdot \cos\theta$

$x_1^2 + x_2^2 + x_3^2 = (x_1 + x_2 + x_3)^2 - 2 \cdot \sum x_1 \cdot x_2$

$= (1 + \cos\theta + \sin\theta)^2 - 2 \cdot (\cos\theta \sin\theta + \cos\theta + \sin\theta)$
$= 1 + \cos^2\theta + \sin^2\theta + 2\cos\theta + 2\cos\theta \sin\theta + 2\sin\theta - 2\cos\theta \sin\theta - 2\cos\theta - 2\sin\theta$
$= 1 + 1 = 2$

15. Number of values of θ in $[0, 2\pi]$ for which at least two roots are equal :

θ के अन्तराल $[0, 2\pi]$ में समीकरण के दो मूलों में अधिकतम संभावित अंतर है :

(A) 6 (B) 5 (C) 4 (D) 3

Ans. B

Sol. $x^3 - x^2 - x^2(\cos\theta + \sin\theta) + x(\cos\theta + \sin\theta) + x\cos\theta \sin\theta - \sin\theta \cos\theta = 0$
$x^2 \cdot (x - 1) - x \cdot (\cos\theta + \sin\theta)(x - 1) + (x - 1)\sin\theta \cdot \cos\theta = 0$

$(x-1)\left[x^2 - x(\cos\theta + si\theta) + \sin\theta \cdot \cos\theta\right] = 0$

$(x - 1)(x - \cos\theta)(x - \sin\theta) = 0$
$x = 1$ or $x = \cos\theta$ or $x = \sin\theta$
At least two roots are equal
$\sin\theta = 1$ or $\cos\theta = 1$ or $\sin\theta = \cos\theta$
in $\theta \in [0, 2\pi]$

$\theta = \frac{\pi}{2}$ $\theta = 2\pi, 0$ $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$

and $\sin\theta = \cos\theta = 1$ (No value of θ)
total values of $\theta = 5$

16. Greatest possible difference between two of the roots if $\theta \in [0, 2\pi]$ is :

यदि θ के अन्तराल $[0, 2\pi]$ में समीकरण के दो मूलों में अधिकतम संभावित अन्तर है :

(A) 2 (B) 1 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

Ans. A

Sol. difference of two roots :

$d_1 = |\sin\theta - \cos\theta| = \sqrt{2}$

$d_2 = |1 - \sin\theta| = 2$
$d_3 = |1 - \cos\theta| = 2$ $\Big]_{max.}$

MATCH THE COLUMN TYPE QUESTION(S)

17. Match the following columns :

Column-I	**Column-II**
(A) If $\alpha, \beta \in (0, \pi)$ and $\alpha \neq \beta$ satisfy the equation $\frac{1-\cos 2\theta}{\sin\theta} = \frac{1}{2}$, then the value of $\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right)$ is equal to	(P) 0
(B) If the expression $\frac{x^2 + (2m+3)x + (m^2+3)}{\sqrt{x^2 + (2m+1)x + m^2 + 2}}$ is non-negative $\forall\ x \in R$, then the possible values of m can be equal to	(Q) 8
(C) If the parabola $y = 5x^2 + x - 3$ lies above the parabola $y = 2x^2 + 6x - 1$, then integral values of x can be equal to	(R) 1
(D) The number of real solutions of the equation $x^{2\log_x(x+3)} = 16$ is equal to	(S) -1
	(T) 2

निम्न Columns को सुमेलित कीजिए :

Column-I	**Column-II**
(A) यदि $\alpha, \beta \in (0, \pi)$ तथा $\alpha \neq \beta$ समीकरण $\frac{1-\cos 2\theta}{\sin\theta} = \frac{1}{2}$ को संतुष्ट करता है तथा $\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right)$ का मान बराबर है	(P) 0
(B) यदि व्यंजक $\frac{x^2 + (2m+3)x + (m^2+3)}{\sqrt{x^2 + (2m+1)x + m^2 + 2}}$ $\forall\ x \in R$ ऋणात्मक नहीं है, तब m का संभव मान है	(Q) 8
(C) यदि परवलय $y = 5x^2 + x - 3$ परवलय $y = 2x^2 + 6x - 1$ के ऊपर स्थित है, तो x का पूर्णांक मान है	(R) 1
(D) समीकरण $x^{2\log_x(x+3)} = 16$ के वास्तविक हलों की संख्या बराबर है	(S) -1
	(T) 2

(A) A-P,S; B-Q; C-Q,S; D-P

(B) A-P; B-Q,S; C-Q; D-P,S

(C) A-Q; B-P,S; C-Q,S; D-P

(D) A-P; B-Q,S; C-P,S; D-Q

Ans. (C)

Sol. (A) If $\alpha,\beta \in (0,\pi)$ and $\alpha \neq \beta$

$$\frac{1-\cos 2\theta}{\sin\theta}=\frac{1}{2}$$

$$\frac{2\sin^2\theta}{\sin\theta}=\frac{1}{2}$$

$$\sin\theta=\frac{1}{4}$$

$$\frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}=\frac{1}{4}$$

$$\tan^2\frac{\theta}{2}-8\tan\frac{\theta}{2}+1=0\left\langle\begin{matrix}\alpha\\ \beta\end{matrix}\right.$$

$$\tan\left(\frac{\alpha}{2}\right)+\tan\left(\frac{\beta}{2}\right)=8$$

$A \rightarrow Q$

(B) $$\frac{x^2+(2m+3)x+(m^2+3)}{\sqrt{x^2+(2m+1)x+m^2+2}}\geq 0$$

$x^2 + (2m + 3)x + (m^2 + 3) \geq 0$

$D \leq 0$

$(2m + 3)^2 - 4(m^2 + 3) \leq 0$

$4m^2 + 9 + 12m - 4m^2 - 12 \leq 0$

$12m - 3 \leq 0$

$$m \leq \frac{1}{4}$$

$x^2 + (2m + 1)x + m^2 + 2 > 0$

$D < 0$

$(2m + 1)^2 - 4(m^2 + 2) < 0$

$4m^2 + 1 + 4m - 4m^2 - 8 < 0$

$4m < 7$

$$m < \frac{7}{4}$$

Possible value of $m \in \left(-\infty, \frac{1}{4}\right]$

(C) $5x^2 + x - 3 > 2x^2 + 6x - 1$

$3x^2 - 5x - 2 > 0$

$3x^2 - 6x + x - 2 > 0$

$3x(x - 2) + (x - 2) > 0$

$(3x + 1)(x - 2) > 0$

$x \in \left(-\infty, -\frac{1}{3}\right) \cup (2, \infty)$

(D) $x^{2\log_x(x+3)} = 16$

$(x + 3)^2 = 16; \qquad x > 0, x \neq 1$

$x^2 + 6x - 7 = 0$

$x = 1, -7$

$\therefore$ Number of real solution = 0

$D \rightarrow P$

18. Match the following columns:

Column-I	**Column-II**
(A) The expression, $x^{\ln y - \ln z} \cdot y^{\ln z - \ln x} \cdot z^{\ln x - \ln y}$ when simplified reduces to	(P) $\frac{1}{8}$
(B) Let $p = \log_5 \log_5(3)$. If $3^{C+5^{-p}} = 45$, the value of C equals	(Q) $\frac{1}{4}$
(C) The real value of x for which the statement $\log_6 9 - \log_9 27 + \log_8 x = \log_{64} x - \log_6 4$ holds true, is	(R) $\frac{1}{2}$
(D) $\frac{1}{2}\log_{\sqrt{5}} \sin\left(\frac{\pi}{5}\right) . \log_{\sqrt{\sin\frac{\pi}{5}}} 5$ simplifies to	(S) 1
	(T) 2

निम्न Columns को सुमेलित कीजिए :

Column-I	**Column-II**
(A) व्यंजक $x^{\ln y - \ln z} \cdot y^{\ln z - \ln x} \cdot z^{\ln x - \ln y}$ का मान होगा	(P) $\frac{1}{8}$
(B) यदि $p = \log_5 \log_5(3)$ तथा $3^{C+5^{-p}} = 45$ तो C का मान होगा	(Q) $\frac{1}{4}$
(C) x के किस मान के लिए समीकरण $\log_6 9 - \log_9 27 + \log_8 x = \log_{64} x - \log_6 4$ संतुष्ट होता है	(R) $\frac{1}{2}$

(D) $\frac{1}{2}\log_{\sqrt{5}}\sin\left(\frac{\pi}{5}\right).\log_{\sqrt{\sin\frac{\pi}{5}}}5$ का मान होगा (S) 1

(T) 2

(A) A-S; B-T; C-P; D-T

(B) A-T; B-S; C-P; D-T

(C) A-P; B-T; C-T; D-S

(D) A-T; B-P; C-T; D-S

Ans. (A)

Sol. (A) Let $P = x^{\ln y-\ln z}\cdot y^{\ln z-\ln x}\cdot z^{\ln x-\ln y}$

$\log P = (\log y-\log z)\cdot\log x+(\log z-\log x)\cdot\log y+(\log x-\log y)\cdot\log z$

$\log P = \log x\cdot\log y-\log z\cdot\log x+\log y\cdot\log z-\log x\cdot\log y+\log z\cdot\log x-\log y\cdot\log z$

$\log P = 0$

$P = 1$

(B) $P = \log_5\log_5(3)$

And $3^{c+5^{-P}} = 45$

$(c+5^{-P})\log 3 = \log 45$

$c+5^{-P} = \log_3 45$

$c = \log_3 45-5^{-P}$

$= \log_3 45-5^{-\log_5\log_5(3)}$

$= \log_3 45-\left(\log_5 3\right)^{-1}$

$= \log_3\frac{45}{5} = \log_3 9 = 2$

(C) $\log_6 9-\log_9 27+\log_8 x = \log_{64} x-\log_6 4$

$\log_6 36-\frac{3}{2}+\log_8 x-\frac{1}{2}\log_8 x = 0$

$2-\frac{3}{2}+\frac{1}{2}\log_8 x = 0$

$\frac{1}{2}\log_8 x = -\frac{1}{2}$

$x = \frac{1}{8}$

(D) $\frac{1}{2}\log_{\sqrt{5}}\sin\left(\frac{\pi}{5}\right)\cdot\log_{\sqrt{\sin\frac{\pi}{5}}}5$

$= \frac{1}{2}\cdot 2\cdot\log_5\sin\frac{\pi}{5}\cdot 2\log_{\sin\frac{\pi}{5}}5$

$= 2\times 1 = 2$

19. Match the following columns :

	Column-I	Column-II
A.	If equation $x^2 + px + q = 0$ $(p, q \in R)$ and $x^3 + 3x^2 + 5x + 3 = 0$ has two common roots then $(p + q)$	P. 1
B.	If $f(x) = 4x^2 + 3x - 7$ and α is common roots of equation $x^2 - 3x + 2 = 0$ and $x^2 + 2x - 3 = 0$ then value of $f(\alpha)$ =	Q. 2
C.	Number of real roots of equation $\log_3 (x^2 + 9) + \log_3 (x^4 + 81) = 6$	R. 0
D.	Number of real roots of equation $\log_x(5x - 2) = 3$	S. 5

(A) A-R; B-S; C-Q; D-P
(B) A-S; B-R; C-P; D-Q
(C) A-Q; B-P; C-S; D-R
(D) A-P; B-Q; C-S; D-R

निम्न Columns को सुमेलित कीजिए :

	Column-I	Column-II
A.	यदि समीकरण $x^2 + px + q = 0$ $(p, q \in R)$ तथा $x^3 + 3x^2 + 5x + 3 = 0$ के दो मूल उभयनिष्ठ है, तो $(p + q)$ का मान होगा।	P. 1
B.	यदि $f(x) = 4x^2 + 3x - 7$ तथा α समीकरण $x^2 - 3x + 2 = 0$ तथा $x^2 + 2x - 3 = 0$ का उभयनिष्ठ मूल हो, तो $f(\alpha)$ बराबर है	Q. 2
C.	समीकरण $\log_3 (x^2 + 9) + \log_3 (x^4 + 81) = 6$ के वास्तविक हलों की संख्या है	R. 0
D.	समीकरण $\log_x(5x - 2) = 3$ के वास्तविक हलों की संख्या है	S. 5

(A) A-R; B-S; C-Q; D-P
(B) A-S; B-R; C-P; D-Q
(C) A-Q; B-P; C-S; D-R
(D) A-P; B-Q; C-S; D-R

Ans. B

Sol. (A) $x^2 + px + q = 0$, $(p, q \in R)$
$x^3 + 3x^2 + 5x + 3 = 0$
$(x + 1)(x^2 + 2x + 3) = 0$
$D < 0$
$\therefore$ roots of $x2 + 2x + 3 = 0$ will be imaginary and occurs in conjugate pair
$\therefore$ Equation $x^2 + px + q = 0$
and $x^2 + 2x + 3 = 0$ have both common root.
$\therefore$ $p = 2, q = 3$
$p + q = 2 + 3 = 5$
$A \rightarrow S$
(B) if $f(x) = 4x^2 + 3x - 7$
And α is common root of equation.
$x^2 - 3x + 2 = 0$, $x^2 + 2x - 3 = 0$
$(x - 2)(x - 1) = 0$ $(x + 3)(x - 1) = 0$
$x = 2, 1$ $x = -3, 1$
$\Rightarrow \alpha = 1$

$\therefore$ $f(\alpha) = f(1) = 4 + 3 - 7 = 0$

(C) $\log_3 (x^2 + 9) + \log_3 (x^4 + 81) = 6$

$\log_3 (x^2 + 9)(x^4 + 81) = 6$

$(x^2 + 9)\cdot(x^4 + 81) = 3^6$

$x^6 + 81x^2 + 9x^4 + 729 = 729$

$x^2\cdot[x^4 + 9x^2 + 81] = 0$

$x = 0$

$x^4 + 9x^2 + 81 = 0$

put $x^2 = t$

$t^2 + 9t + 81 = 0$

$\because$ $D < 0$

$\therefore$ roots are not real.

$\therefore$ Only one root is real.

$C \rightarrow P$

(D) $\log_x (5x - 2) = 3$; $x \neq 1$, $5x - 2 > 0$

$x^3 - 5x + 2 = 0$ $x > \frac{2}{5}$

$(x - 2)\cdot(x^2 + 2x - 1) = 0$

$x = 2, -1 - \sqrt{3}, \sqrt{3 - 1}$

Final $x = 2, \sqrt{3} - 1$

$D \rightarrow Q$